Introduction to Cloud Computing

Dr. Ndeh Ningo

Introduction to Cloud Computing

Foundations of Cloud Computing

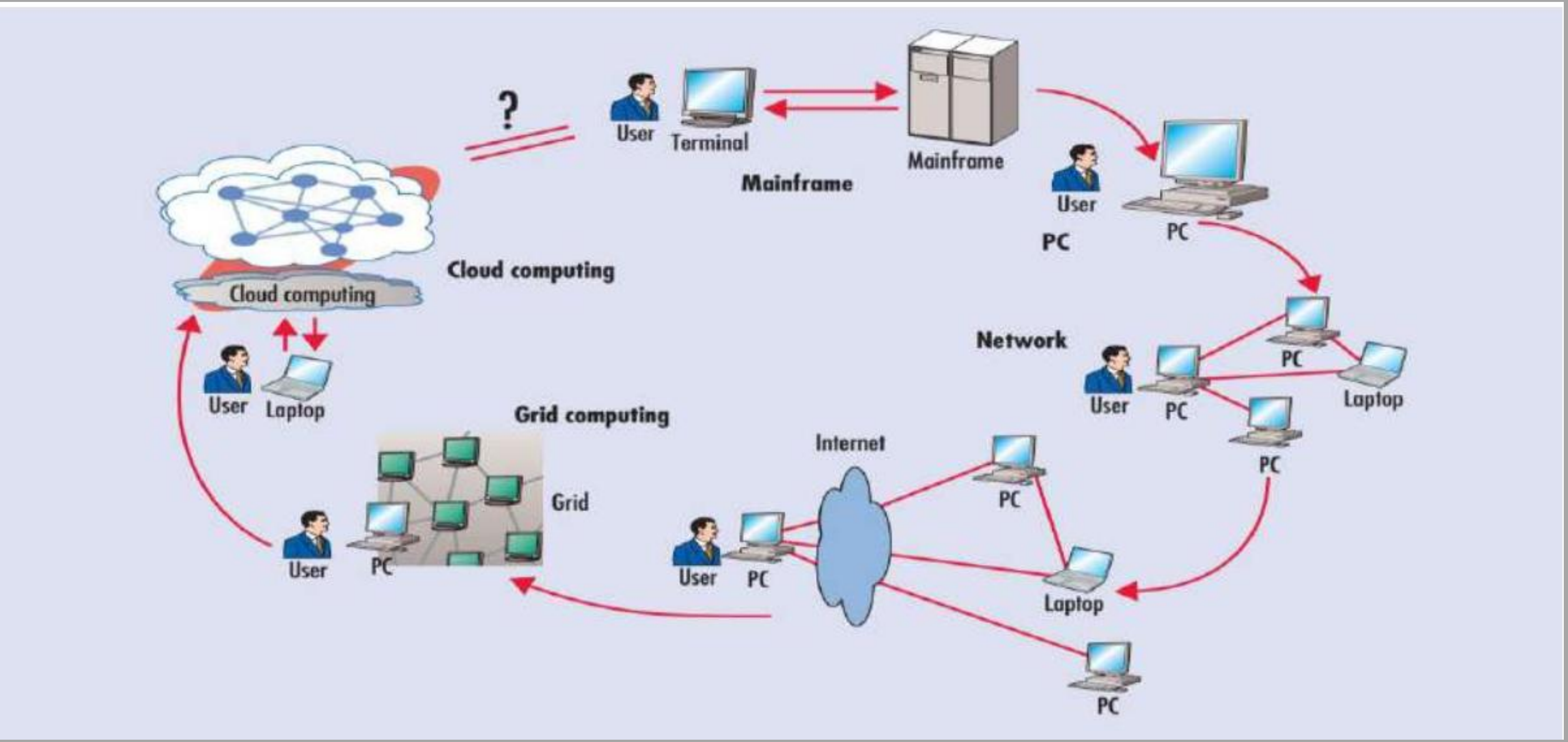
Introduction to Cloud Computing

Foundations of Cloud Computing

Historically, there have been six phases of computing archetypes:

1) dummy terminals,
2) mainframes,
3) Personal Computers,
4) network computing,
5) Grid computing, and
6) cloud computing.

Introduction to Cloud Computing

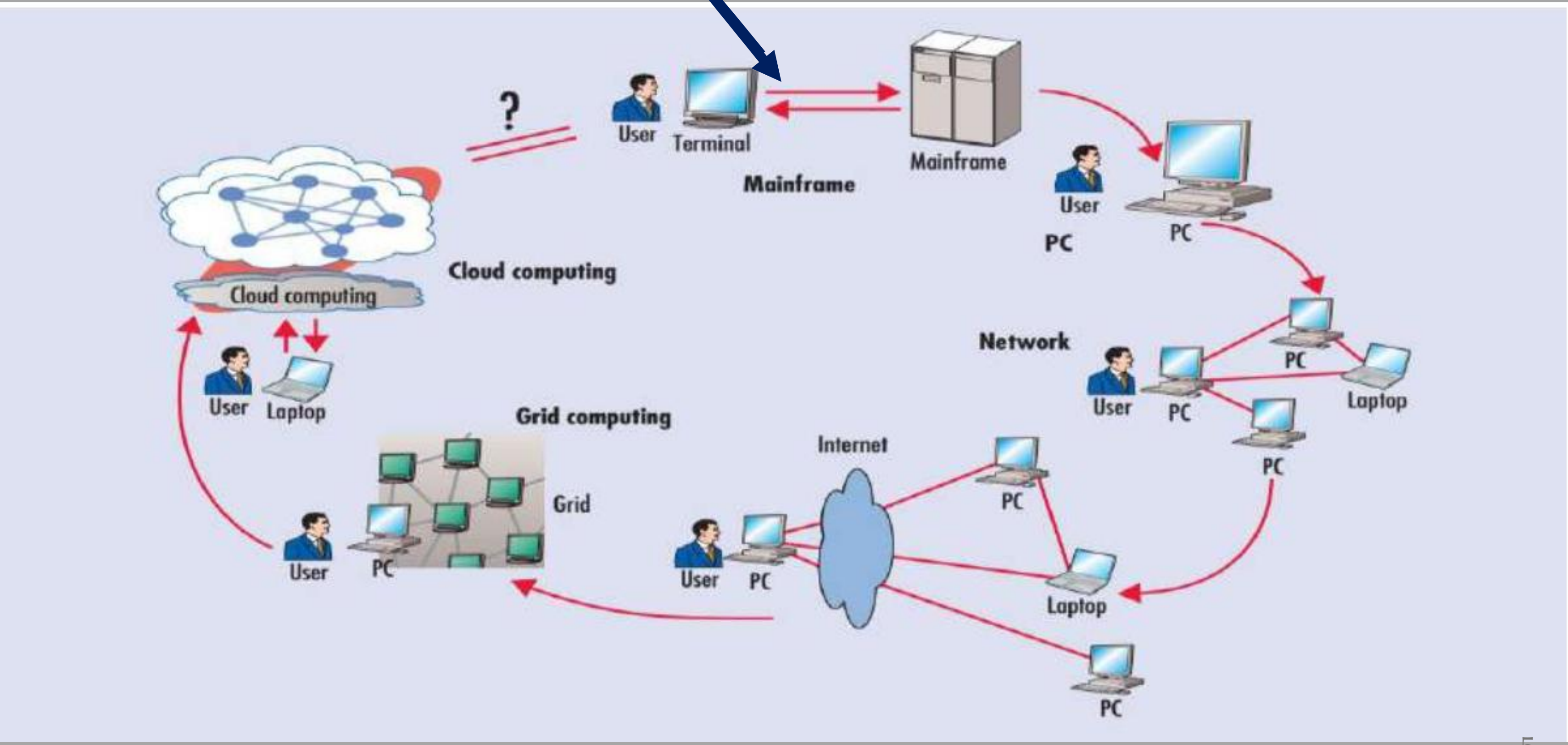


Foundations of Cloud Computing

Introduction to Cloud Computing

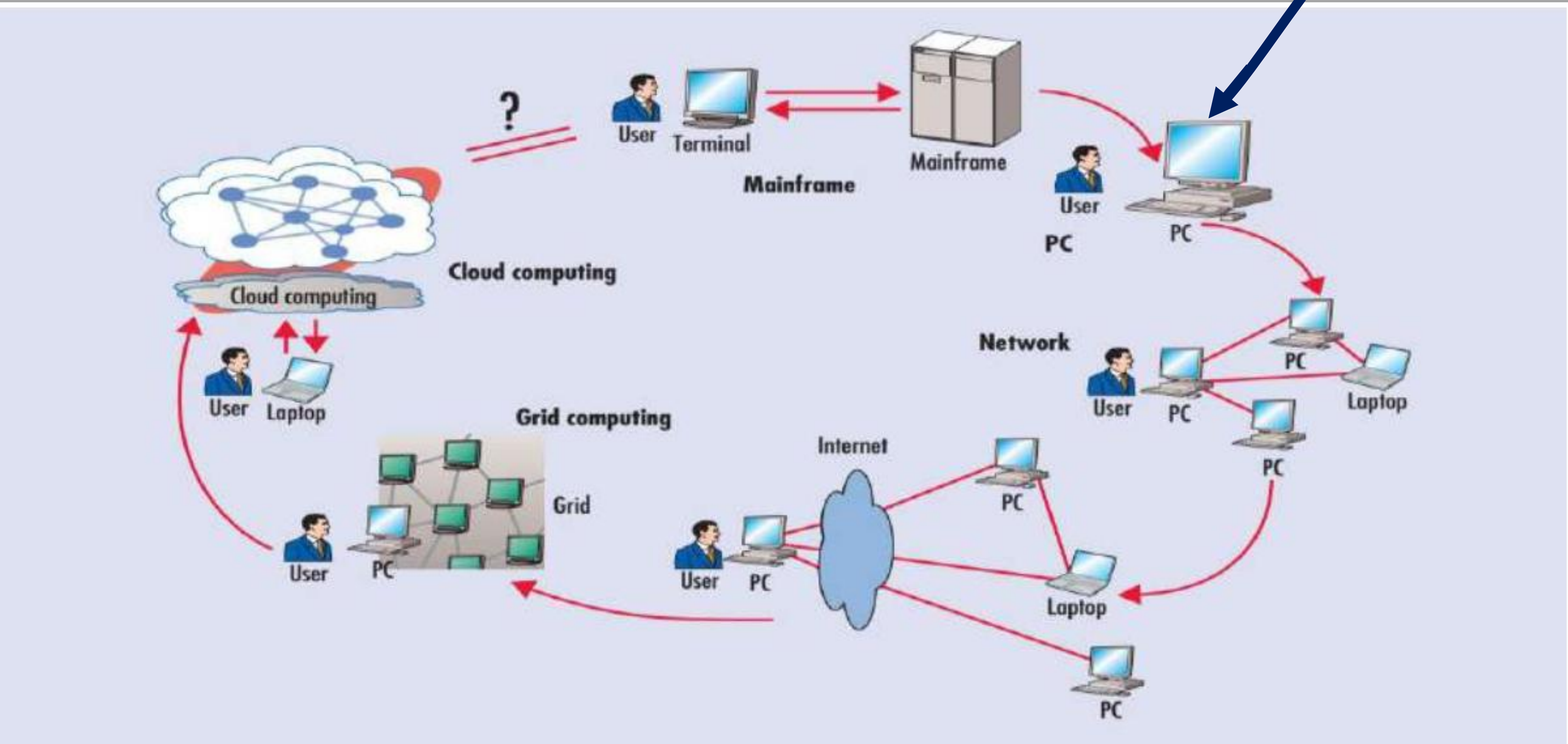
Foundations of Cloud Computing

Phase 1: Dummy terminals are used to connect to powerful hosts shared by many users. At that time, the terminals were basically keyboards and monitors.



Introduction to Cloud Computing

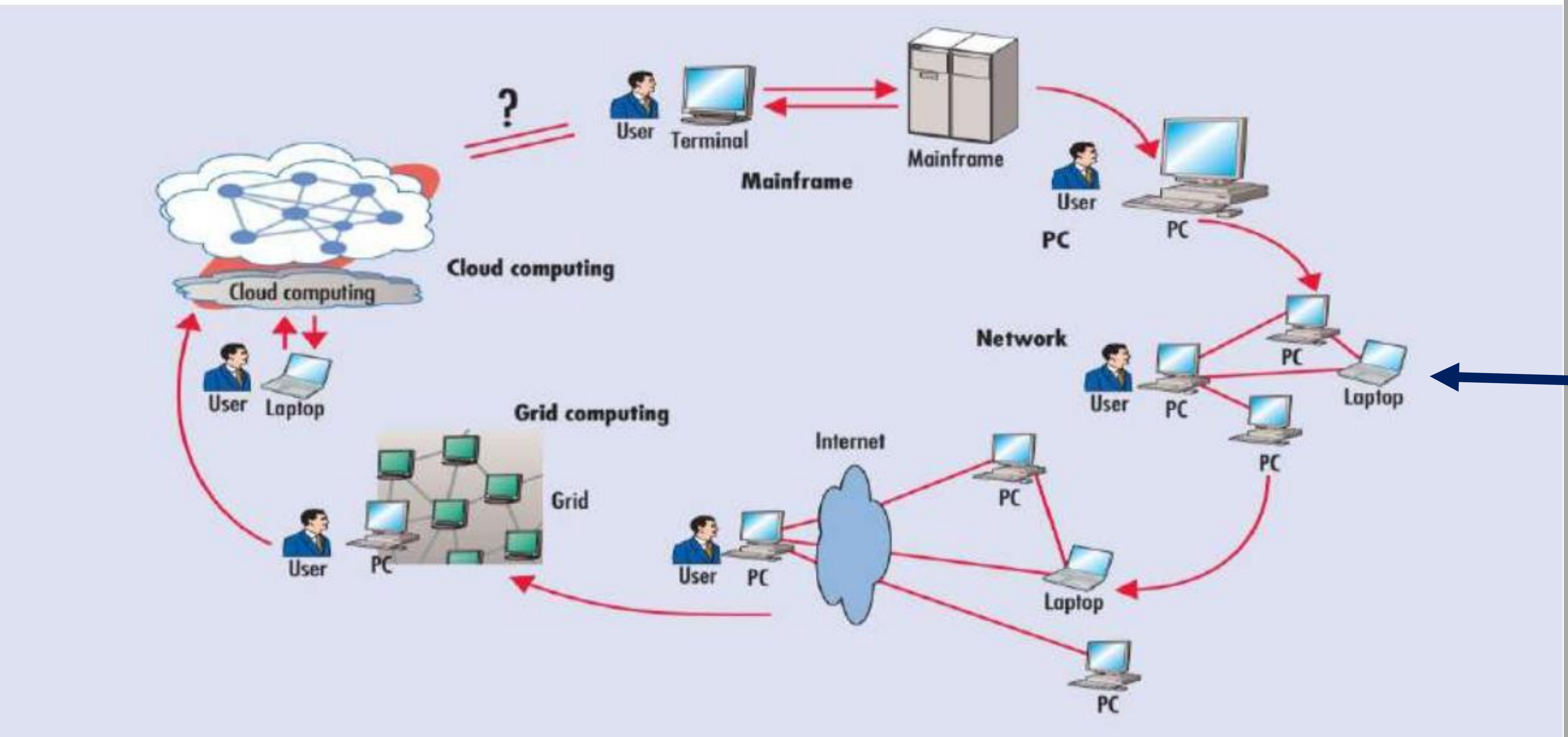
Foundations of Cloud



Phase 2: Independent **personal computers** (PCs) became powerful enough to satisfy users' daily work, which means that you didn't have to share a mainframe with anyone else.

Introduction to Cloud Computing

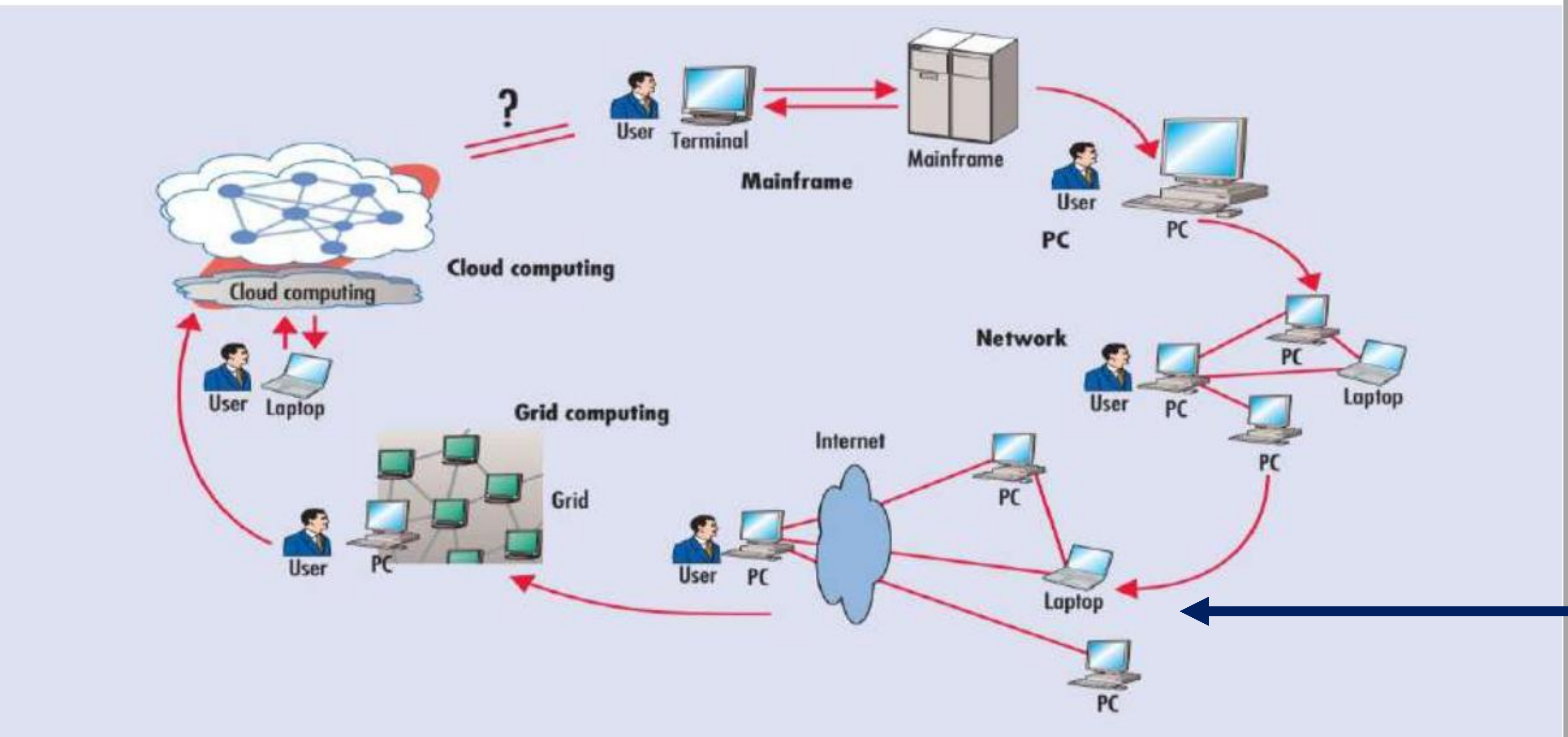
Foundations of Cloud Computing



Phase 3: A **computer network** that allowed multiple computers to connect to each other. You could work on a PC and connect to other computers through local networks (LAN) to share re-sources.

Introduction to Cloud Computing

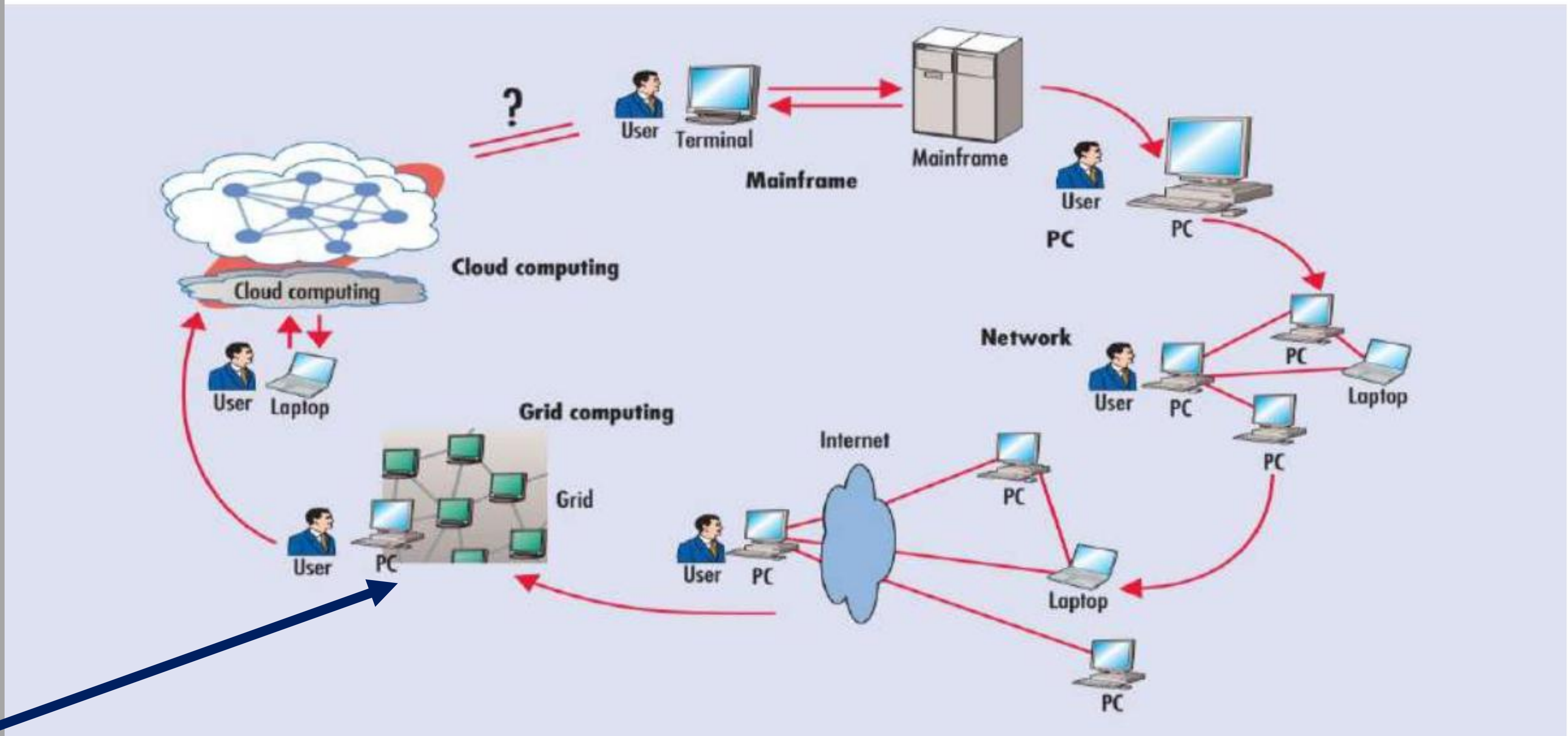
Foundations of Cloud Computing



Phase 4: **Local networks were connected to other local networks** to establish a more global network - users could now connect to the **Internet** to utilize remote applications and resources.

Introduction to Cloud Computing

Phase 5: Brought us the concept of an **electronic grid** to facilitate shared computing power and storage resources (**distributed computing**). People used PCs to access a grid of computers transparently.

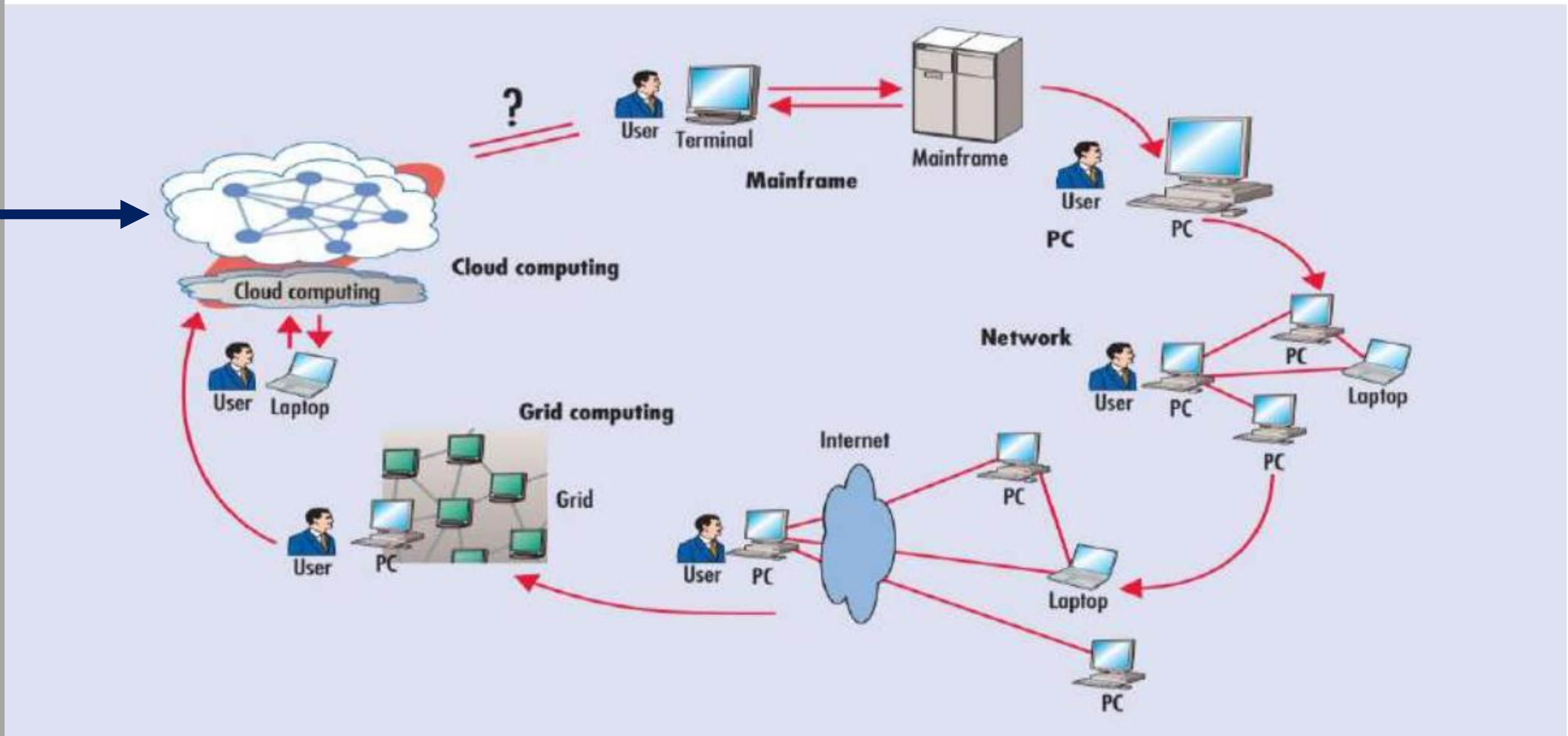


Foundations of Cloud Computing

Introduction to Cloud Computing

Foundations of Cloud Computing

Phase 6: We can leverage all available resources on the Internet in an extremely scalable and simple way by **cloud computing**.



Introduction to Cloud Computing

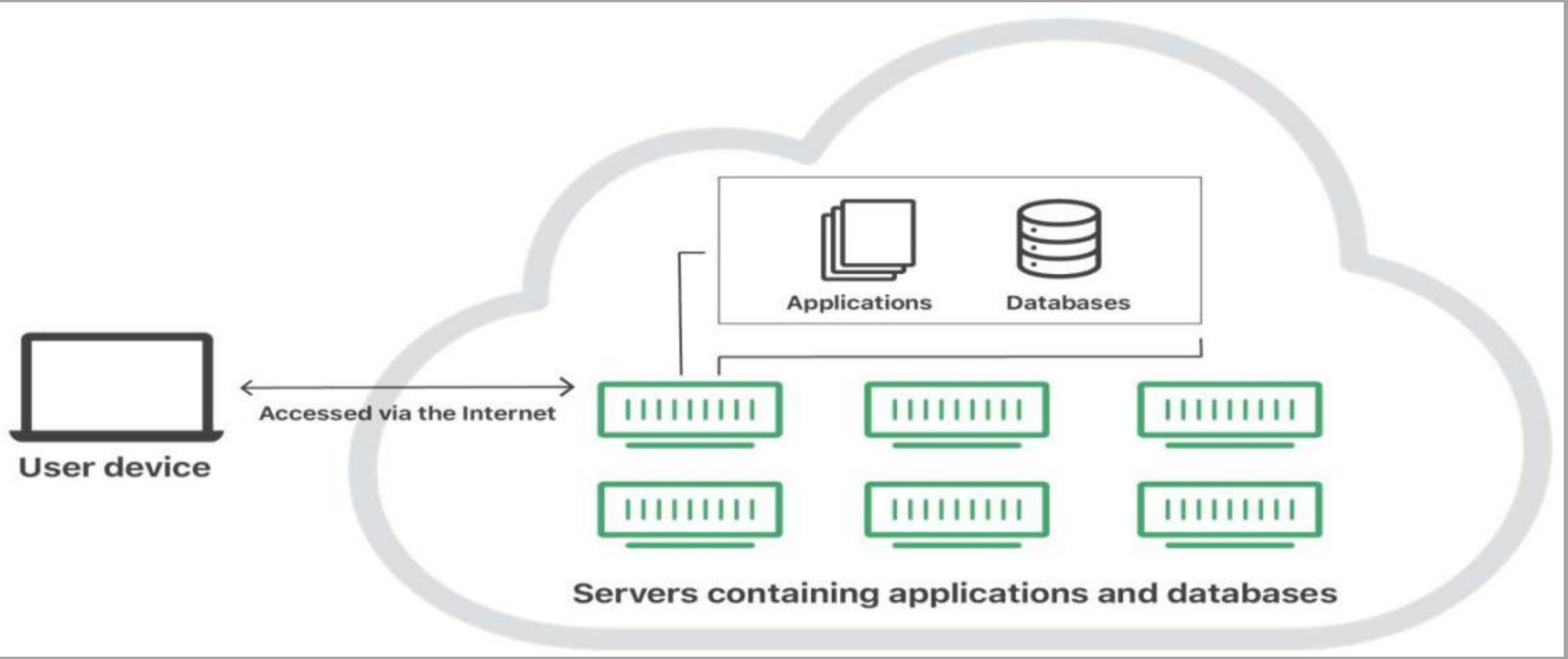
What is Cloud Computing

Introduction to Cloud Computing

Definition: Cloud

- Cloud refers to servers that are accessed over the Internet, and the software and databases that run on those servers.

Introduction to Cloud Computing



Cloud

Introduction to Cloud Computing

Most common characteristics of a cloud:

- **Clouds are a large pool of easily usable and accessible virtualized resources** (such as hardware, development platforms and/or services). These resources can be dynamically reconfigured to adjust to a variable load (scale), allowing also for an optimum resource utilization.
- This pool of resources is typically exploited by a **pay-per-use model** in which guarantees are offered by the Infrastructure Provider.

Introduction to Cloud Computing

Most common characteristics of a cloud:

1) Pay-per-use (no ongoing commitment, utility prices);
2) Elastic capacity and the illusion of infinite resources;
3) Self-service interface;
4) Resources that is abstracted or virtualized.
5) The elimination of an up-front commitment by cloudusers;

Introduction to Cloud Computing

Definition

Cloud computing is the on-demand delivery of compute power, database, storage, applications, and other IT resources via the internet with pay-as-you-go pricing.

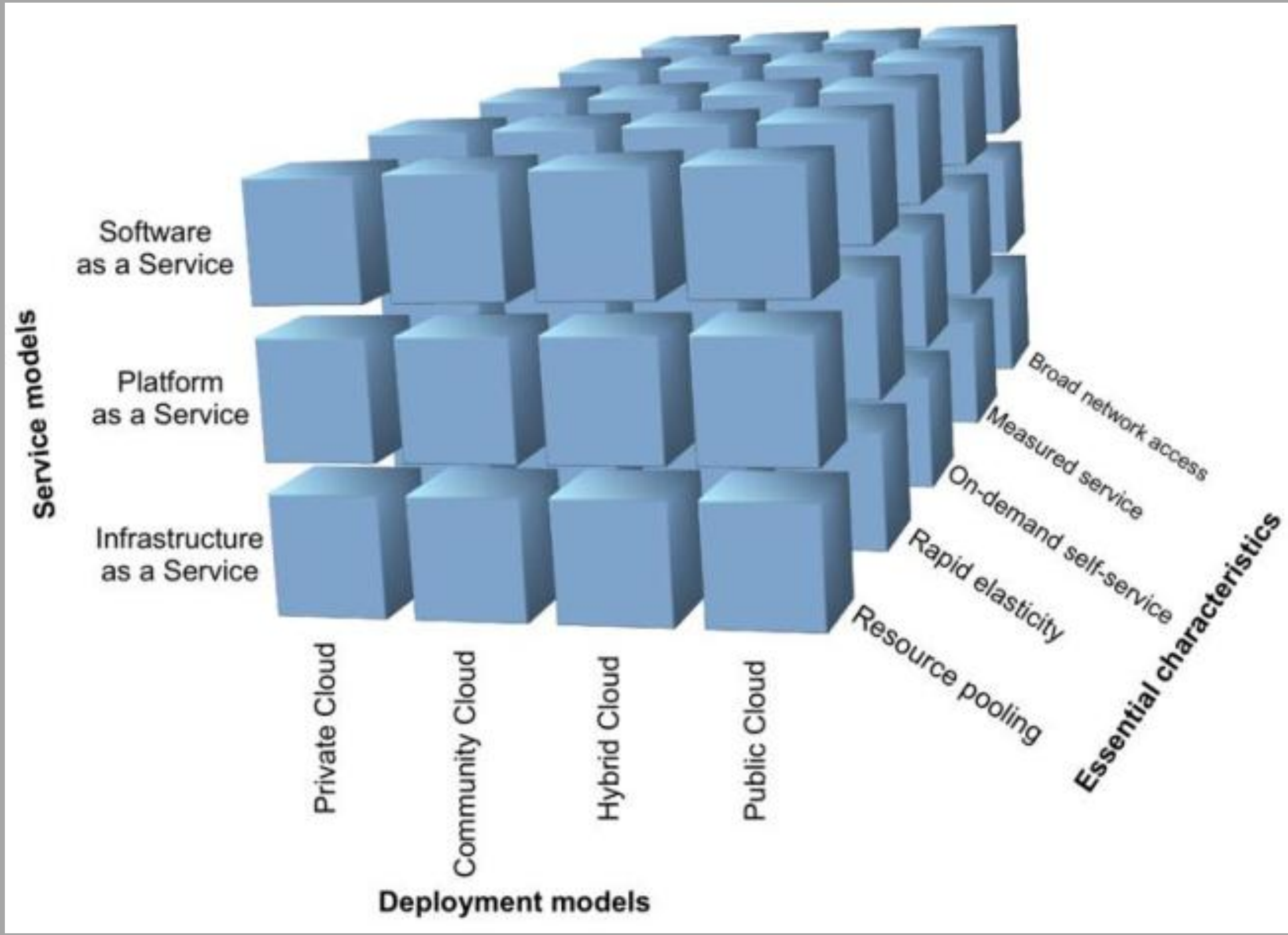
Cloud computing provides a simple way to access servers, storage, databases and a broad set of application services over the internet.

Introduction to Cloud Computing

NIST Definition: Cloud Computing

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model is composed of five essential characteristics, three service models, and four deployment models.

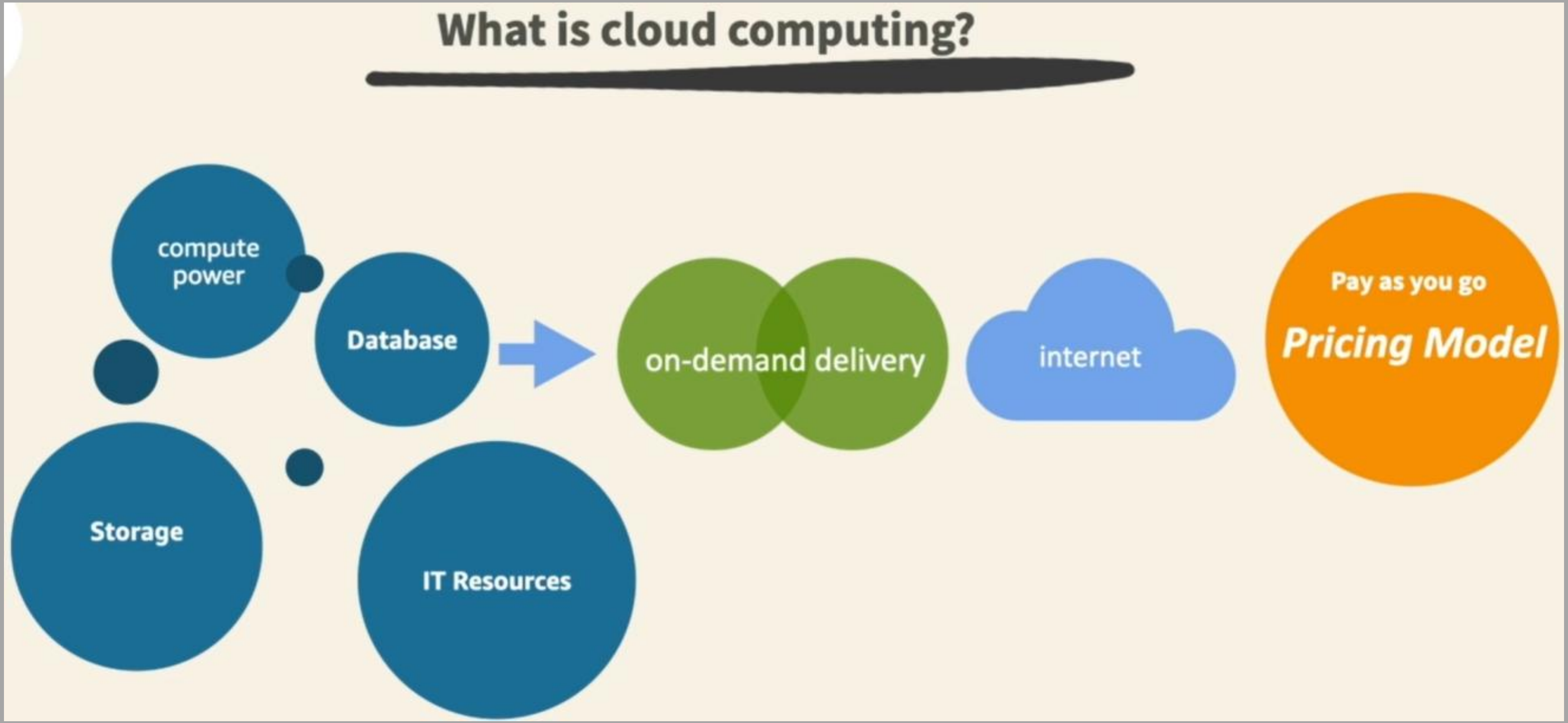
Introduction to Cloud Computing



NIST model in three dimensions consisting of

- service models,
- deployment models, and
- essential features

Introduction to Cloud Computing



Introduction to Cloud Computing

Definition

Cloud computing is the **on-demand delivery** of compute power, database, storage, applications, and other IT resources (through a cloud services platform) via the internet with pay-as-you-go pricing.

- **On-demand delivery** indicates that the cloud provider has the resources the customer needs when the customer needs them.

Introduction to Cloud Computing

Definition

- The customer does not give advance notice that the resources are going to be needed. Suppose there is a suddenly need for 300 virtual servers or for 2,000 terabytes of storage. Well, just a few clicks and launch them and just start using them. When they are no longer needed, just as quickly you can return them and stop paying immediately.

Introduction to Cloud Computing

Definition

Cloud computing is the on-demand delivery of compute power, database, storage, applications, and other IT resources (through a cloud services platform) via the internet with pay-as-you-go pricing.

- Over the Internet seems simple enough, but it implies that you can access those resources using a secure web page console or programmatically. No additional contracts or sales calls are needed

Introduction to Cloud Computing

Definition

Cloud computing is the on-demand delivery of compute power, database, storage, applications, and other IT resources (through a cloud services platform) via the internet with pay-as-you-go pricing.

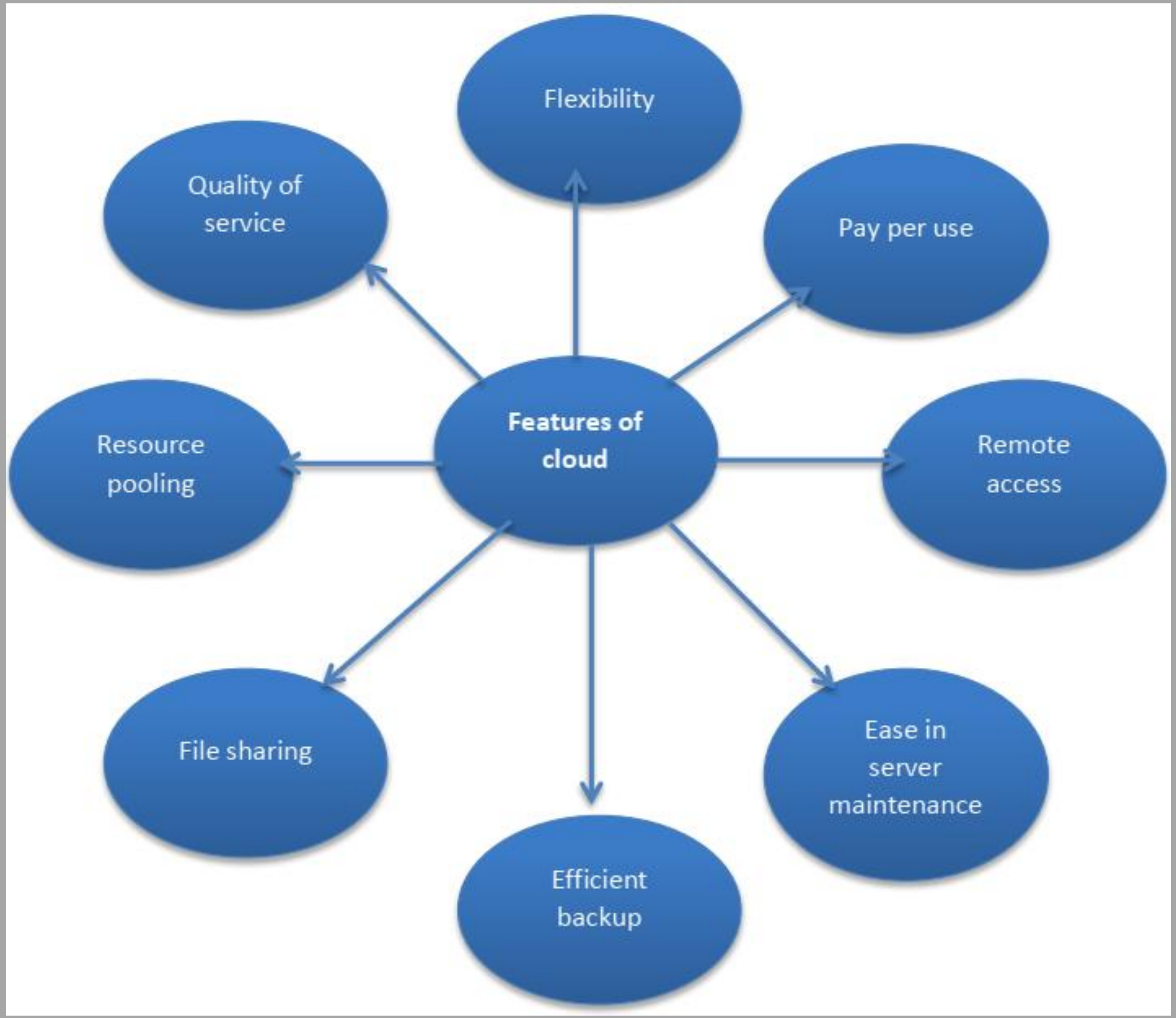
- With pay-as-you-go pricing indicates that the customer only pays for the resources when they are being used.

Introduction to Cloud Computing

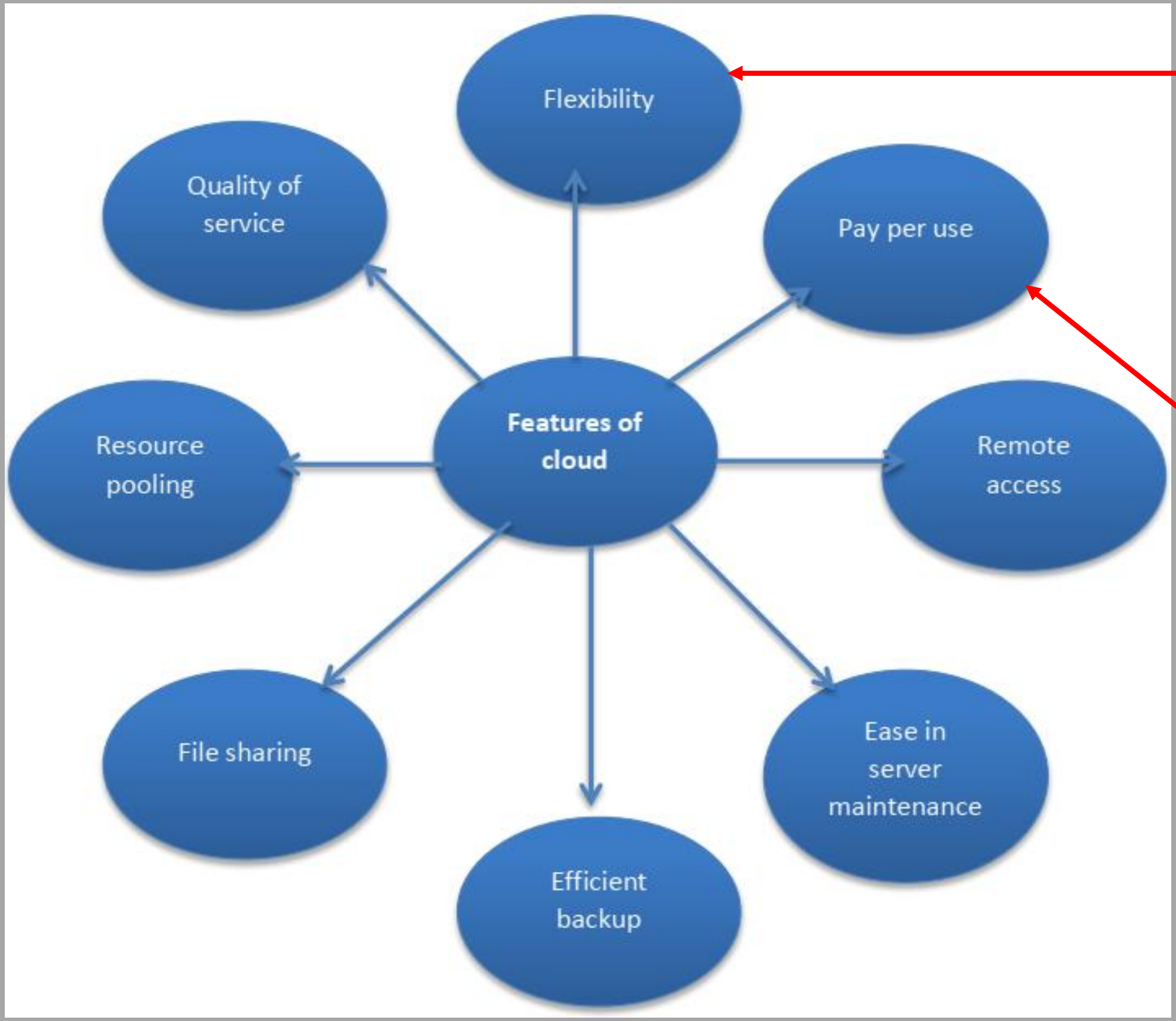
Essentially cloud computing delivers computing as a utility and can be defined as “on demand delivery of infrastructure, applications, and business processes in a security-rich, shared, scalable, and based computer environment over the Internet for a fee”.

In other words, through cloud computing, computing resources, such as computation and storage, are packaged as metered services similar to public utilities such as water and electricity wherein usage is measured and one pays only for the quantity used.

Introduction to Cloud Computing



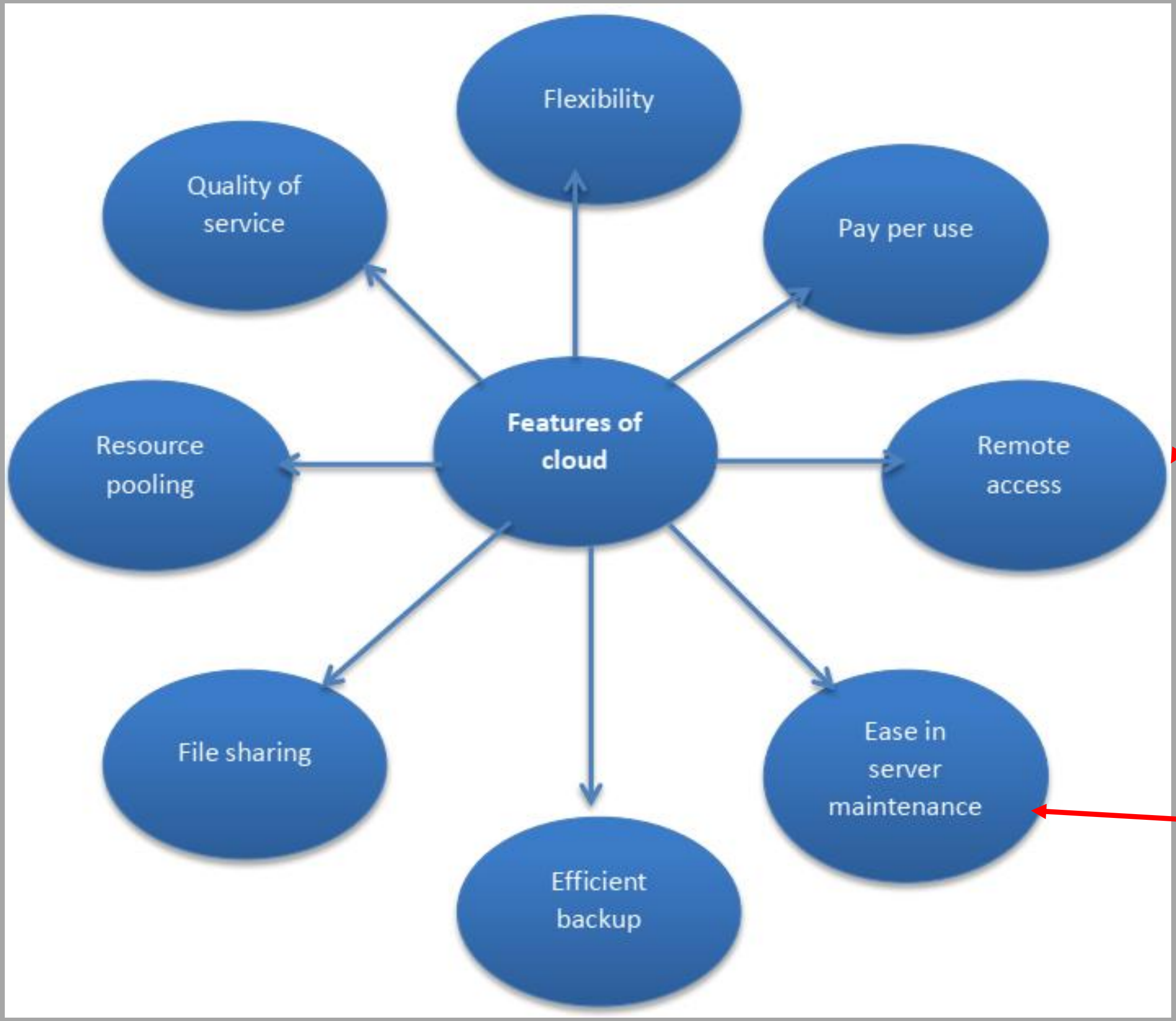
Introduction to Cloud Computing



Flexibility: The number of users using the cloud services remains flexible and can be increased or decreased depending on the current need.

Pay-per-use: Different abstraction models in cloud computing ensures that the users are paying for the cloud services or platform, only on the basis of the services they are using at that moment.

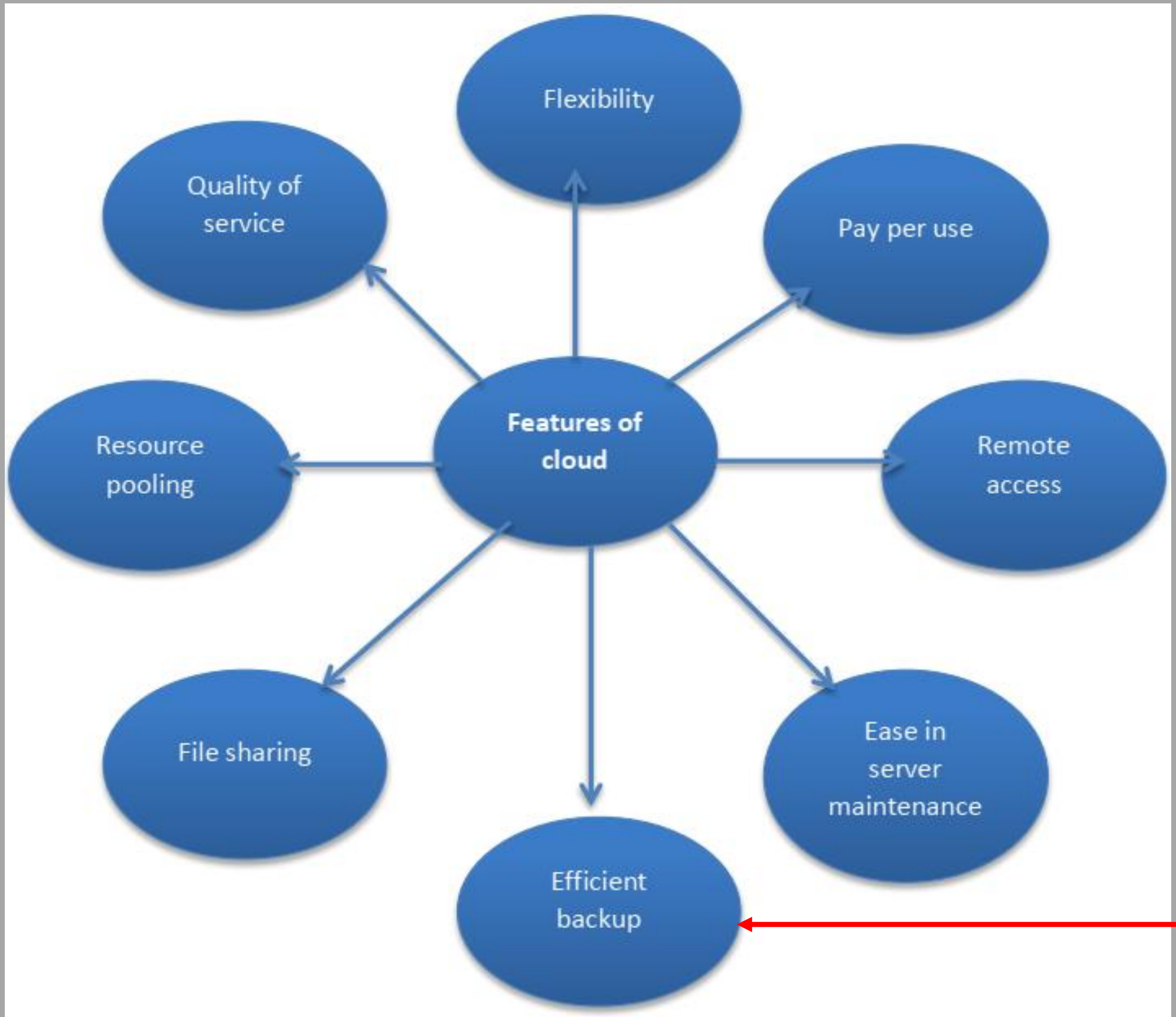
Introduction to Cloud Computing



Accessed remotely: Cloud is essentially location-free. It allows the users to work from anywhere as long they have access to an Internet connection. This allows many employees to work on their comfort from their own homes, which ensures a good work-life balance and an increase in productivity.

Ease in server maintenance: For public cloud, since servers are located off-site, it is the responsibility of the providers to maintain these servers. Users are free of the burden of trying to install the server and updating it as needed.

Introduction to Cloud Computing



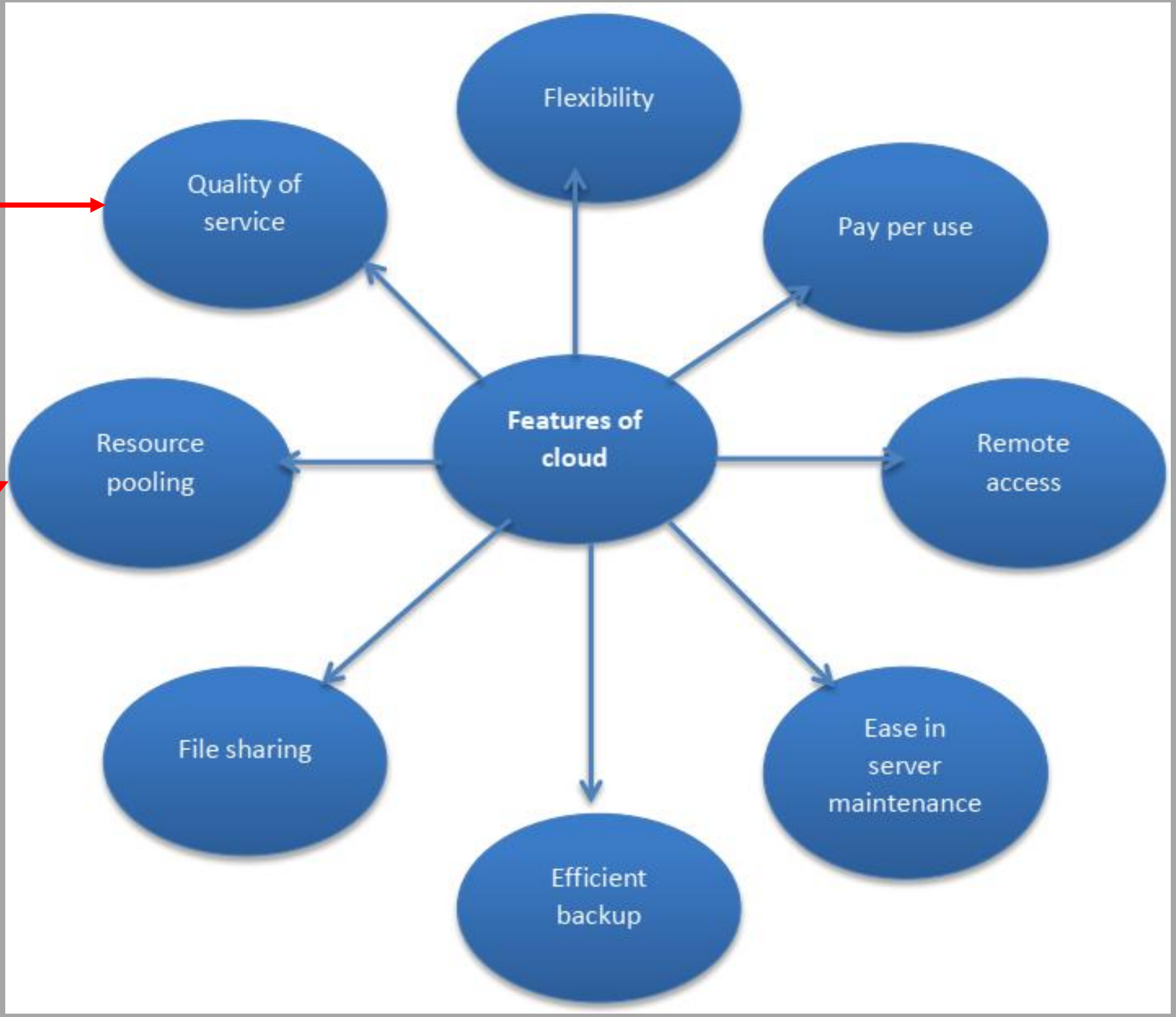
Efficient backup:

- Using cloud-based backups, a lot of time is saved. It also decreases the involved costs.
- Using the cloud also reduces the size of the **data centers** (**DC**).
- Via cloud-based backups the cost towards acquiring servers and software is reduced.

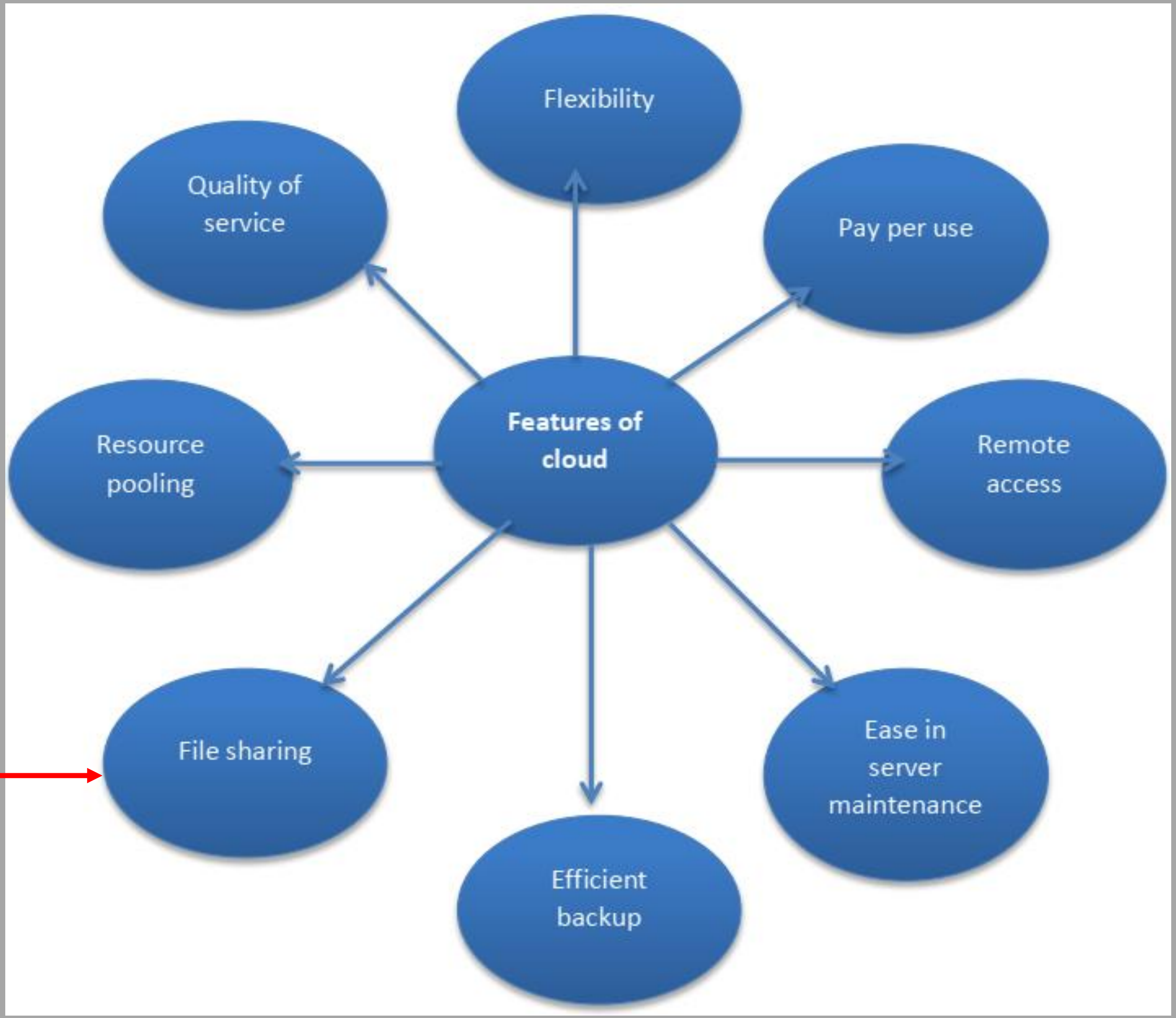
Introduction to Cloud Computing

Quality of service: Cloud computing really assures that the services users get is of the top-notch quality. Service providers do not compromise on the quality. The agreements include round-the-clock availability, adequate resources, performance as well as bandwidth.

Resource pooling: In cloud computing, many resources are brought together to assist larger number of users simultaneously as in the case of public and community cloud. These resources are dynamically allocated and de-allocated on the basis of demand. The system is elastic enough to satisfy huge requirement of resources without exhausting the resources.



Introduction to Cloud Computing



Cloud-based file sharing: Cloud allows streamline teamwork collaboration since data can be shared, accessed and edited from anywhere in the world. Updates can be seen in real-time by all the teammates.

Introduction to Cloud Computing

Key Characteristics of cloud computing

Introduction to Cloud Computing

Key characteristics of cloud computing

- The illusion of infinite computing resources
- The elimination of an up-front commitment by cloud users
- The ability to pay for use…as needed

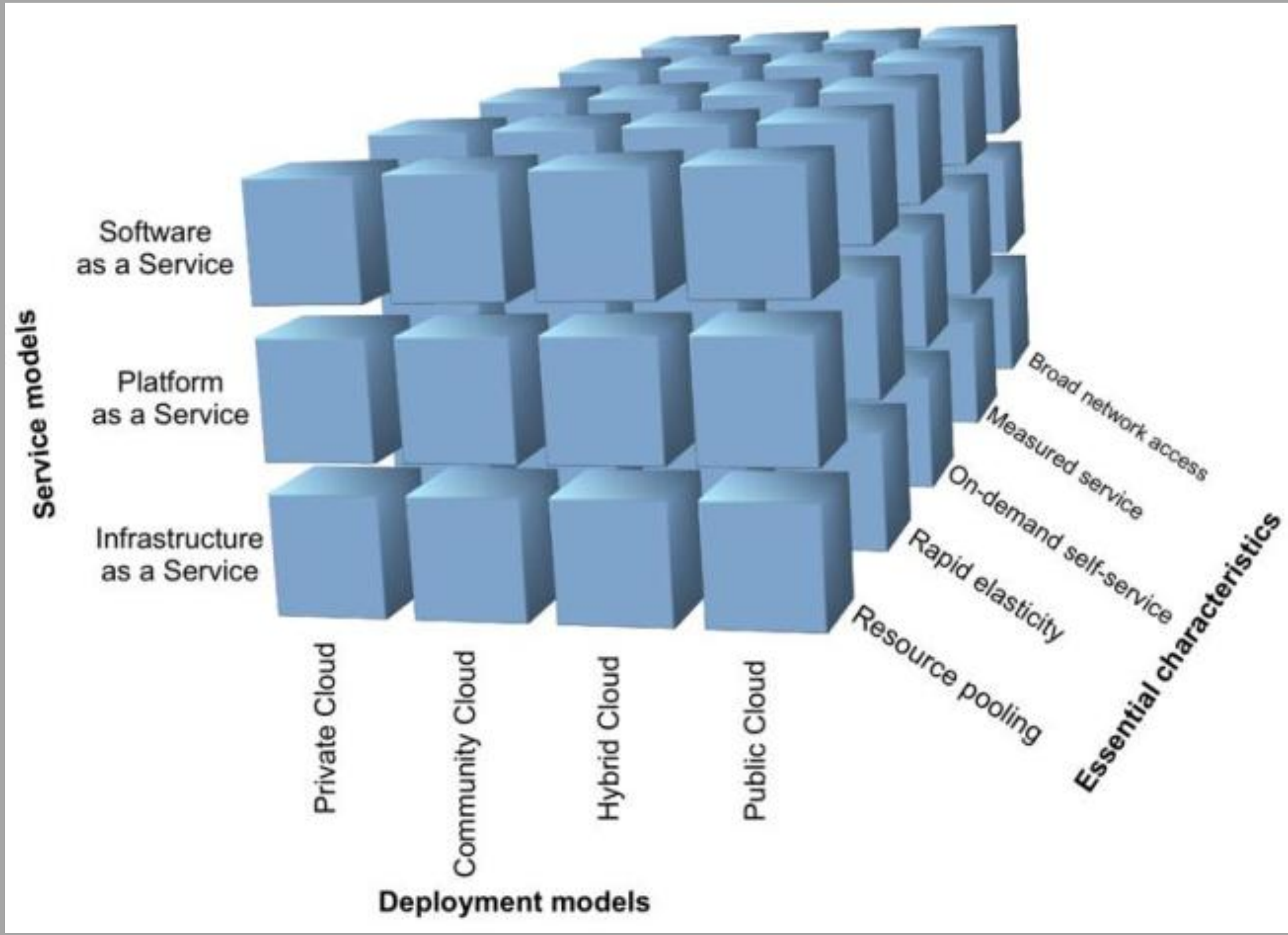
Introduction to Cloud Computing

Let us repeat the NIST definitions of cloud computing from which we can deduce its key characteristics.

NIST Definition: Cloud Computing

Cloud computing is a model for enabling ubiquitous, convenient, on demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model promotes availability and is composed of five essential characteristics, three service models, and four deployment models.

Introduction to Cloud Computing



NIST model in three dimensions consisting of

- service models,
- deployment models, and
- essential features

Introduction to Cloud Computing

Key characteristics of cloud computing

According to the NIST definition, cloud computing has the following five key characteristics:

1) broad network access,
2) on-demand self-service,
3) resource pooling or shared services,
4) rapid elasticity, and
5) measured service.

Every single one of these properties needs to be present for a service to qualify as a cloud service. It is very important to understand this because otherwise the service is just not a cloud service, despite the claims made by would-be cloud service providers.

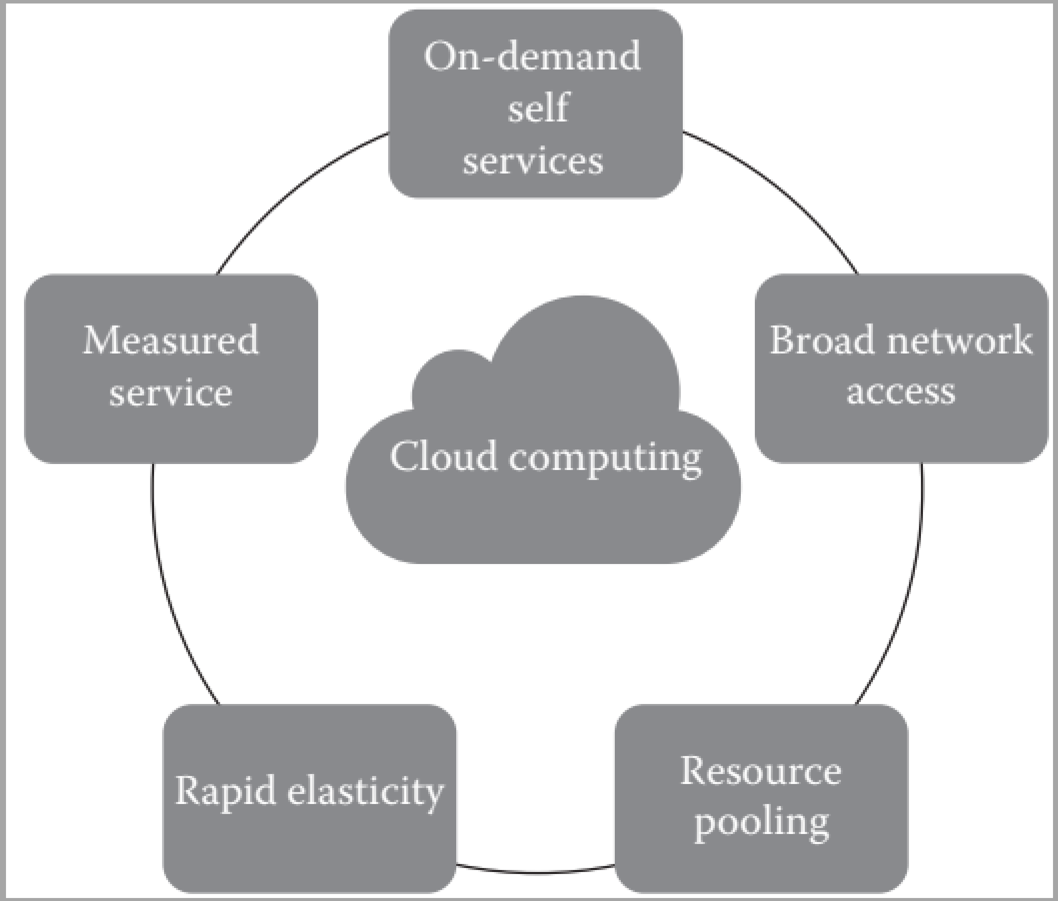
Introduction to Cloud Computing

Characteristic	Description	Parameter
Broad network access	Consume services from anywhere.	Where
On-demand self-service	Consume services when you want.	When
Resource pooling and virtualization	Pool the infrastructure, virtual platforms, and applications.	How
Rapid elasticity	Share pooled resources to enable horizontal scalability.	How
Measured service	Pay for the service you consume as you consume it.	How much

Characteristics of cloud computing

Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

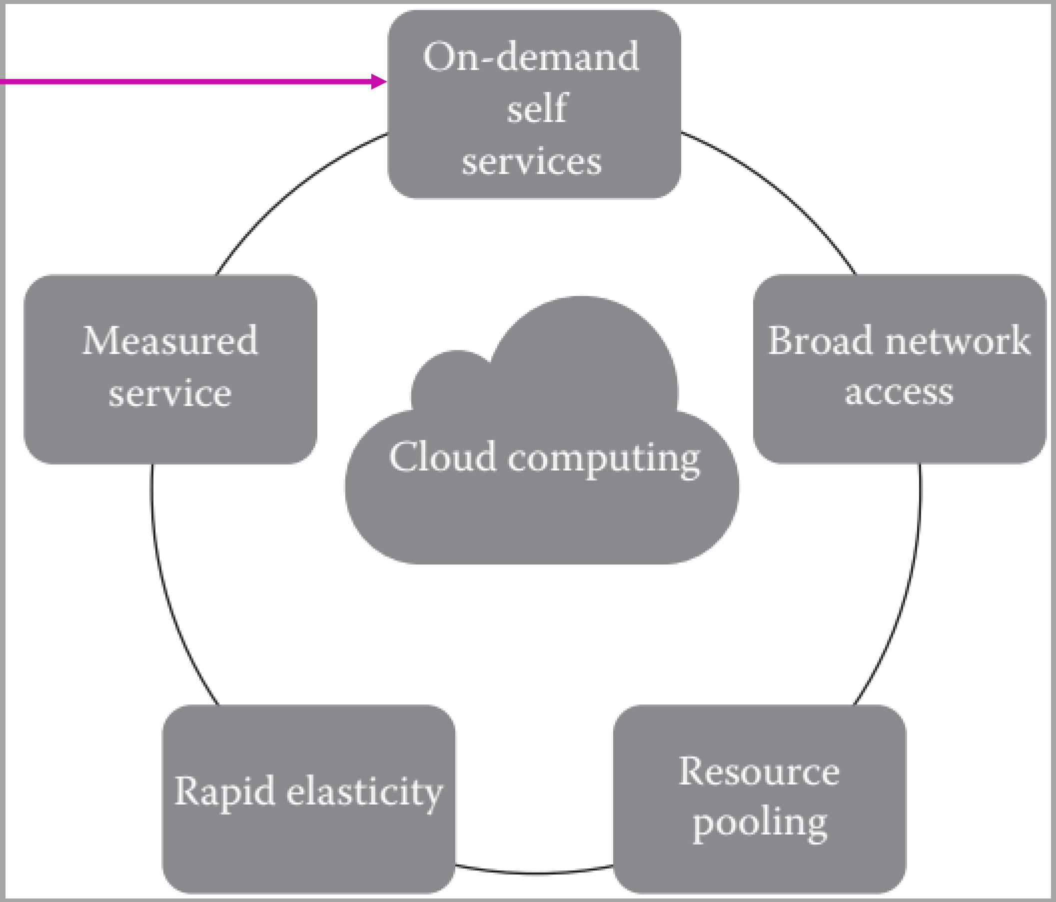


Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

On-Demand Self-Service

- Enables consumers to get computing resources when required, without any human or cloud provider intervention.
- Facilitates consumer to leverage "ready to use" services or enables choosing required services from the service catalog.
- Allows provisioning of resources using self-service interface.
- Self-service interface should be user-friendly.

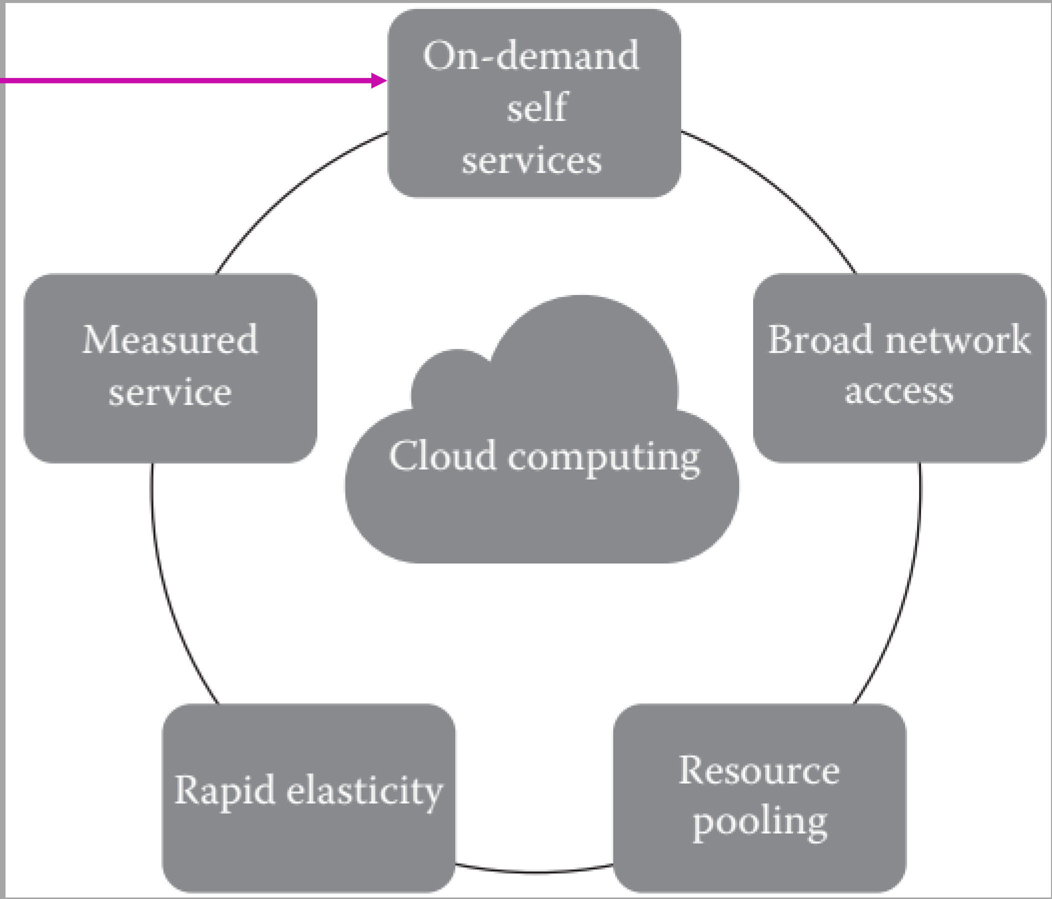


Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

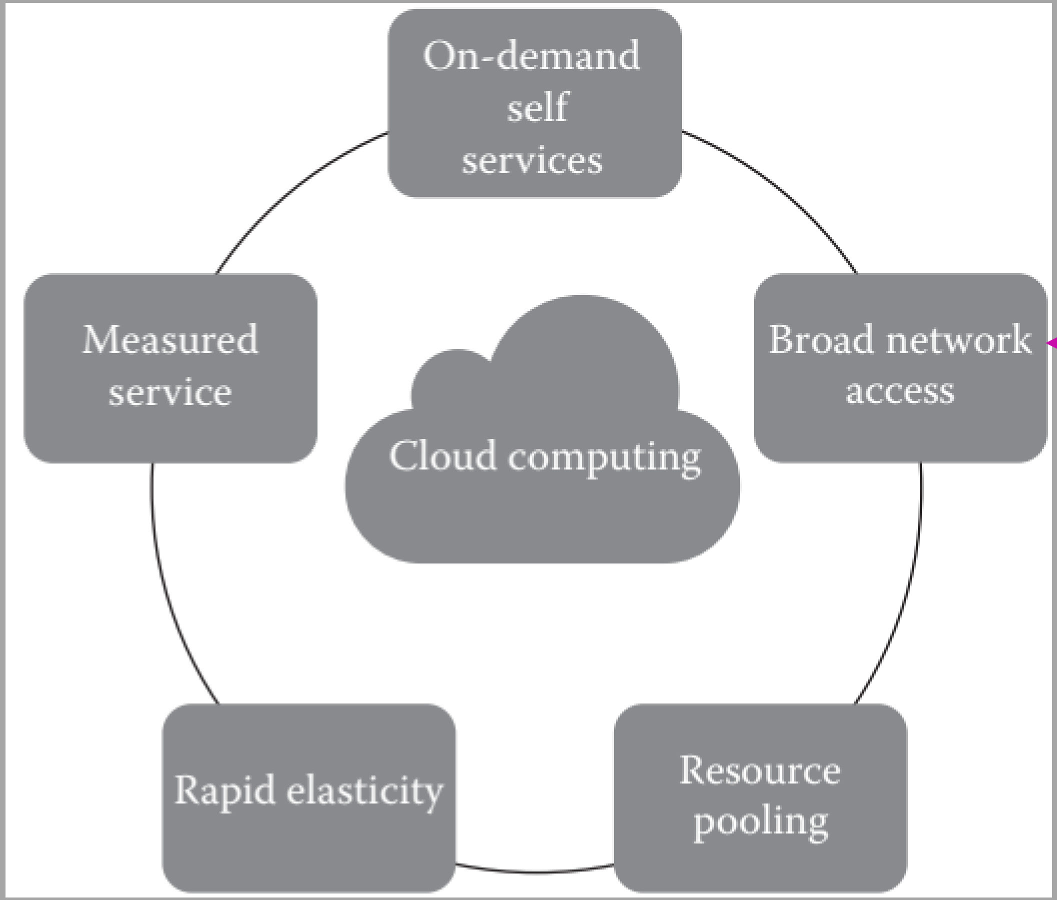
On-Demand Self-Service

On-demand self-service highlights the need for the end user to be able to provision computing resources by him/herself. On-demand self-service resulted in agility and flexibility.



Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

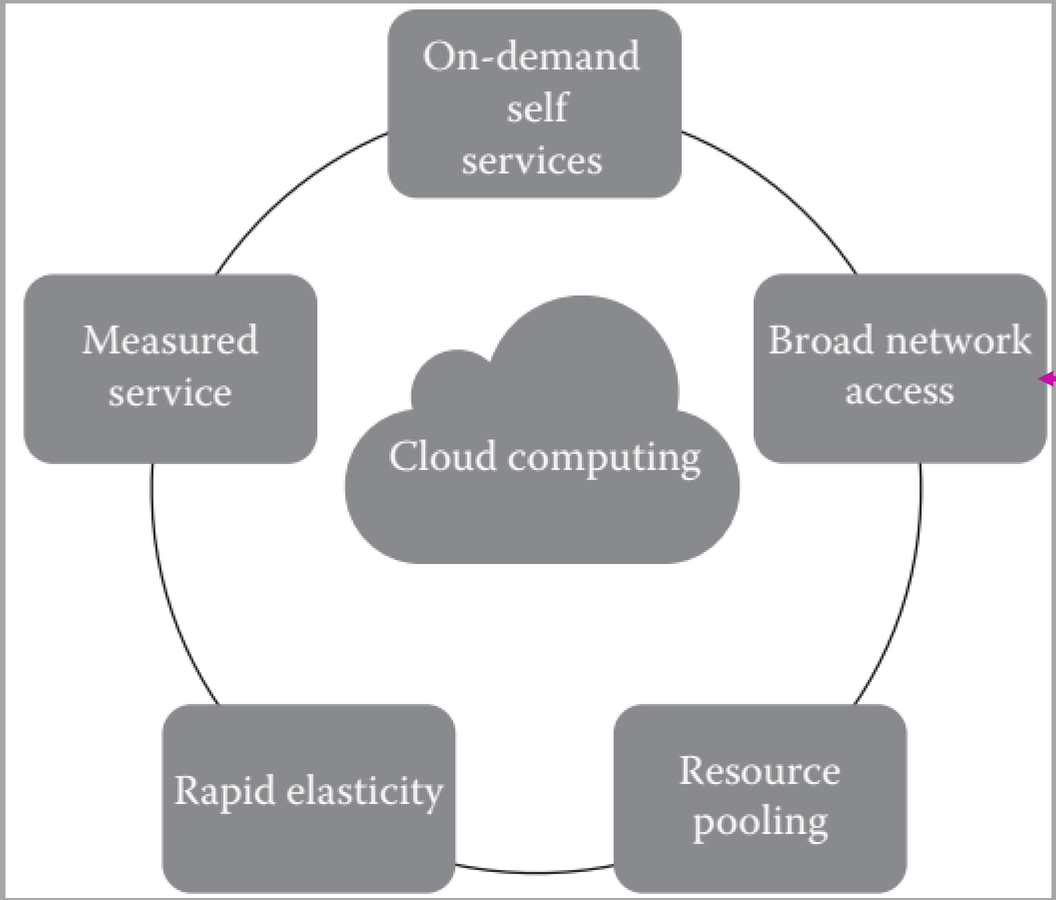


Broad Network Access

- Cloud services are accessed via the network, usually the internet, from a broad range of client platforms such as:
  - Desktop computer
  - Laptop
  - Mobile phone
  - Thin Client
- Eliminates the need for accessing a particular client platform to access the services.
- Enables accessing the services from anywhere across the globe at anytime.

Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

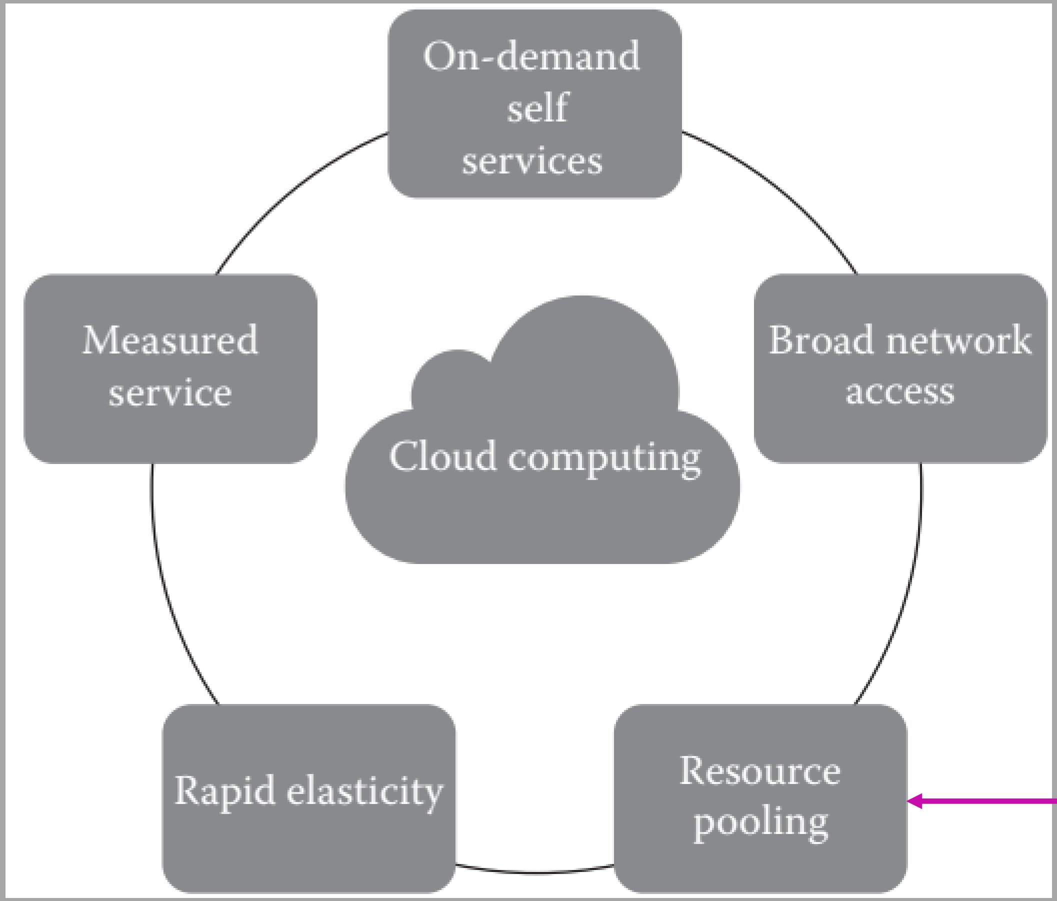


Broad Network Access

- **Broad network access** in practice refers basically to services being available through the Internet since most cloud computing is public cloud.
- Basically, anyone with a laptop, table, or cell phone should be able to access the computing resources.

Introduction to Cloud Computing

Essential Characteristics of Cloud Computing



Resource Pooling

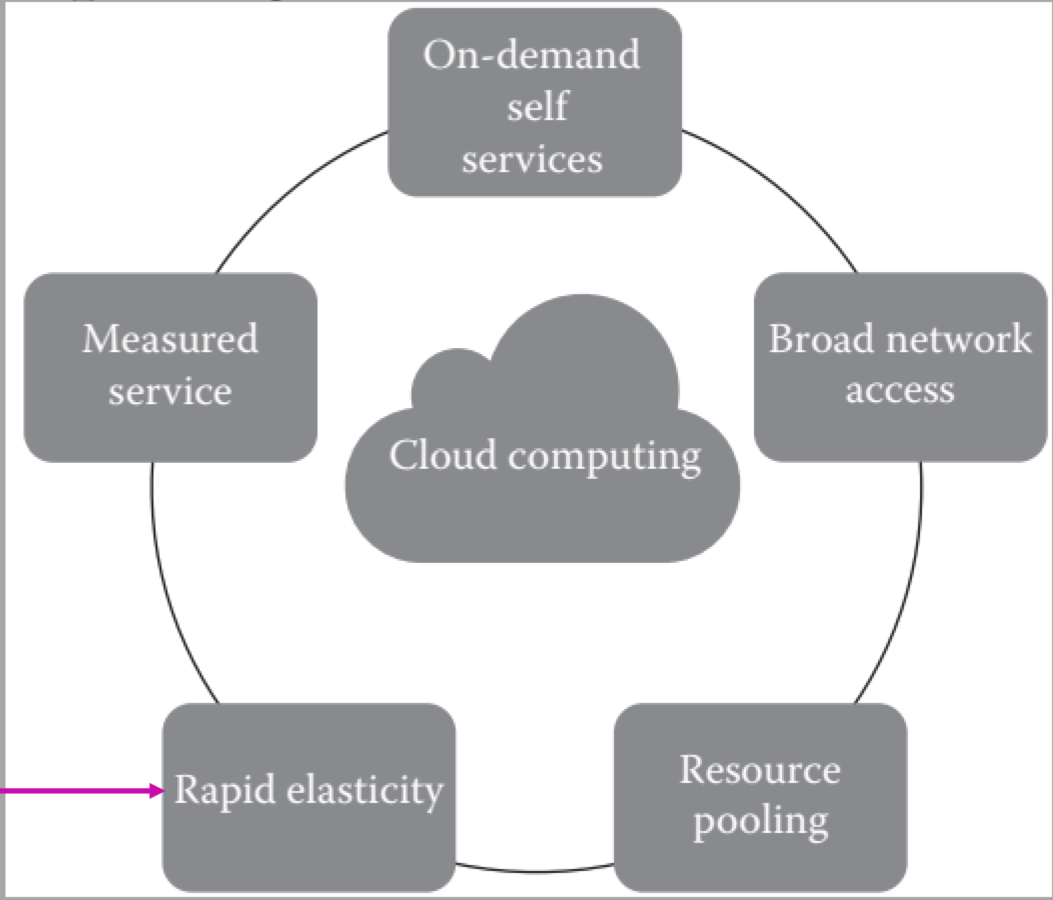
- IT resources (compute, storage, network) are pooled to serve multiple consumers
  - Based on multi-tenant model
- Consumer has no knowledge about the exact location of the resources provided.
- Resources are dynamically assigned and reassigned based on the consumer demand.

Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

Rapid Elasticity

- Ability to scale IT resources rapidly, as required, to fulfill the changing needs without interruption of service
  - Resources can be both scaled up and scaled down or scaled out and scaled in dynamically
- To the consumer, the Cloud appears to be infinite
  - Consumers can start with minimal computing power and can expand their environment to any size.

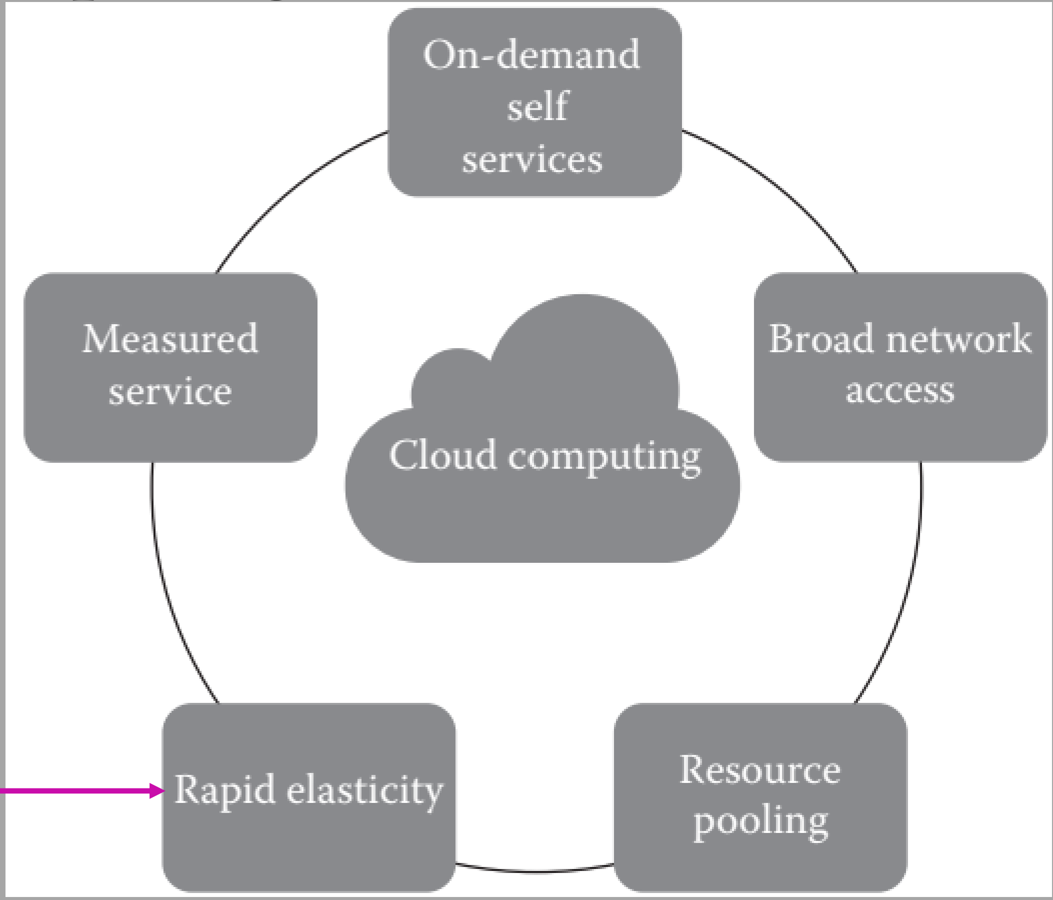


Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

Rapid Elasticity

- Elastic characteristic means that cloud computing has the ability to assign resources when they are needed by the customers and to remove them when they are no longer needed.
- Cloud computing services should have IT resources that can be scaled out and in quickly and on a need basis, so that they can be automatically provisioned and released.
- The usage, capacity, and cost can be scaled up or down with no additional contract or penalty.

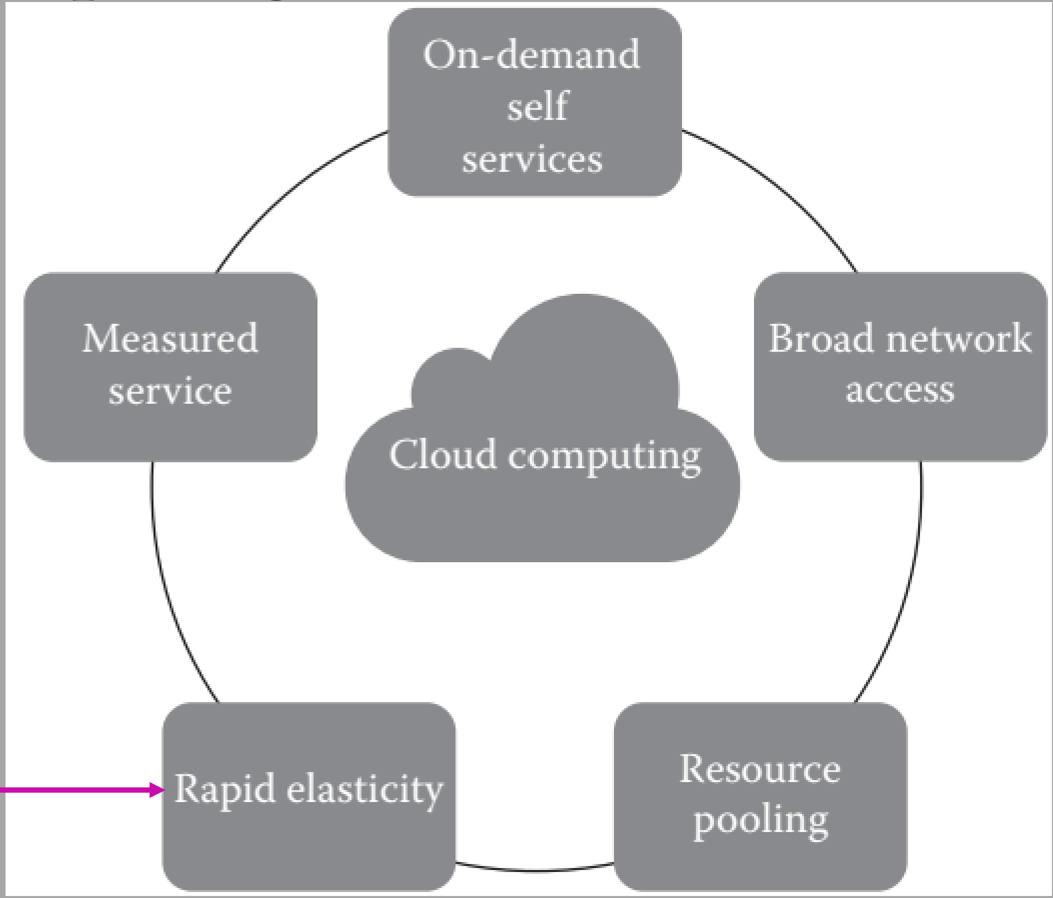


Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

Rapid Elasticity

- **Rapid elasticity means that** if a service like a website suddenly has a lot more users because of a flash sale, for example, it needs to be able to scale up the computing power to handle this quickly. Then, when the sale is over, it should scale back down like an elastic band that stretches and retracts when force is no longer applied.

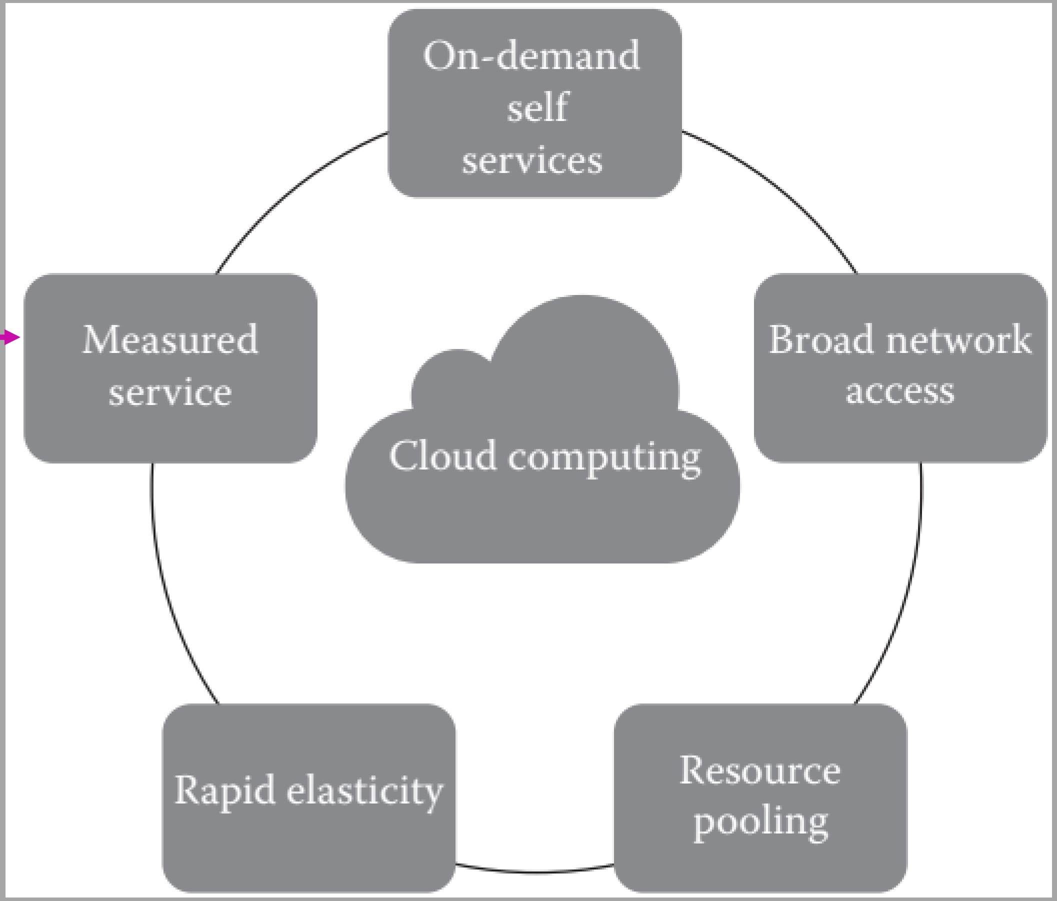


Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

Metered Service

- Consumers are billed based on the metered usage of cloud resources
  - Cost incurred on a pay-per-use basis;
  - Pricing/billing model is tied to the required service levels
- Resource usage is monitored and reported, which provides transparency for charges back to both cloud service provider and consumer about the utilized service.

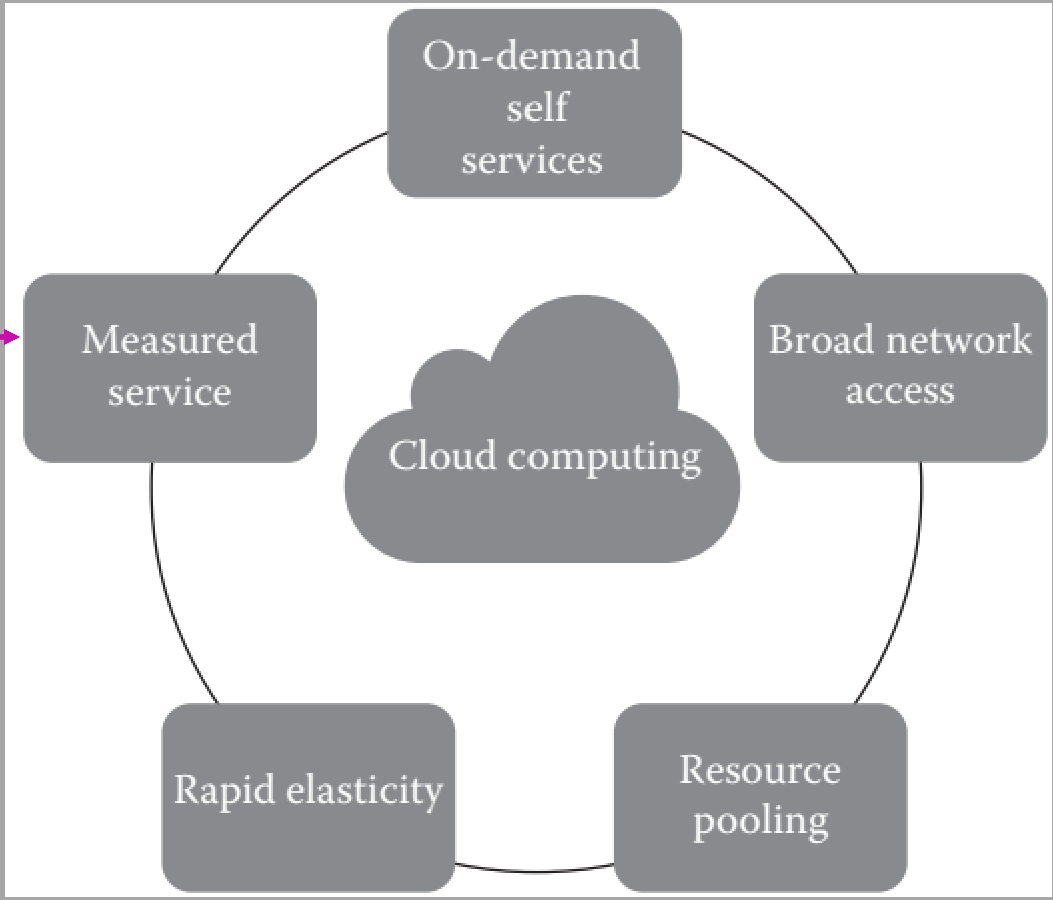


Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

Metered Service

- **Measured service** basically means that the customer should only pay for what they use to avoid paying for a machine standing idle in a data center. This should be transparent to the customer. In practice the unit used to measure can differ greatly. It could be time, storage, processing capacity, number of users, number of instructions, or any other measurable property of relevance to the service. The most common are time and number of users.



Introduction to Cloud Computing

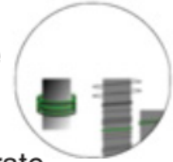
Essential Characteristics of Cloud Computing

Putting it all together

1. On-demand self-service. A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.

2. Broad network access. Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).

3. Resource pooling. The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. There is a sense of location independence in that the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or data center).



4. Rapid elasticity. Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.

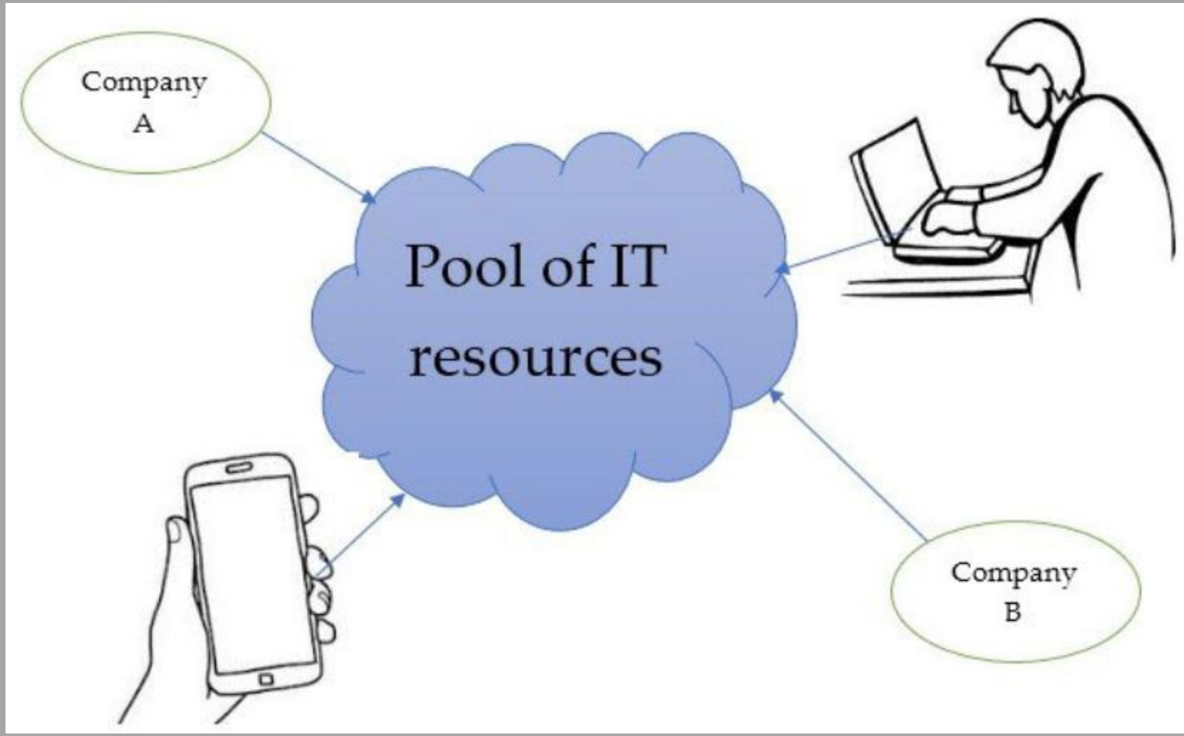
5. Measured service. Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

Multitenancy

- Cloud computing supports a multi-tenant model. By multi-tenancy, we mean that multiple customers can share the same resource or application while maintaining privacy and security.
- With this, only one server instance can run among a large number of customers and updates can be deployed to them easily.
- This is also cost efficient.

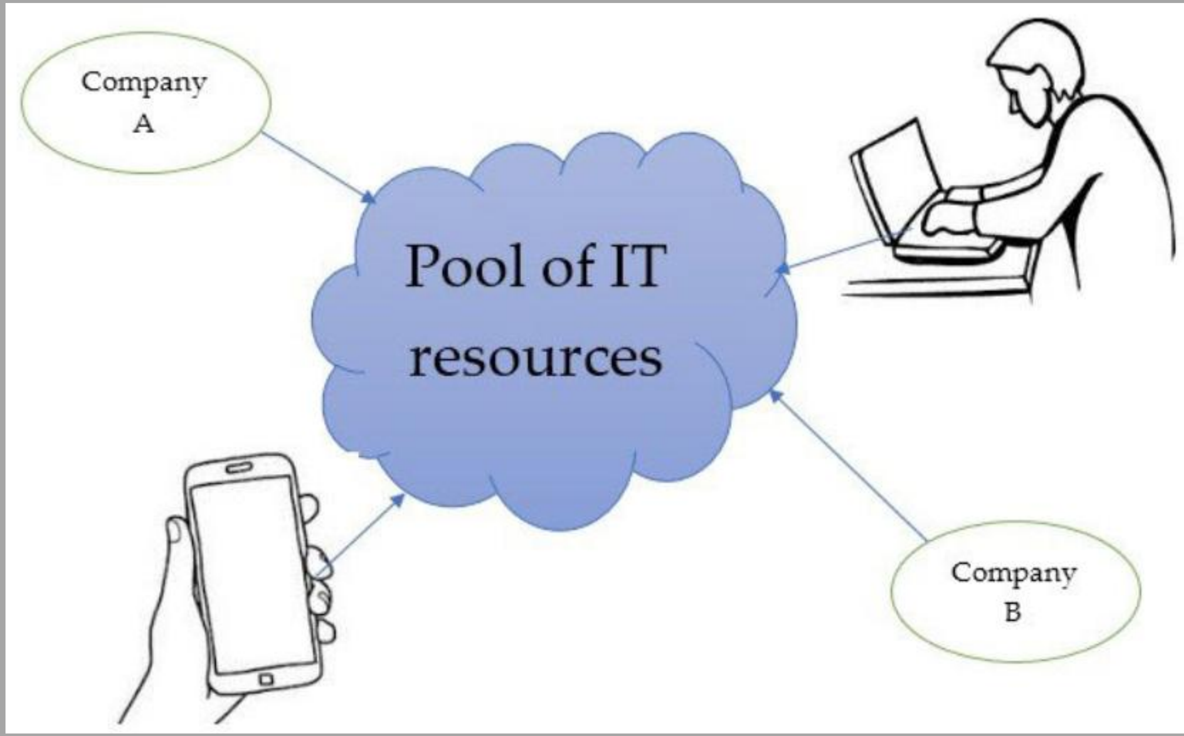


Introduction to Cloud Computing

Essential Characteristics of Cloud Computing

Multitenancy

- Having multiple customers and applications running within the same environment but in a way that they are isolated from each other and not visible to each other but share the same resources.

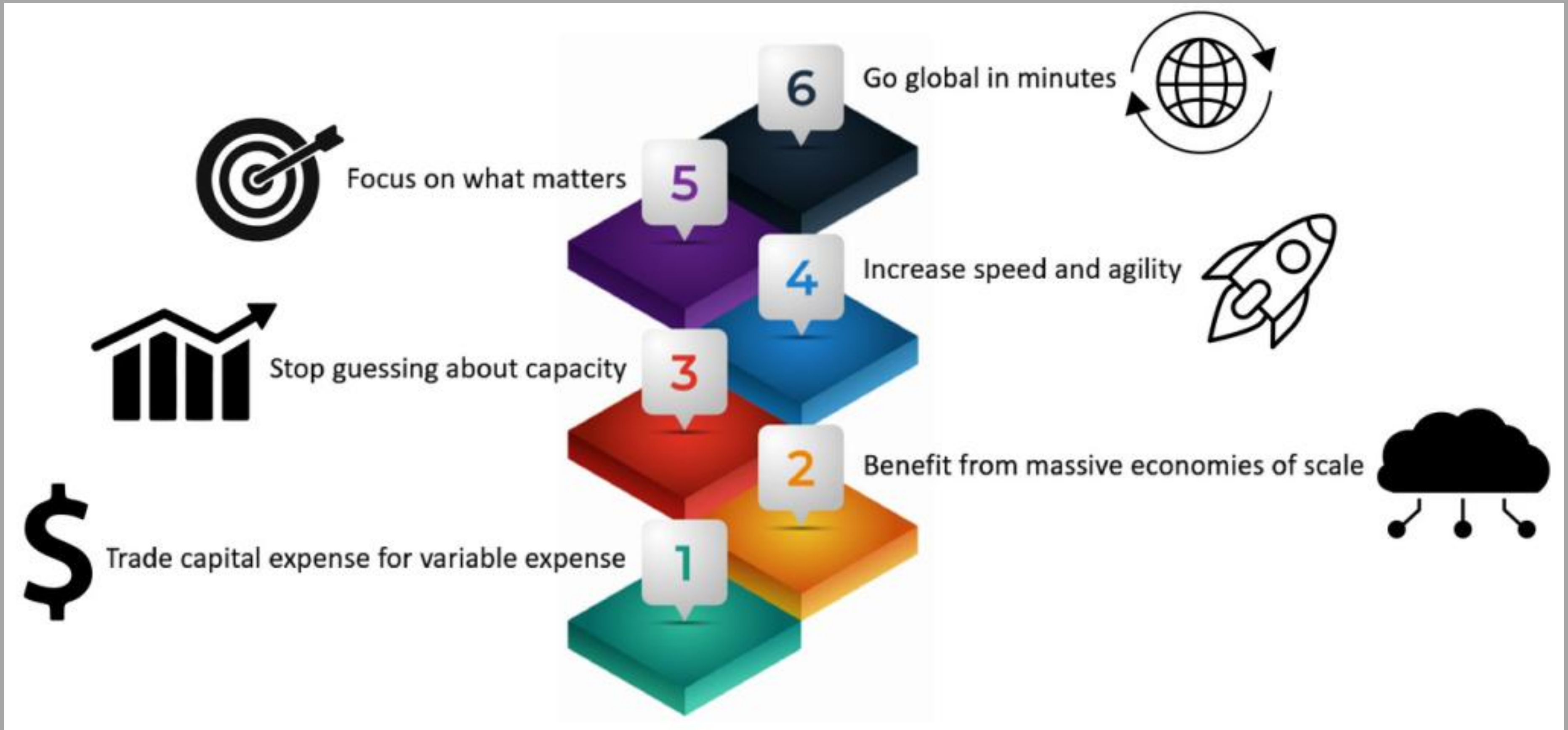


Introduction to Cloud Computing

Advantages of Cloud Computing

Introduction to Cloud Computing

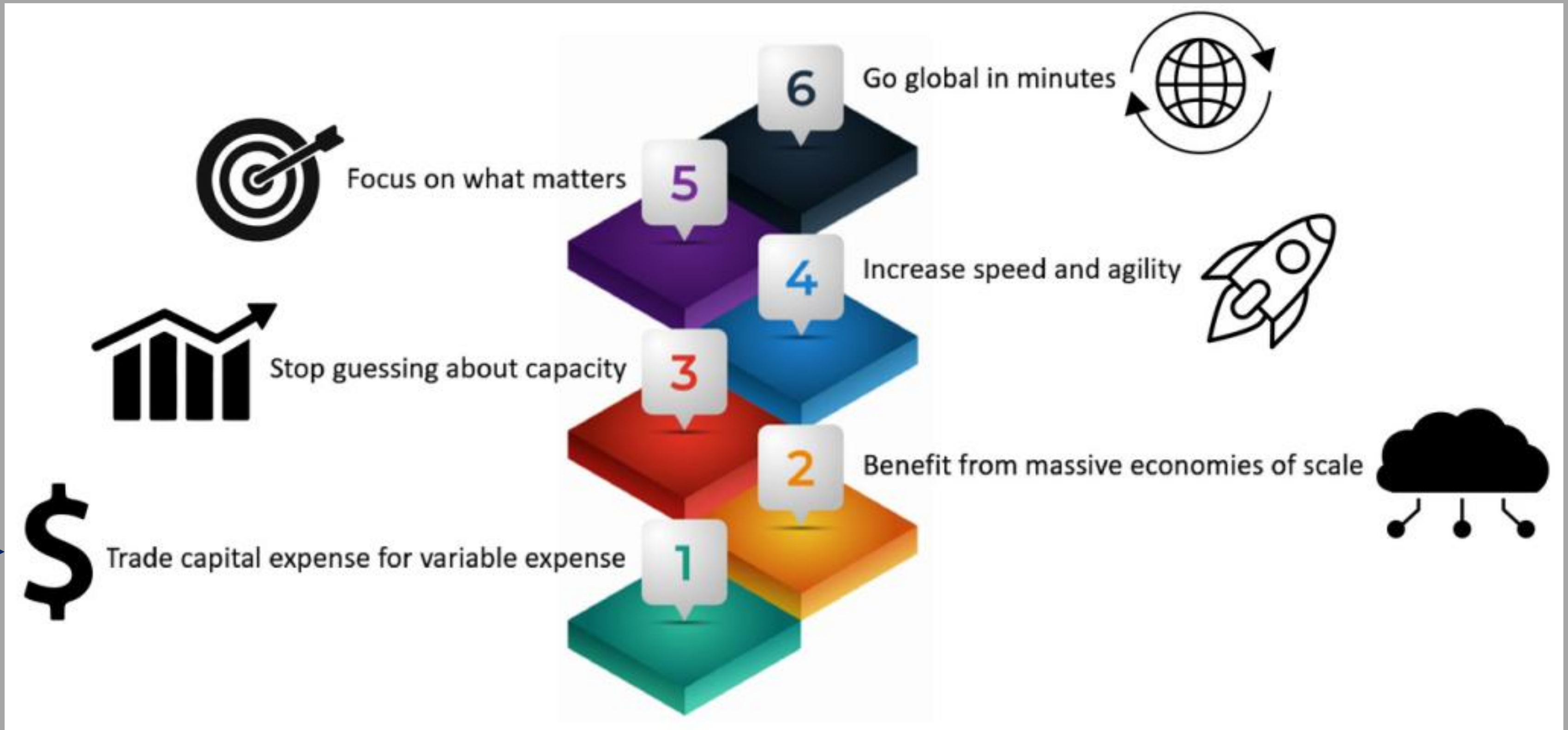
Advantages of Cloud Computing



Introduction to Cloud Computing

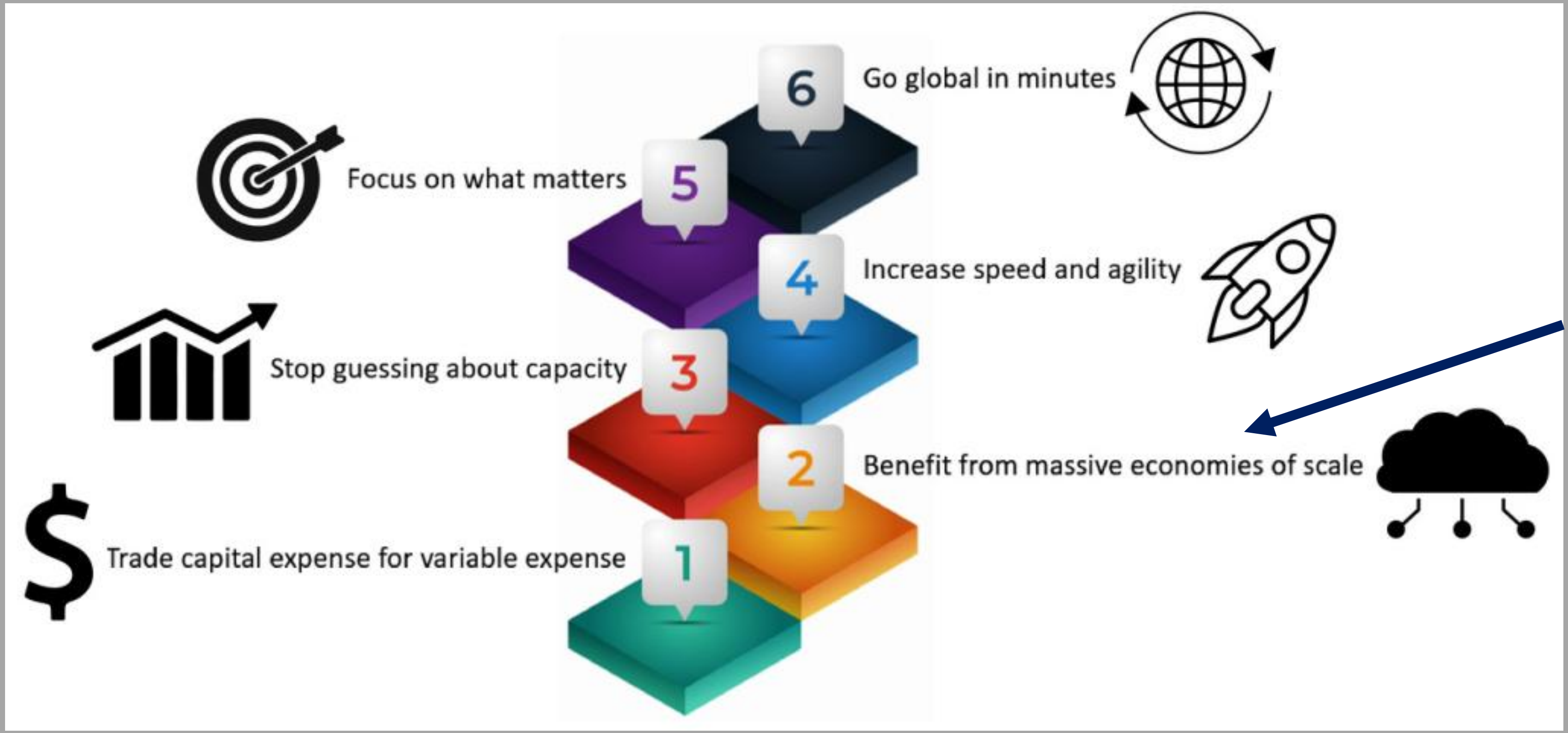
Advantages of Cloud Computing

Trade fixed expense for variable expense – Instead of having to invest heavily in data centers and servers before you know how you're going to use them, you can pay only when you consume computing resources, and pay only for how much you consume.



Introduction to Cloud Computing

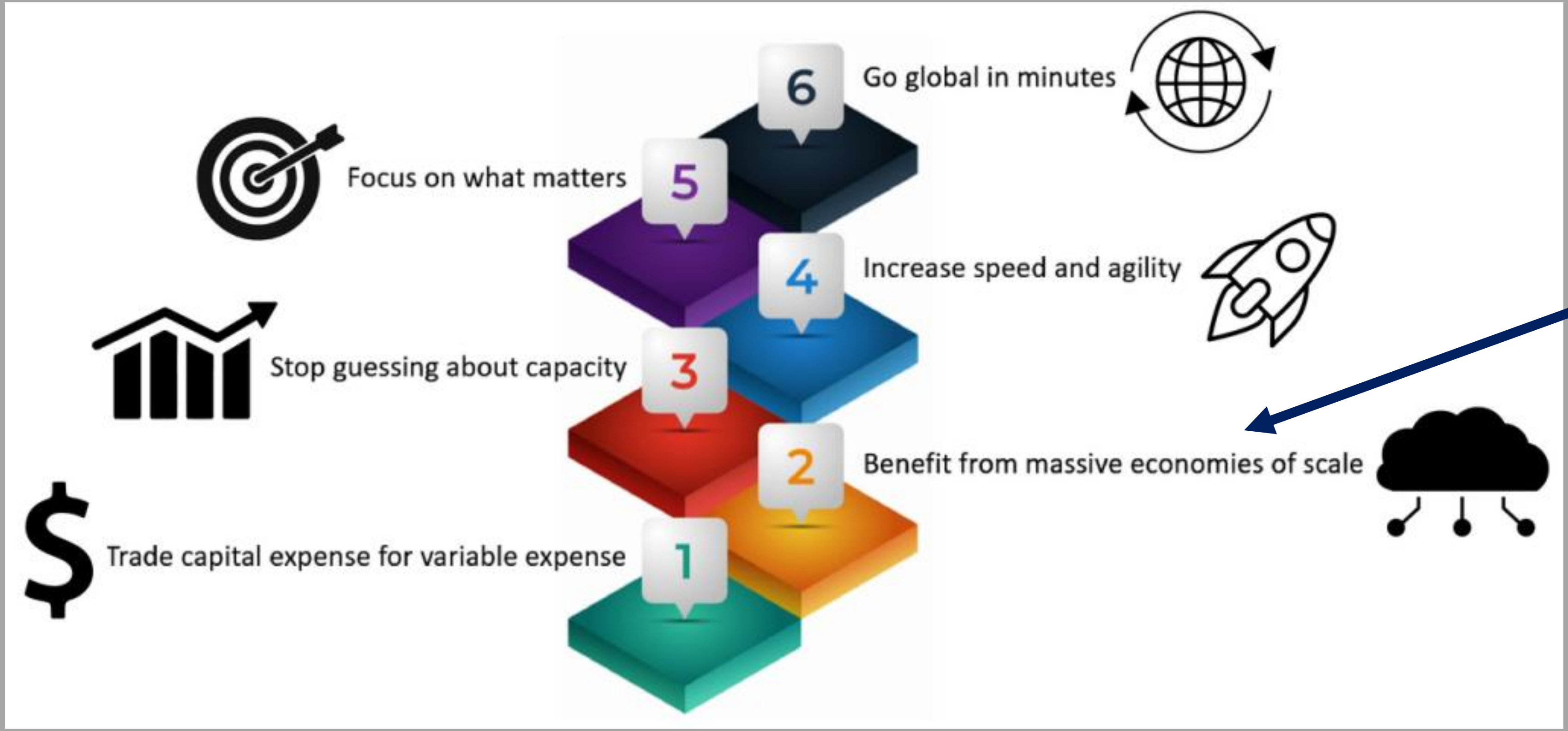
Advantages of Cloud Computing

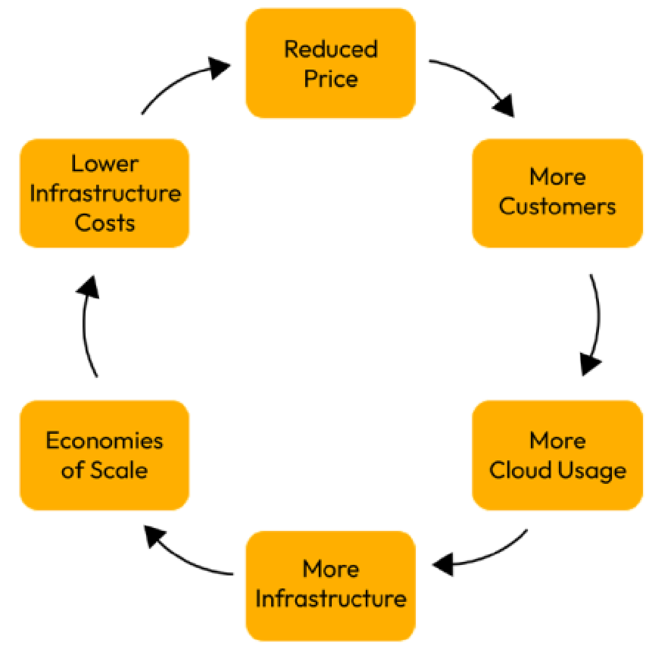


Benefit from massive economies of scale – By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay as-you-go prices.

Introduction to Cloud Computing

Advantages of Cloud Computing





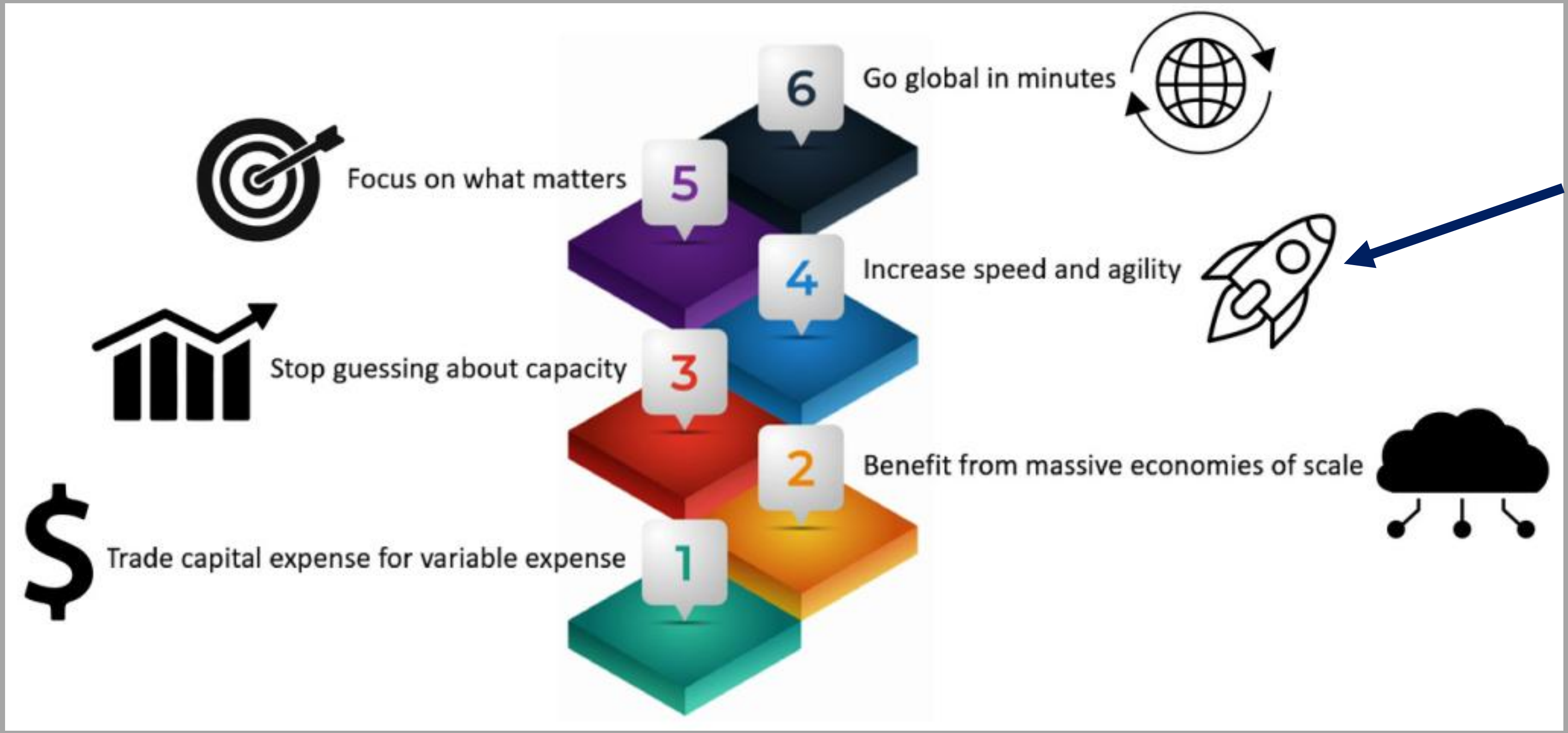
Introduction to Cloud Computing

Advantages of Cloud Computing

Stop guessing capacity – Eliminate guessing on your infrastructure capacity needs. When a decision is made about capacity prior to deploying an application, one can often end up either sitting on expensive idle resources or dealing with limited capacity. With cloud computing, these problems go away and one can access as much or as little capacity as one needs, and scale up and down as required with only a few minutes' notice.

Introduction to Cloud Computing

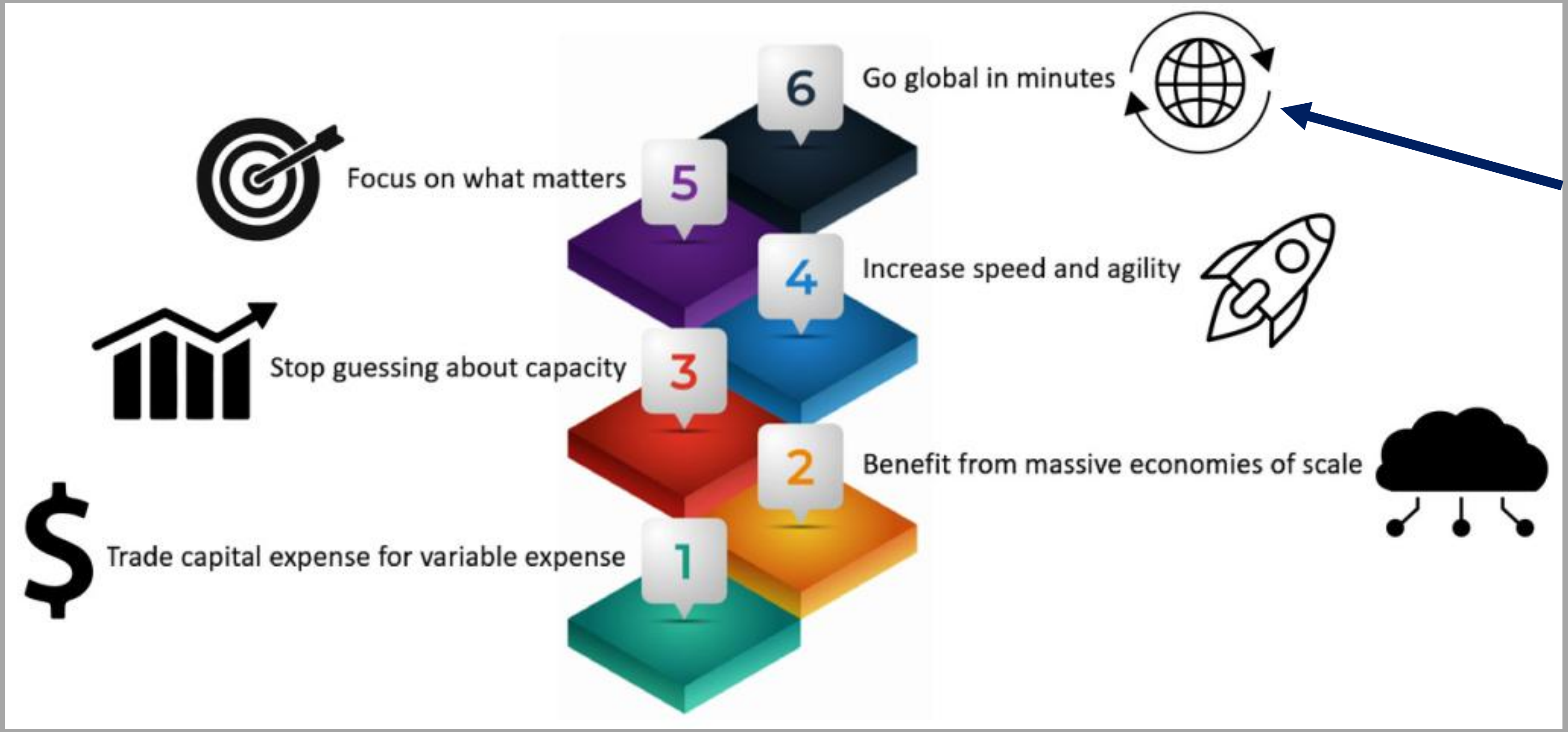
Advantages of Cloud Computing



Increase speed and agility – In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization, since the cost and time it takes to experiment and develop is significantly lower.

Introduction to Cloud Computing

Advantages of Cloud Computing



Go global in minutes – Easily deploy your application in multiple regions around the world with just a few clicks. This means you can provide lower latency and a better experience for your customers at minimal cost.

Introduction to Cloud Computing

Advantages of Cloud Computing

Stop spending money running and maintaining data centers – Focus on projects that differentiate the business, not the infrastructure. Cloud computing results in focusing on your customers, rather than on the heavy lifting of racking, stacking, and powering servers.

Introduction to Cloud Computing

Other Advantages of Cloud Computing

Introduction to Cloud Computing

Advantages of Cloud Computing

1) Economical

- One of the biggest benefits of cloud computing is that there is no need to invest a large sum of money to buy IT infrastructure because cloud computing provides access to various computing resources at a low cost.
- Further, there is no need to buy and maintain expensive infrastructure, hardware, facilities, or invest money to recruit an expert staff and IT team to manage the set up. There are no administrative, operational, and upfront costs.

Introduction to Cloud Computing

Advantages of Cloud Computing

2) Unlimited storage capacity and universal access

- Cloud service providers provide ample amount of data storage capacity, enabling the expansion of storage capacity whenever required.
- Cloud computing enables work and access to data and services from anywhere at anytime with an internet connection. In other words, cloud computing provides universal access to data and other IT resources.

Introduction to Cloud Computing

Advantages of Cloud Computing

3) Scaling

- Scaling means the ability to dynamically increase or decrease the IT resources to meet the customer demands.
- Scaling can be up and down (vertical scaling) or in and out (horizontal scaling)

Introduction to Cloud Computing

Advantages of Cloud Computing

3) Scaling

Let us use more precise terms to distinguish between two fundamentally different ways to scale compute resources— **horizontal scale and vertical scale**.

- **Horizontal scaling** adds more compute powers by adding more servers. Horizontal scaling uses *scaling out* and *scaling in* to refer to compute resource increases and decreases. Horizontal scaling is often used by a cloud that joins more virtual machines together to handle increasing workloads.

Introduction to Cloud Computing

Advantages of Cloud Computing

3) Scaling

- **Horizontal scaling** adds more compute powers by adding more servers. Horizontal scaling uses *scaling out* and *scaling in* to refer to compute resource increases and decreases. Horizontal scaling is often used by a cloud that joins more virtual machines together to handle increasing workloads.

Introduction to Cloud Computing

Advantages of Cloud Computing

3) Scaling

Let us use more precise terms to distinguish between two fundamentally different ways to scale compute resources— **horizontal scale and vertical scale**.

- **Vertical scaling** adds more compute power by making existing servers more powerful. Vertical scaling uses *scaling up* and *scaling down* to refer to server capacity increases and decreases. Scaling up is often used in an on-premises data center through hardware upgrades.

Introduction to Cloud Computing

Advantages of Cloud Computing

3) Scaling

- **Vertical scaling** adds more compute power by making existing servers more powerful. Vertical scaling uses *scaling up* and *scaling down* to refer to server capacity increases and decreases. Scaling up is often used in an on-premises data center through hardware upgrades.

Introduction to Cloud Computing

Advantages of Cloud Computing

4) Redundancy

- Redundancy results in the creation of duplicates of data, which are maintained by service providers on different machines in different locations so as to provide universal access and recovery of data in case any failure occurs at any data center or the data from one machine gets lost.
- The redundancy feature of cloud storage allows automatic data backups and access from any device at any time and any place.

Introduction to Cloud Computing

Advantages of Cloud Computing

5) **Increased security and reliability**
   - Data theft is a major concern for all organizations. Storing data in a cloud system provides security of the data, thereby protecting the data from any internal and external threats and preventing loss of data.

Introduction to Cloud Computing

Advantages of Cloud Computing

6) Ensures backup and recovery of data

- Storing data in cloud systems offers the feature of high availability because the data is stored on many machines in different locations. Therefore, if data on one machine gets lost or damaged, it be easily retrieved from a duplicate on another machine in the cloud.

Introduction to Cloud Computing

Advantages of Cloud Computing	
No.	**Advantages**
1	Pay for what is used • Scale up = pay more • Scale down = pay less
2	No space is required for servers (at provider)
3	No experts required for hardware and software maintenance
4	Better data security
5	Disaster recovery
6	High flexibility
7	Automatic software updates
8	Teams can co;;aborate from widespread locations
9	Data can be accessed and shared anywhere over the internet
10	Rapid implementation

Introduction to Cloud Computing

Disadvantages of Cloud Computing

Introduction to Cloud Computing

Disadvantages of Cloud Computing

1) Internet connection

- Cloud computing resources can be accessed only with the help of an internet connection. In the absence of an internet connection or in the presence of a poor connection, then access to the cloud and its services and resources will be impossible (no connection) or very limited (poor connection). So access to the cloud is totally dependent on the availability and quality of the internet connection.

Introduction to Cloud Computing

Disadvantages of Cloud Computing

2) Security issues

- Although cloud service providers provide a high level of security of the cloud infrastructure, the customer is responsible for his data in the cloud. Transferring data through the cloud is risky because the data may be exposed to hackers who can steal or corrupt it or to malware.

Introduction to Cloud Computing

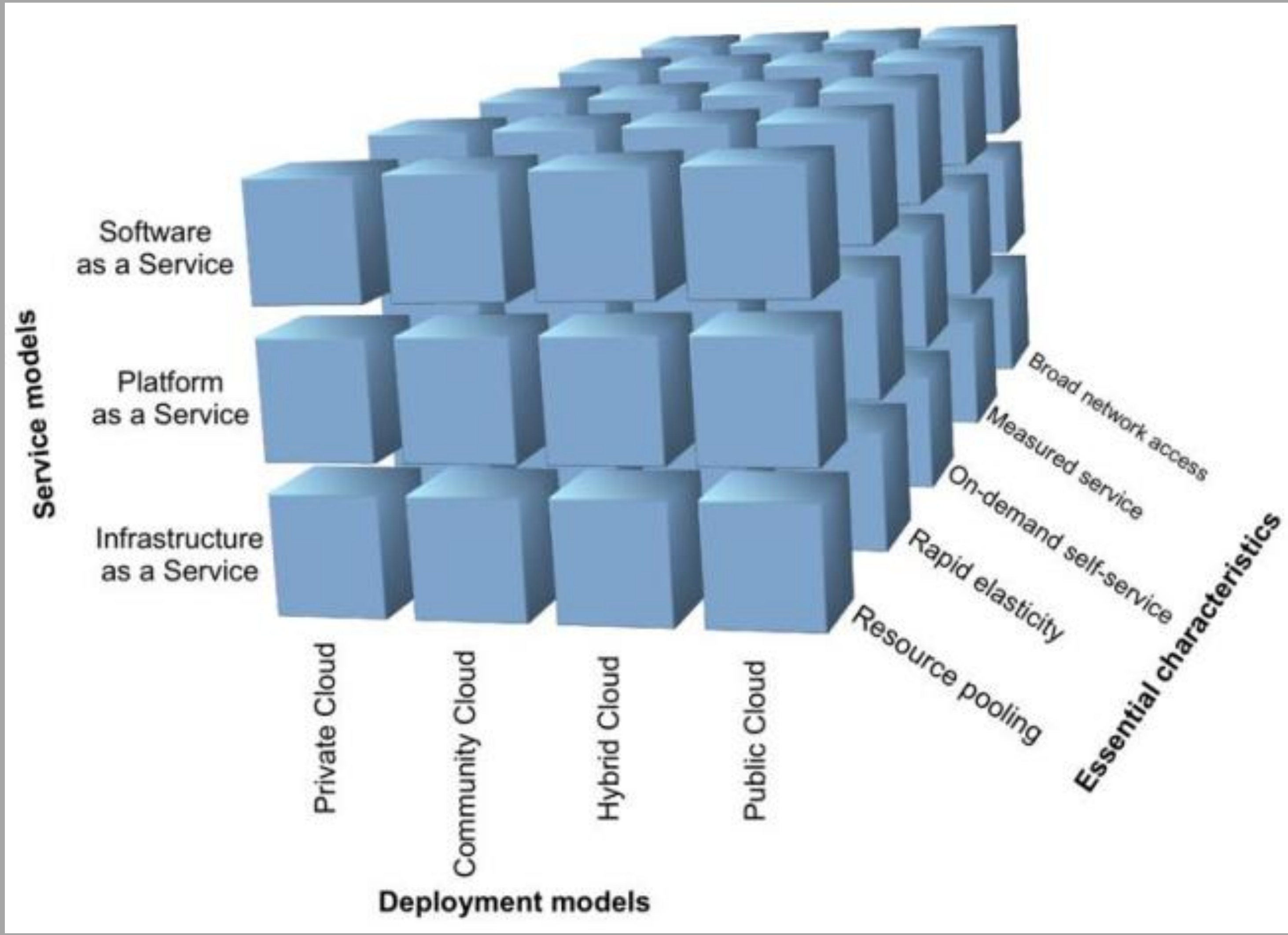
Disadvantages of Cloud Computing

3) Downtime

- Downtime is the worst disadvantage of cloud computing. Downtime means the time during which a machine or computer is out of use or not available because of overload of traffic on servers from various clients and or technical problems facing the cloud service providers.
- Further, because access to cloud resources depends on the availability and quality of the internet connection, any disruption in connectivity will result in downtime.
- Similarly, the internet speed of a cloud provider, the demand for service maintenance will also cause downtime.

Deployment and Service Models of Cloud Computing

Introduction to Cloud Computing



NIST model in three dimensions consisting of

- service models,
- deployment models, and
- essential features

Introduction to Cloud Computing

Types of cloud computing

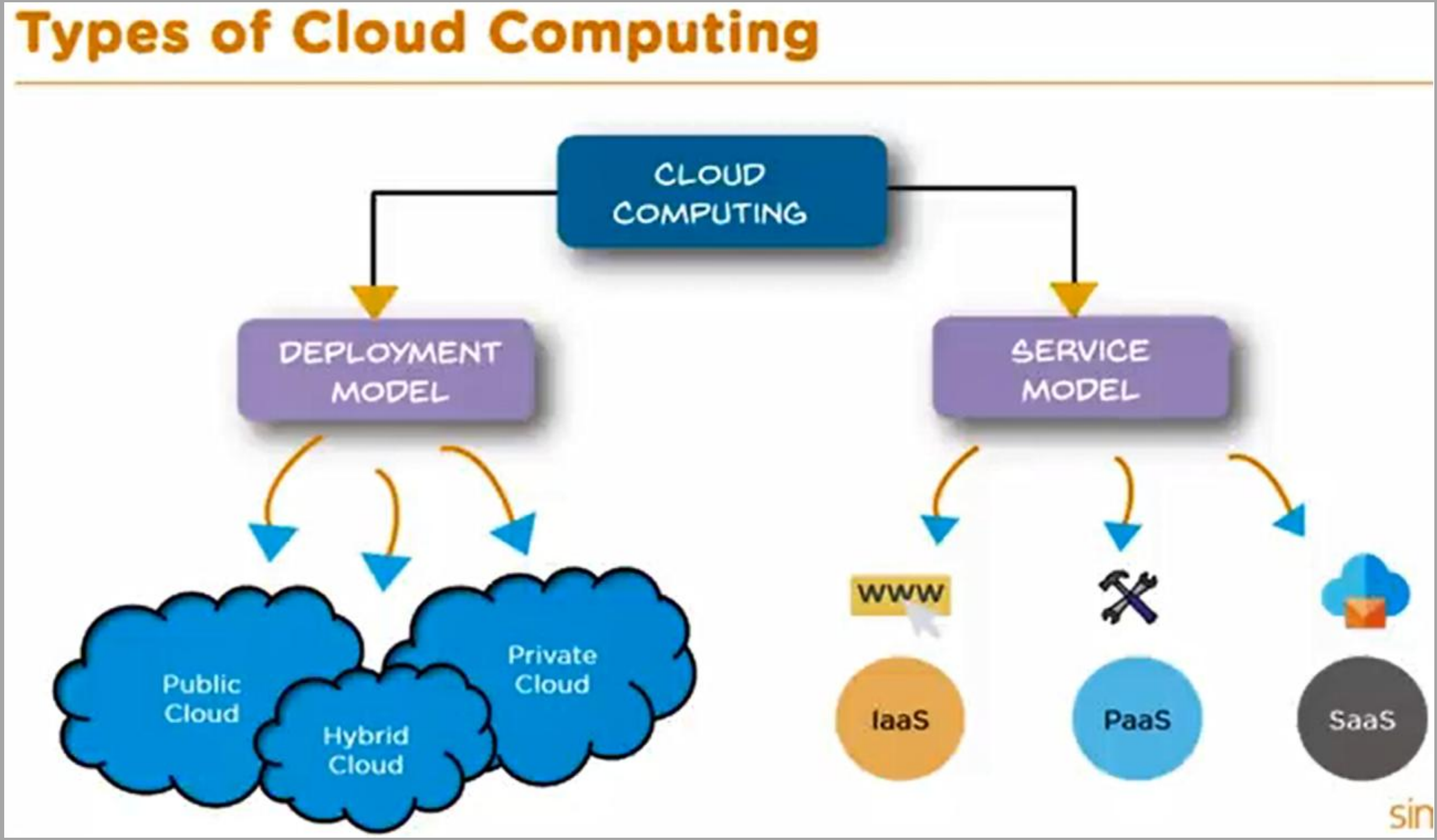
- Cloud computing provides developers and IT departments with the ability to focus on what matters most and avoid undifferentiated work such as procurement, maintenance, and capacity planning.
- As cloud computing has grown in popularity, several different service models and deployment strategies have emerged to help meet specific needs of different users.

Introduction to Cloud Computing

Types of cloud computing

- Understanding the differences between Infrastructure as a Service, Platform as a Service, and Software as a Service, as well as what deployment strategies can be used, can help in deciding what set of services is right for defined needs.

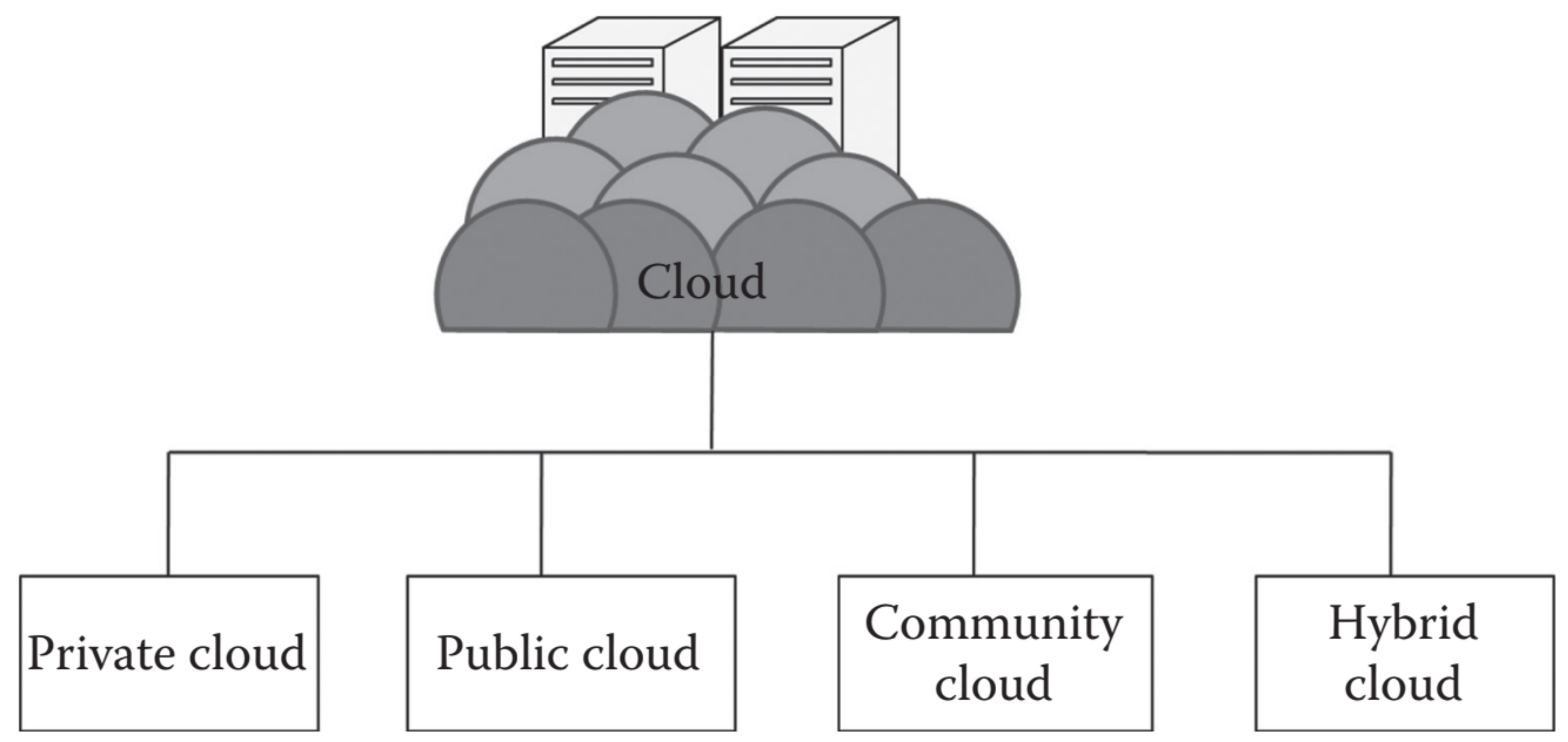
Introduction to Cloud Computing



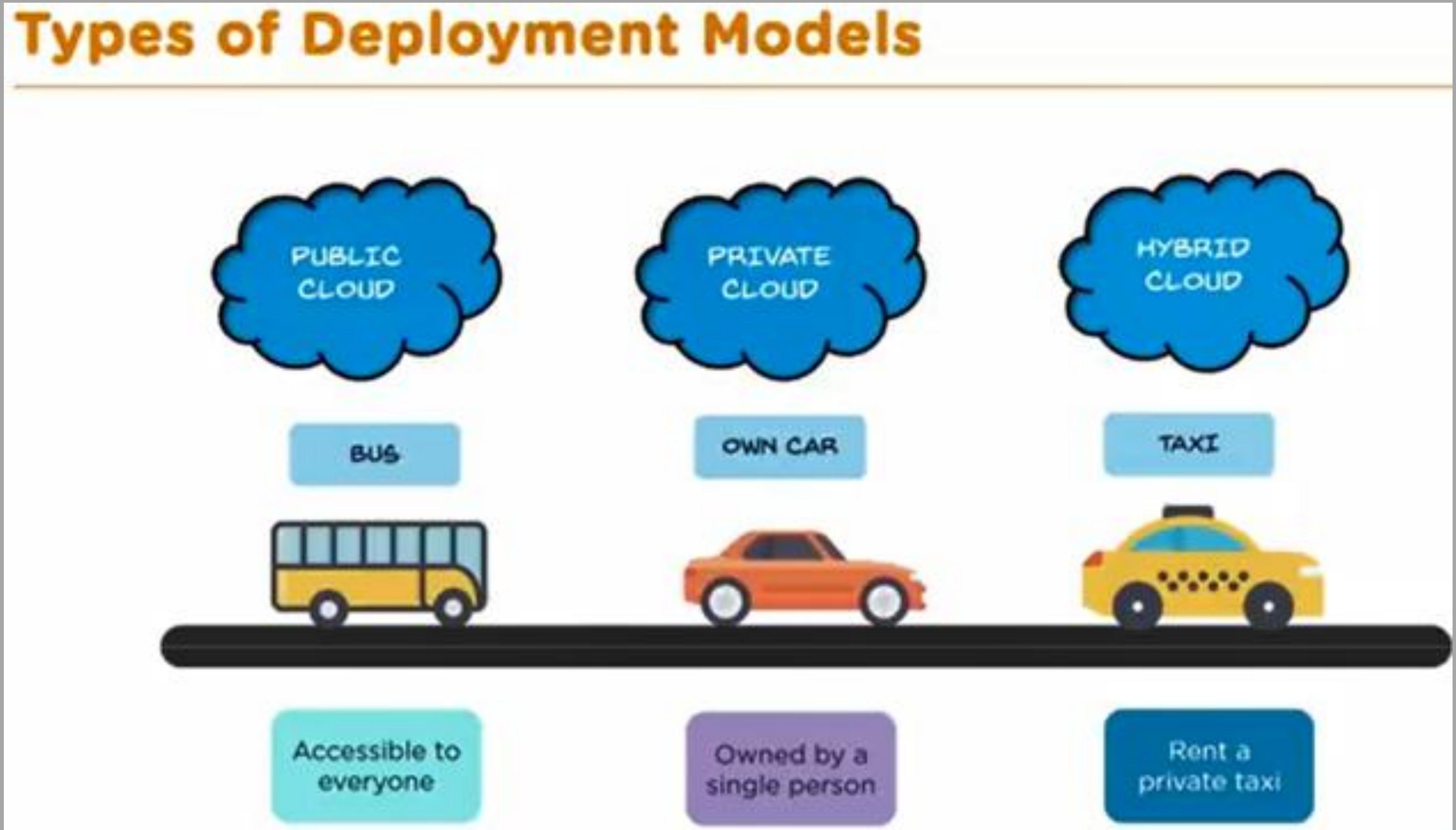
Deployment Models of Cloud Computing

Introduction to Cloud Computing

There are four main types of cloud deployment and hosting models in common use, each of which can host any of the three main cloud service models.



Introduction to Cloud Computing

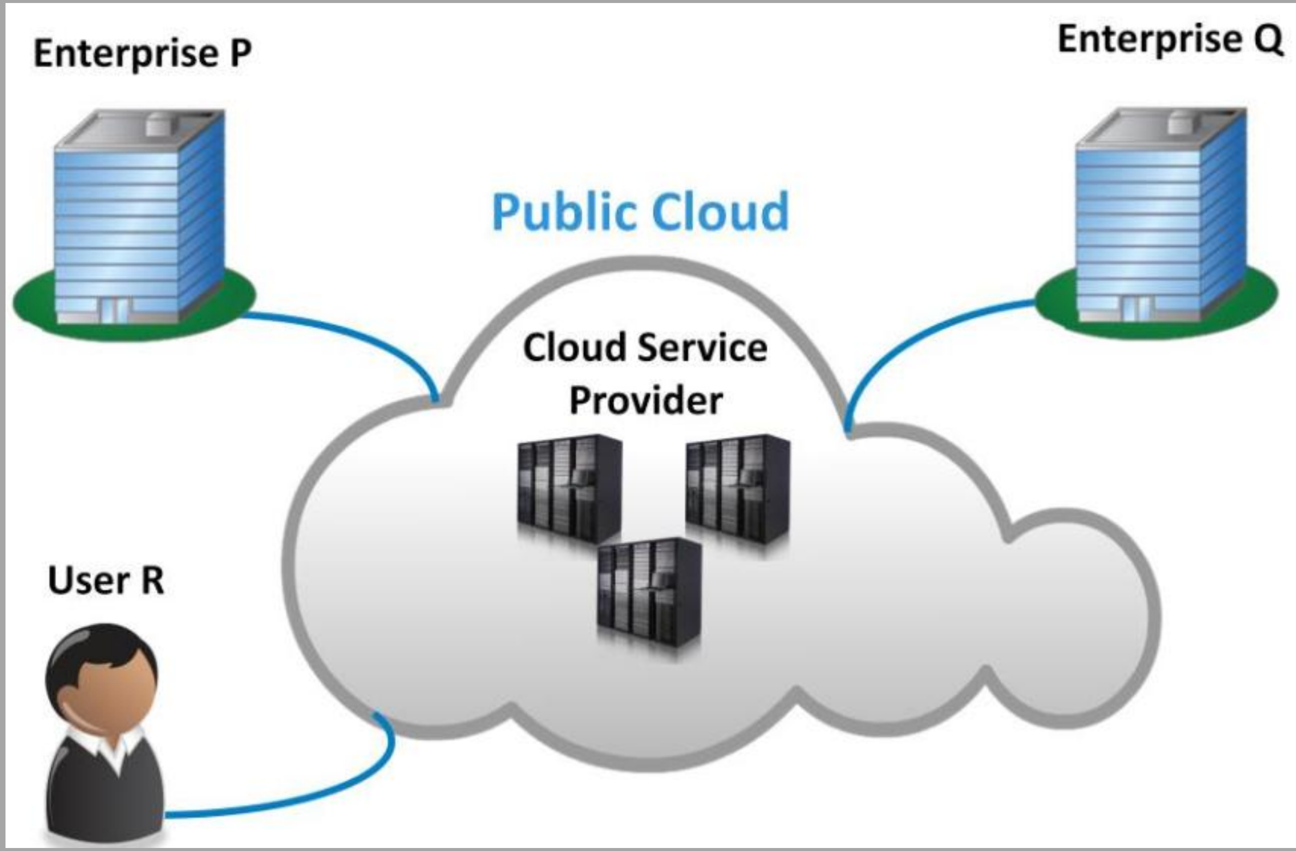


Let us use the example of three types of transport vehicles to illustrate the different cloud computing deployment models

Introduction to Cloud Computing

Cloud Deployment Models: Public Cloud

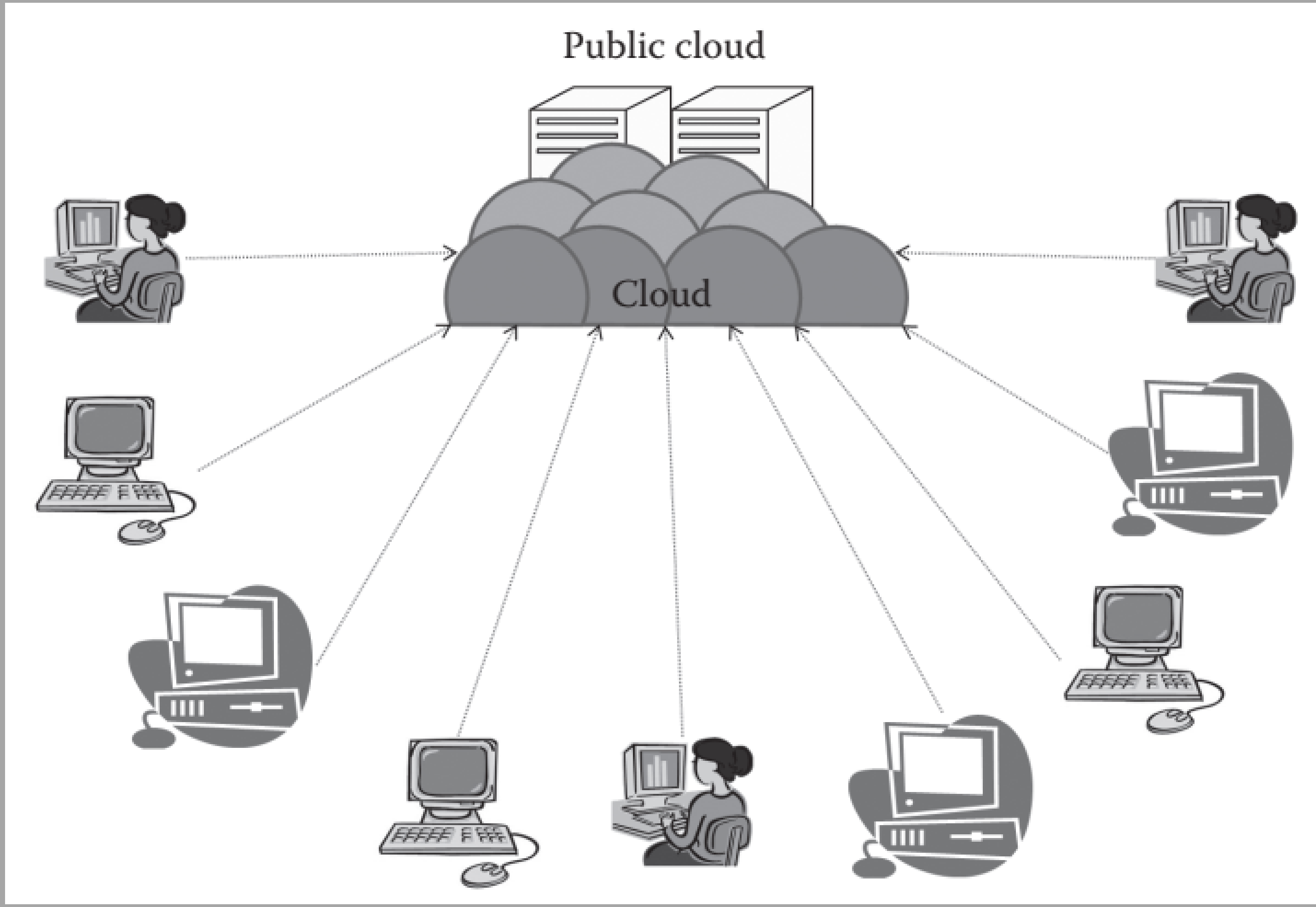
Introduction to Cloud Computing



Cloud Deployment Models: Public Cloud

- In a Public Cloud, the Cloud service provider makes its IT resources available to the general public or organizations. The Cloud services are accessible to everyone via standard Internet connections.
- In a public Cloud, a service provider makes IT resources, such as applications, storage capacity, or server compute cycles, available to any consumer. This model can be thought of as an "on-demand" and as a "pay-as-you-go" environment, where there are no on-site infrastructure or management requirements.

Introduction to Cloud Computing

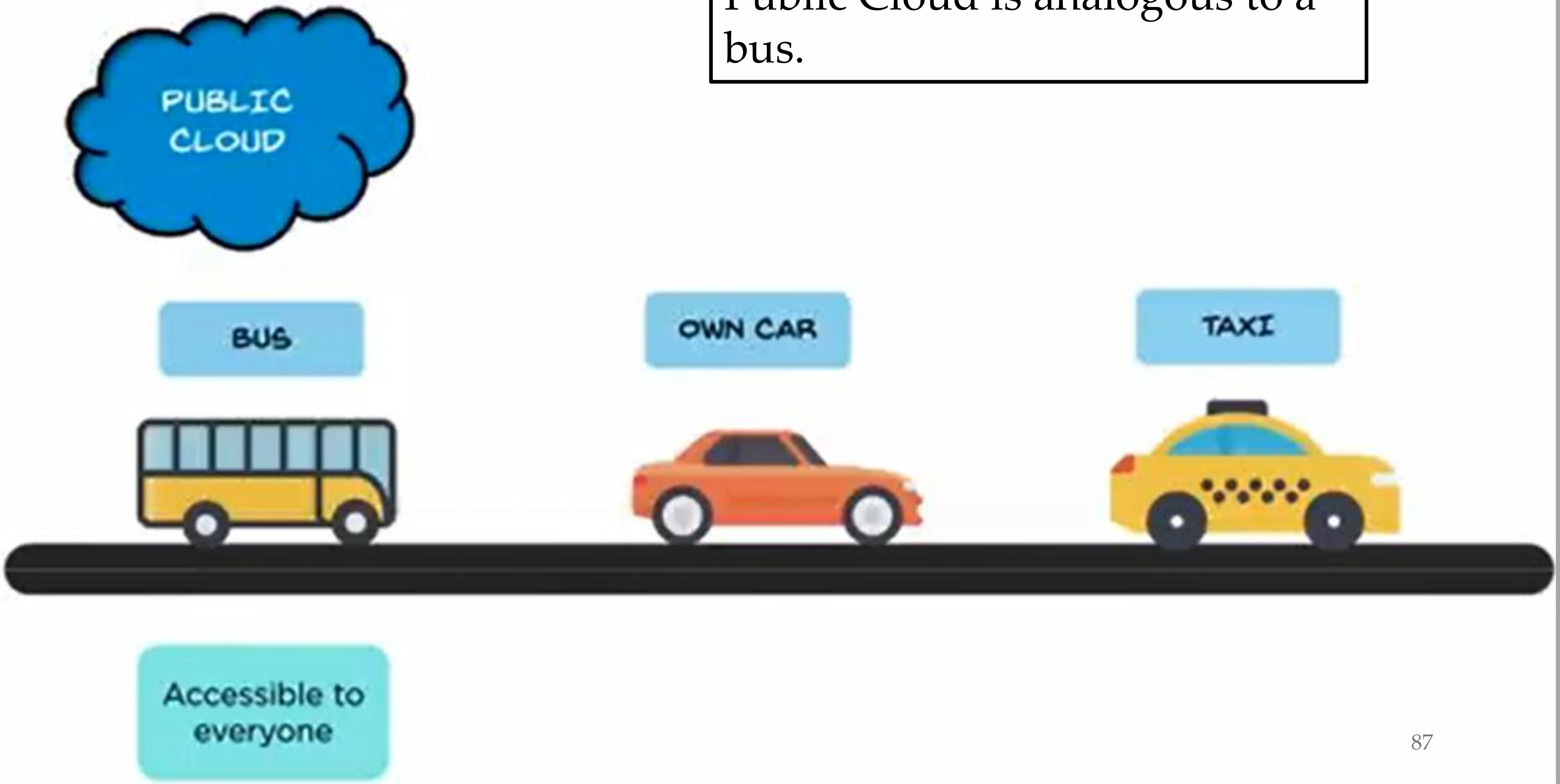


Cloud Deployment Models: Public Cloud

- Public cloud consists of users from all over the world. A user can simply purchase resources on an hourly basis and work with the resources. There is no need of any prebuilt infrastructure for using the public cloud. These resources are available in the cloud provider's premises. Usually, cloud providers accept all the requests, nnd hence, the resources in the service providers' end are considered *infinite* in one aspect.
- Highly scalable
- Affordable
- Less secure / private
- Highly available

Types of Deployment Models

Public Cloud is analogous to a bus.



Introduction to Cloud Computing

Cloud Deployment Models: Public Cloud

Advantages of Public Cloud

The Public Cloud has three main advantages:

1) Economic – much lower cost than a private cloud;
2) Expertise – access to a staff with expertise on many topics;
3) Advanced services – offerings not available elsewhere.

Introduction to Cloud Computing

Cloud Deployment Models: Public Cloud

Advantages of Public Cloud

1) Economic advantage

Because they serve multiple tenants, cloud providers operate data centers that are much larger than private cloud data centers owned and operated by individual organizations. Therefore, public cloud providers benefit from economy of scale because

- they can spread costs among more customers, and
- can negotiate larger discounts on commodity servers and network equipment.
- a public cloud provider shares IT staff expertise across all tenants, the cost of staff for a given tenant is less than the cost of staff for a private data center

Introduction to Cloud Computing

Cloud Deployment Models: Public Cloud

Advantages of Public Cloud

2) Expertise advantage

- Because it is much larger than the IT department in a single company, a public cloud provider can afford to maintain a staff with specialized expertise. Thus, customers have access to expertise on a much wider range of topics than a local IT staff can provide.
- In addition to traditional data center software, such as operating systems, relational database services, and technologies for virtualized servers, cloud providers now include expertise on the broad topic of *Artificial Intelligence* (*AI*) and a technology that AI uses, *machine learning* (*ML*) software.

Introduction to Cloud Computing

Cloud Deployment Models

Advantages of Public Cloud

3) Advanced services advantage

- Initially, providers investigated AI/ML as a way to automate the management of data center hardware and software. *AIops* to refer to the use of AI/ML to automate data center operations.
- More recently, cloud providers have begun offering AIops as a service to customers. In addition to monitoring and automating the deployment and operation of customer apps, providers offer advanced services that use AI/ML software to analyze business data.

Introduction to Cloud Computing

Cloud Deployment Models

Advantages of Public Cloud

3) Advanced services advantage

- In addition to AI-based services, providers offer services that update a customer’s operating systems and other standard software whenever the vendor releases an update. Major cloud providers also offer platforms that allow a customer to build and deploy new apps without understanding the details of the underlying system. In particular, a software engineer can focus on building an app without learning how to create and replicate virtualized servers, establish network connectivity, or handle remote storage.

Introduction to Cloud Computing

Cloud Deployment Models: Public Cloud

Advantages of Public Cloud

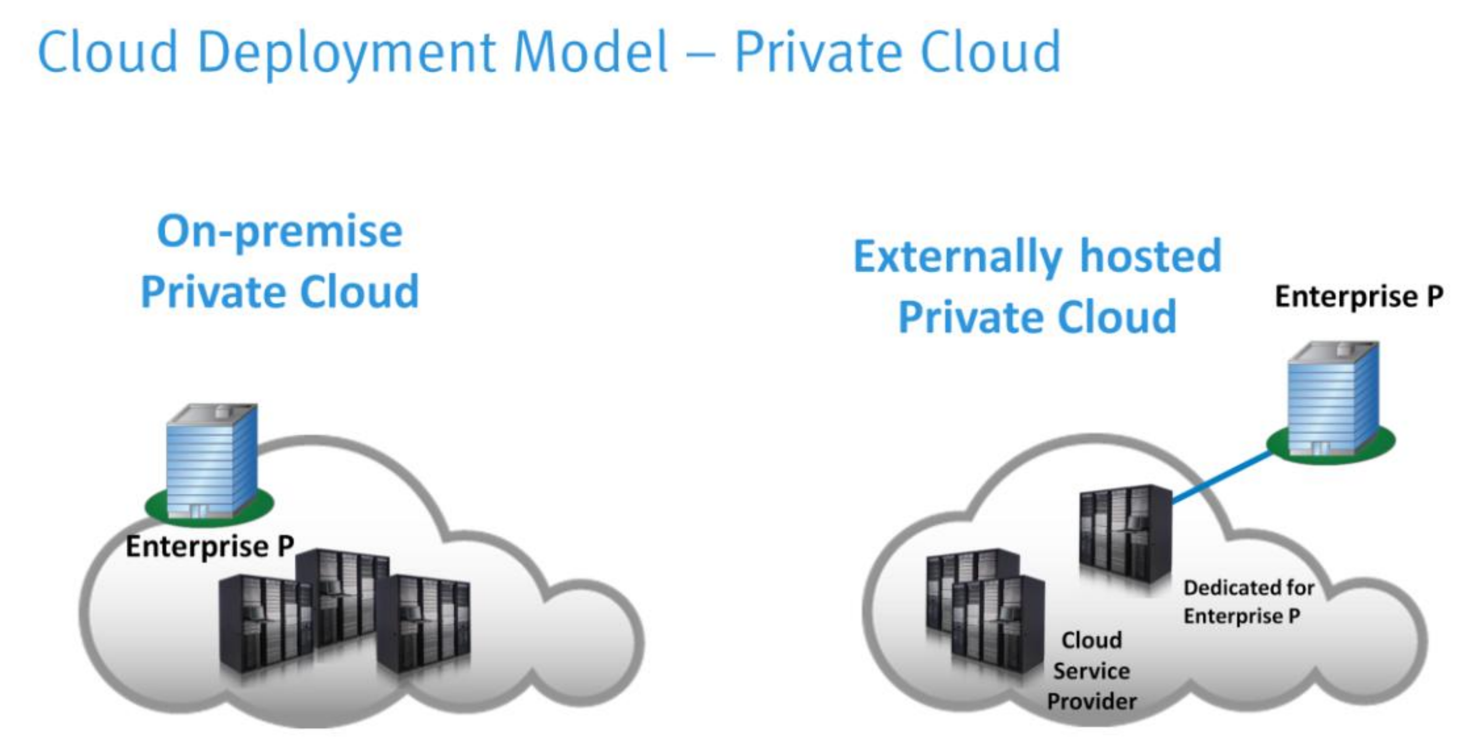
To summarize:

By sharing IT expertise across multiple tenants, negotiating large discounts on commodity servers, a public cloud provider can operate a cloud data center for substantially less than it costs to operate a private cloud data center.

Introduction to Cloud Computing

Cloud Deployment Models:
Private Cloud

Introduction to Cloud Computing

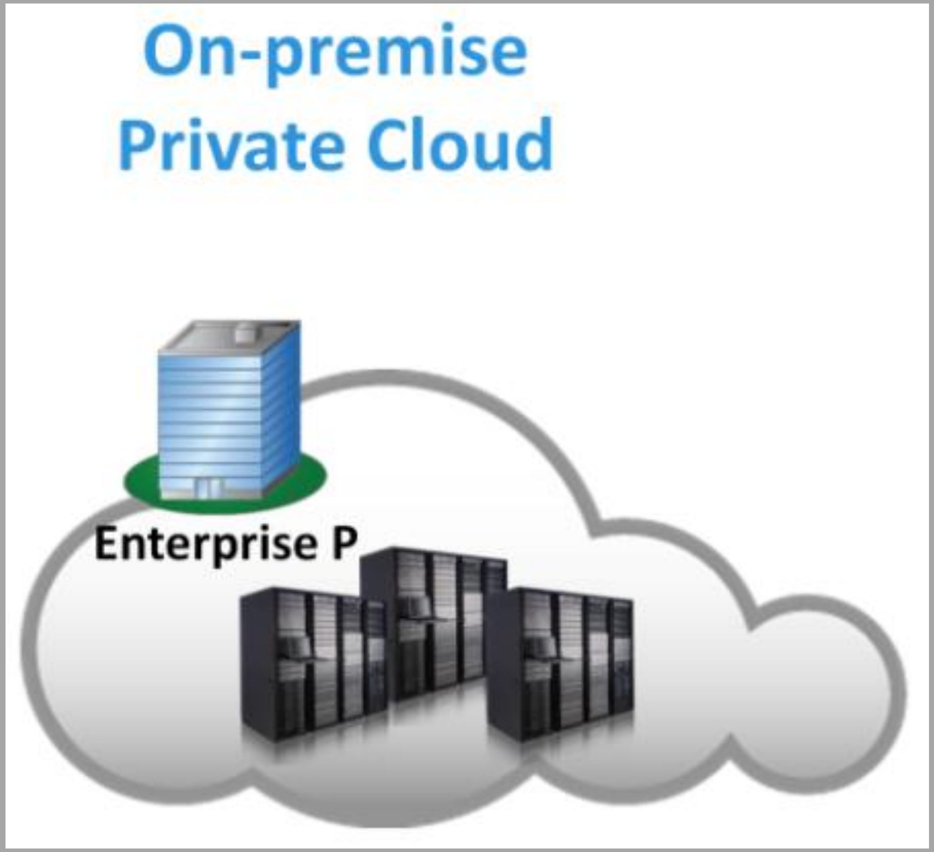


Cloud Deployment Models: Private Cloud

In a private Cloud, the Cloud infrastructure is operated solely for one organization and is not shared with other organizations. This Cloud model offers the greatest level of security and control. There are two variations to a private Cloud:

1) On-premise private cloud;
2) Externally hosted private cloud.

Introduction to Cloud Computing

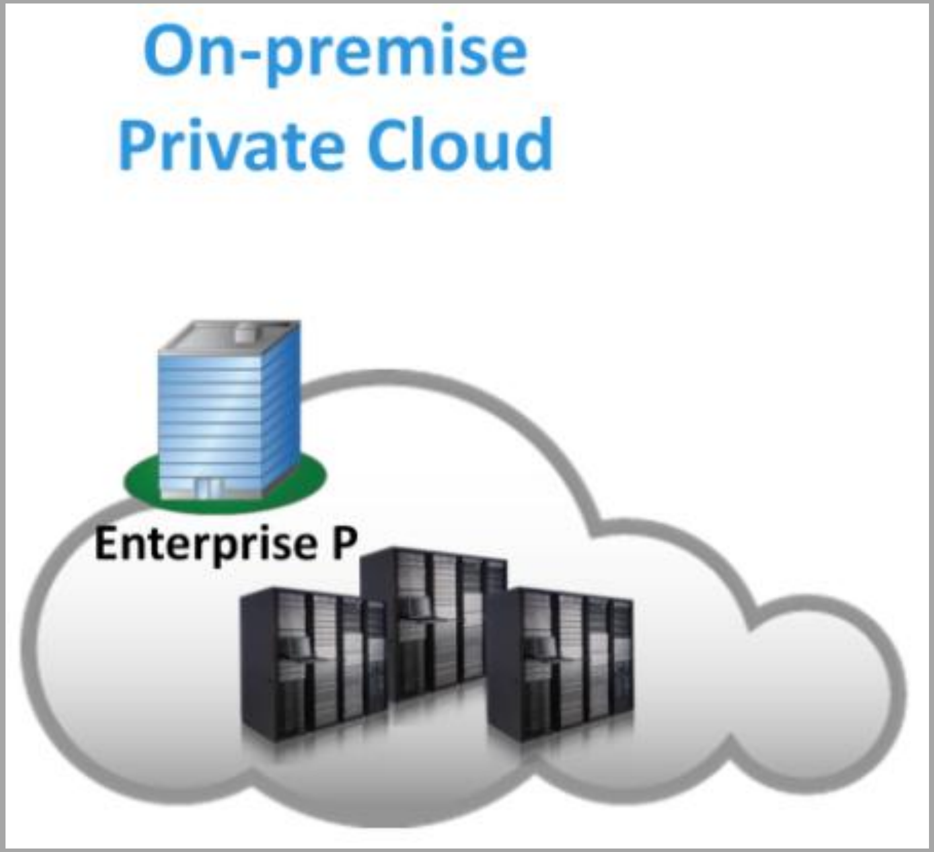


Cloud Deployment Models

On-premise Private Cloud:

- On-premise private Clouds, also known as internal Clouds, are hosted by an organization within their own data centers. This model provides a more standardized process and protection, but is limited in terms of size and scalability
- Organizations would also need to incur the capital and operational costs for the physical resources. This is best suited for applications which require complete control and configurability of the infrastructure and security by the organizaton.

Introduction to Cloud Computing

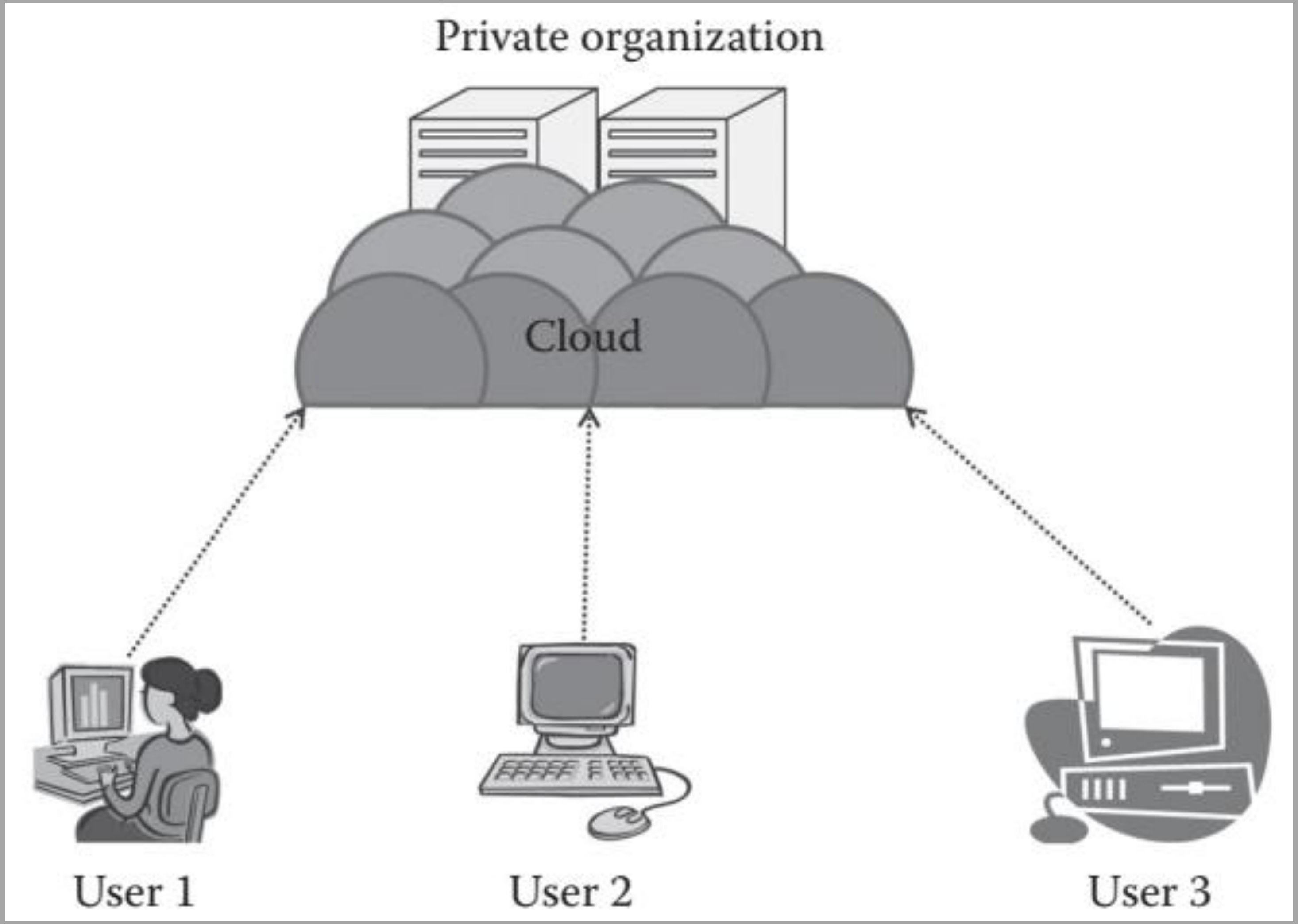


Cloud Deployment Models

On-premise Private Cloud:

- In on-premise private Cloud, organizations will have to run their own hardware, storage, networking, hypervisor, and Cloud software.

Introduction to Cloud Computing

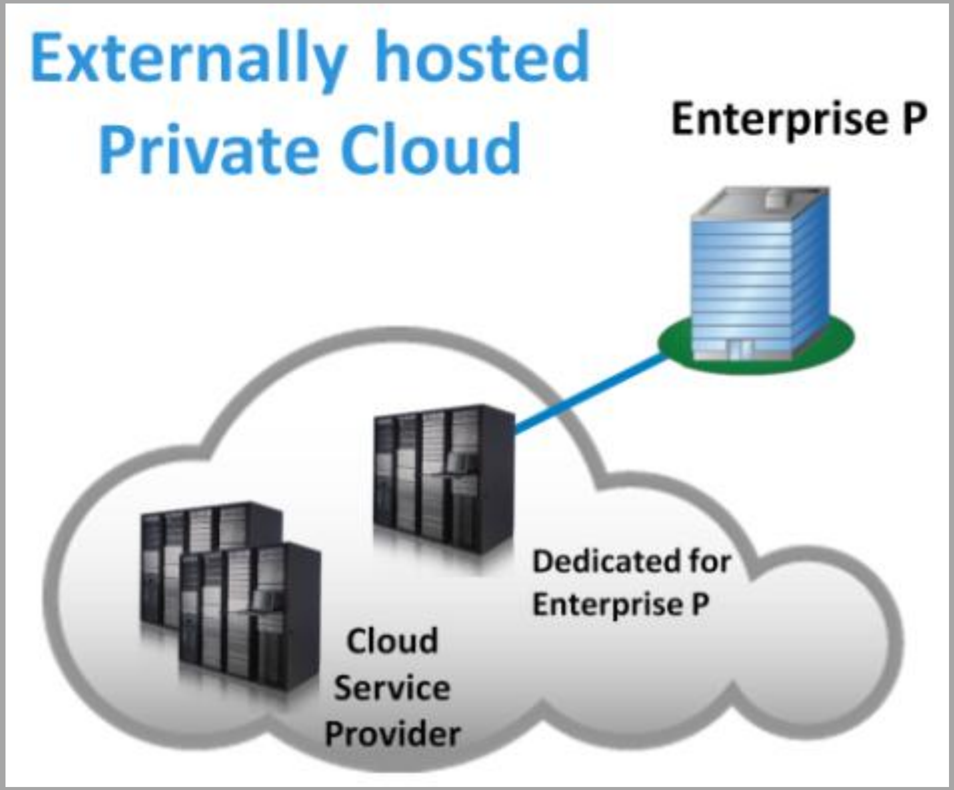


Cloud Deployment Models

On-premise Private Cloud:

- In on-premise private Cloud, organizations will have to run their own hardware, storage, networking, hypervisor, and Cloud software.

Introduction to Cloud Computing

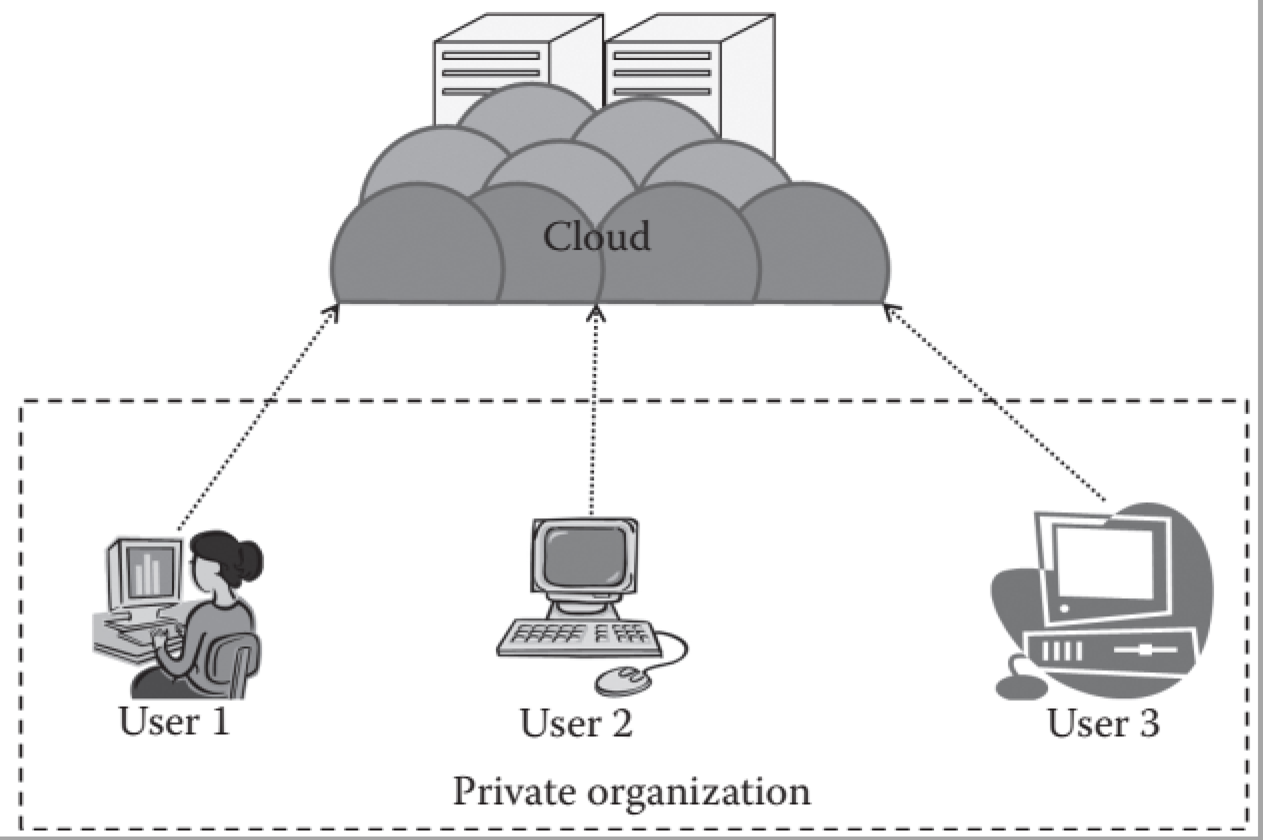


Cloud Deployment Models

Externally-hosted Private Cloud:

- This type of private Cloud is hosted externally with a Cloud provider, where the provider facilitates an exclusive Cloud environment for a specific organization with full guarantee of privacy or confidentiality. This is best suited for organizations that do not prefer a public Cloud due to data privacy/security concerns.

Introduction to Cloud Computing



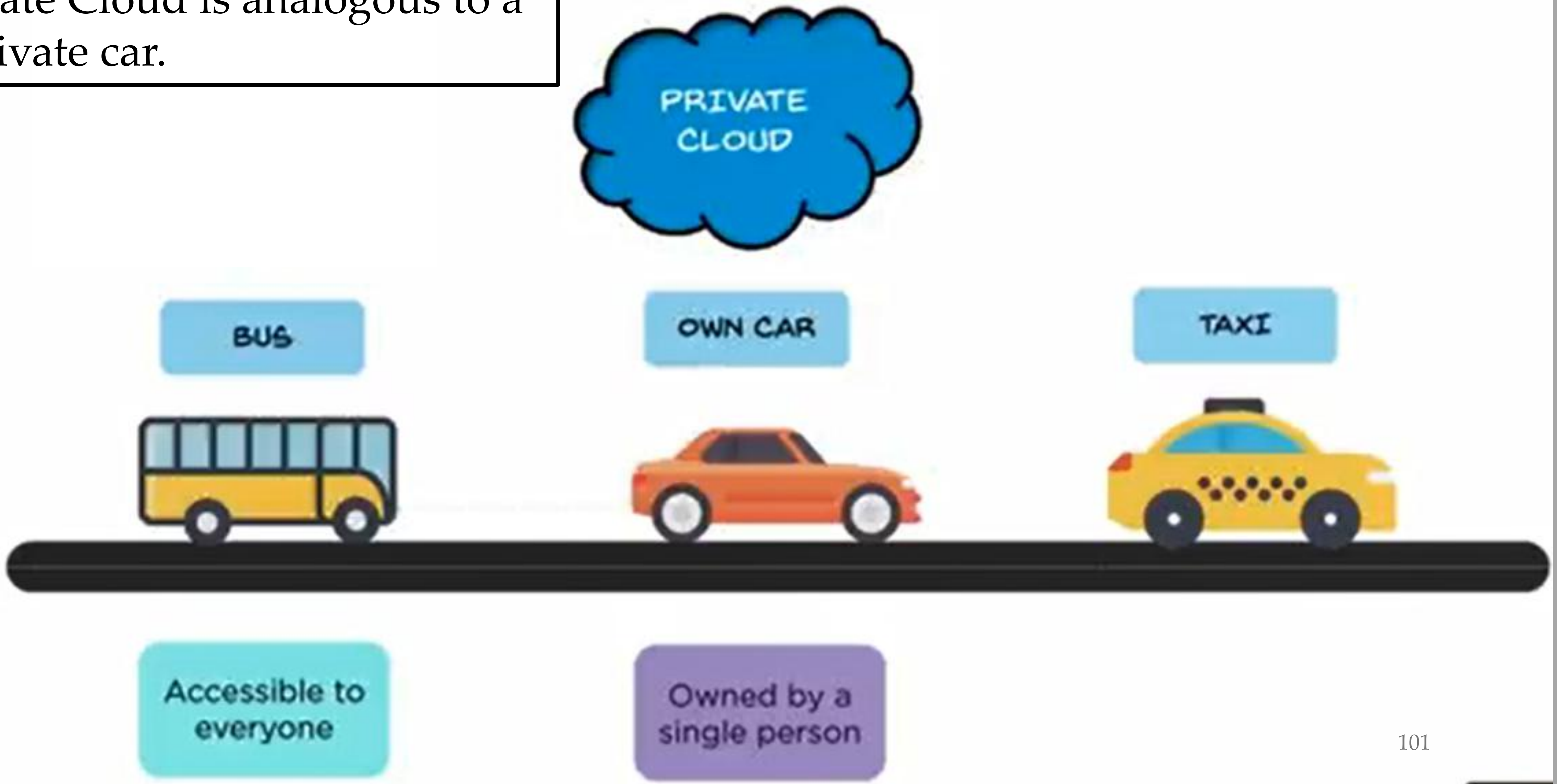
Cloud Deployment Models

Externally-hosted Private Cloud:

- *Network*: The cloud is fully deployed at the third-party site. The cloud's internal network is managed by a third party, and the organizations connect to the third party by means of either a dedicated connection or through the Internet.
- *Security and privacy*: The cloud is less secure than the on-site private cloud. The privacy and security of the data mainly depend on the hosting third party as they have the control of the cloud. But, basically the security threat is from the third party and the internal employee.

Types of Deployment Models

Private Cloud is analogous to a a private car.



Introduction to Cloud Computing

Cloud Deployment Models: Private Cloud

Advantages of Private Cloud

The advantages of the public cloud (lower cost, access to specialized expertise, and advanced services) may give the impression all companies should prefer to use public cloud services. However, private cloud offers advantages:

1) Retention of control and visibility
2) Reduced latency with on-premises facilities
3) Insurance against future rate hikes

Introduction to Cloud Computing

Cloud Deployment Models: Private Cloud

Advantages of Private Cloud

1) **Retention of control and visibility**
   - For organizations in a regulated industry, regulations may require the organization to control the placement and transmission of data as well as the hardware and software used. A private cloud makes it possible for such an organization to comply with the rules.
   - A public cloud customer does not have access to the underlying infrastructure.

Introduction to Cloud Computing

Cloud Deployment Models: Private Cloud

Advantages of Private Cloud

2) Reduced latency with on-premises facilities

- Because they are located within a company, private cloud facilities are said to be *on-premises*. The delay between employees and a private cloud data center can be significantly lower than the delay to a public cloud, especially if an organization is geographically distant from a public cloud site.
- An organization with multiple sites may be able to further reduce latency by placing a private cloud data center at each site. If computation and communication remains primarily within a site, the improvement can be noticeable.

Introduction to Cloud Computing

Cloud Deployment Models: Private Cloud

Advantages of Private Cloud

3) Insurance against future rate hikes

- Private cloud protects a company from provider lock-in which can become a long-term liability because, as the company adopts more of the provider’s services, the cost of changing providers increases. A provider can raise rates without fear of losing customers.

Introduction to Cloud Computing

Cloud Deployment Models: Private Cloud

Advantages of Private Cloud

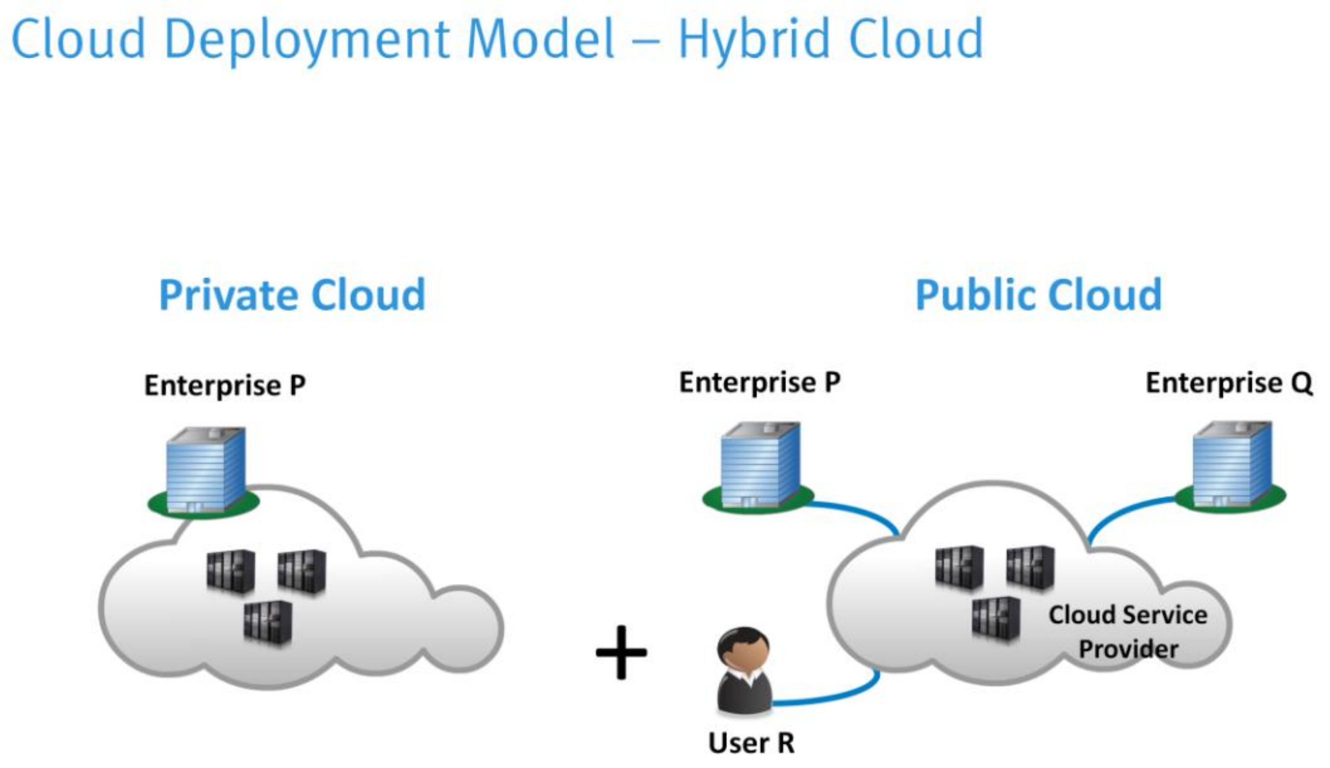
To summarize:

Although the public cloud seems to offer several compelling advantages, the private cloud gives a company more control, can reduce latency, and guards against future rate hikes by a provider.

Introduction to Cloud Computing

Cloud Deployment Models: Hybrid Cloud

Introduction to Cloud Computing

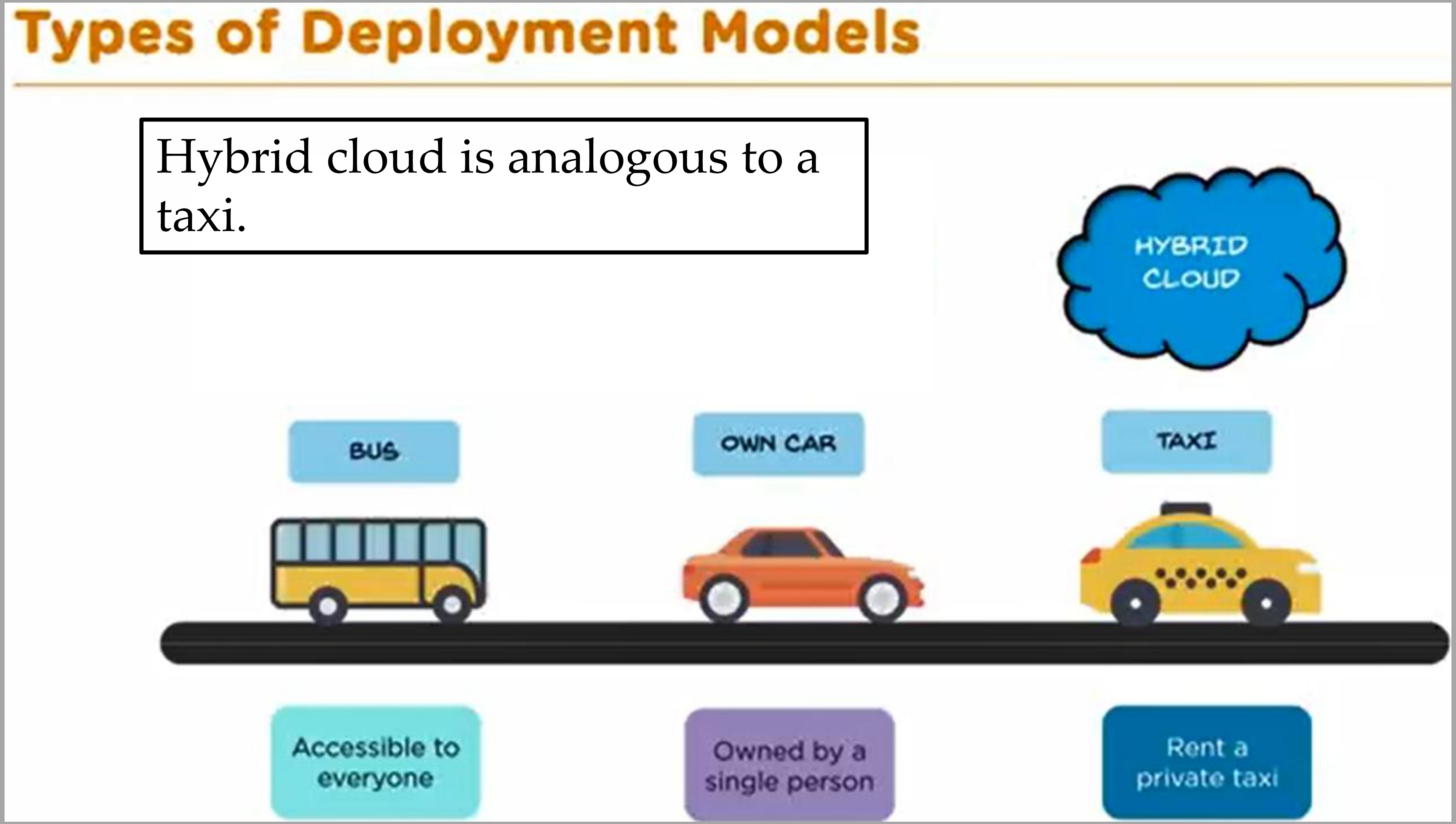


Cloud Deployment Models: Hybrid Cloud

- In hybrid Cloud environment, the organization consumes resources from both private and public Clouds. The ability to augment a private Cloud with the resources of a public Cloud can be utilized to maintain service levels in the face of rapid workload fluctuations.
- Organizations use their computing resources on a private Cloud for normal usage, but access the public Cloud for high/peak load requirements. This ensures that a sudden increase in computing requirement is handled gracefully.

Introduction to Cloud Computing

Hybrid cloud is analogous to a taxi.



Introduction to Cloud Computing

Cloud Deployment Models: Public Cloud

Advantages of Hybrid Cloud

The following are advantages of a hybrid cloud:

1) Control when needed;
2) Computation overflow.

Introduction to Cloud Computing

Cloud Deployment Models: Hybrid Cloud

Advantages of Hybrid Cloud

1) **Control when needed**

   If an organizations must comply with applicable regulations, it can use a hybrid cloud, which allows the organization to store classified data on a private cloud based on servers with controlled access while reducinr overall costs by pushing other data and computations to the public cloud.

Introduction to Cloud Computing

Cloud Deployment Models: Hybrid Cloud

Advantages of Hybrid Cloud

2) Computational overflow

Suppose an organization uses a private cloud for most of their computing needs. If the private cloud does not have enough sufficient resources to handle the load during its peak business period, it could consider adding additional servers, which will remain underutilized most of the time. As an alternative approach, the organization can send the excess computational tasks to a public cloud during the peak period.

Introduction to Cloud Computing

Cloud Deployment Models: Hybrid Cloud

Advantages of Hybrid Cloud

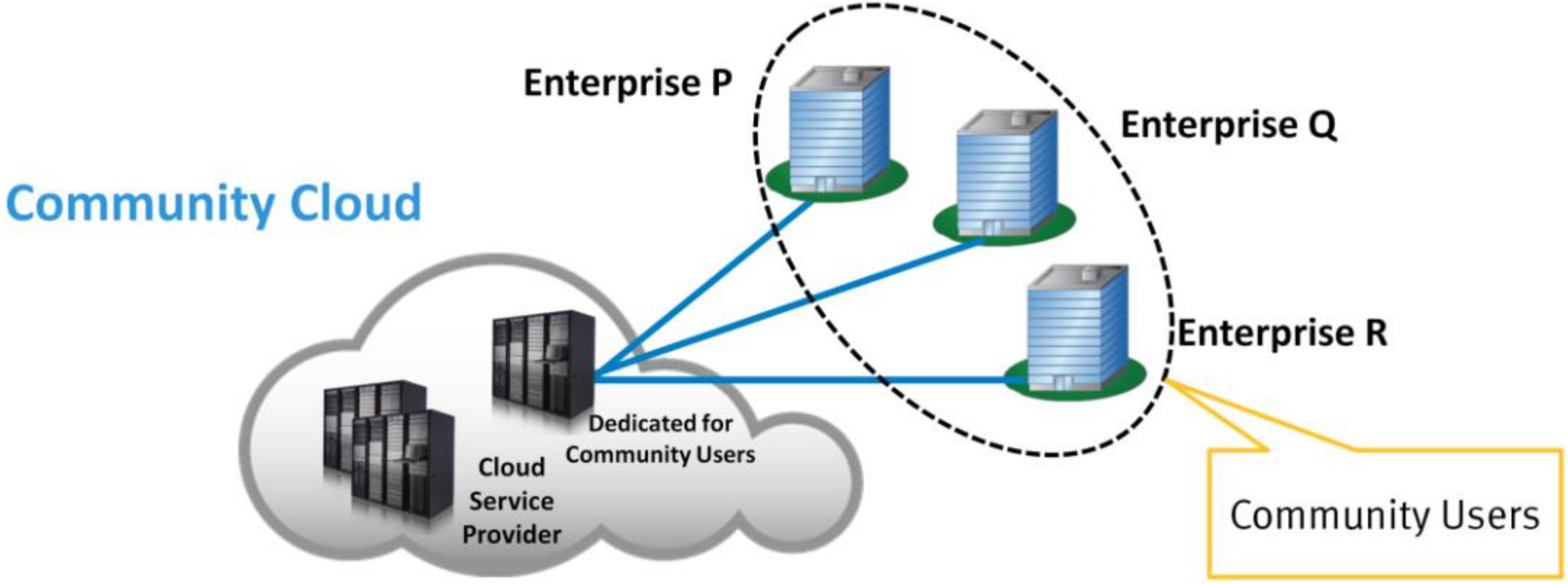
To summarize:

Using a hybrid cloud allows a company to enforce restrictions on data and computations in their private cloud and reduce costs by pushing the rest to a public cloud; the hybrid approach can also be used to handle overflow.

Introduction to Cloud Computing

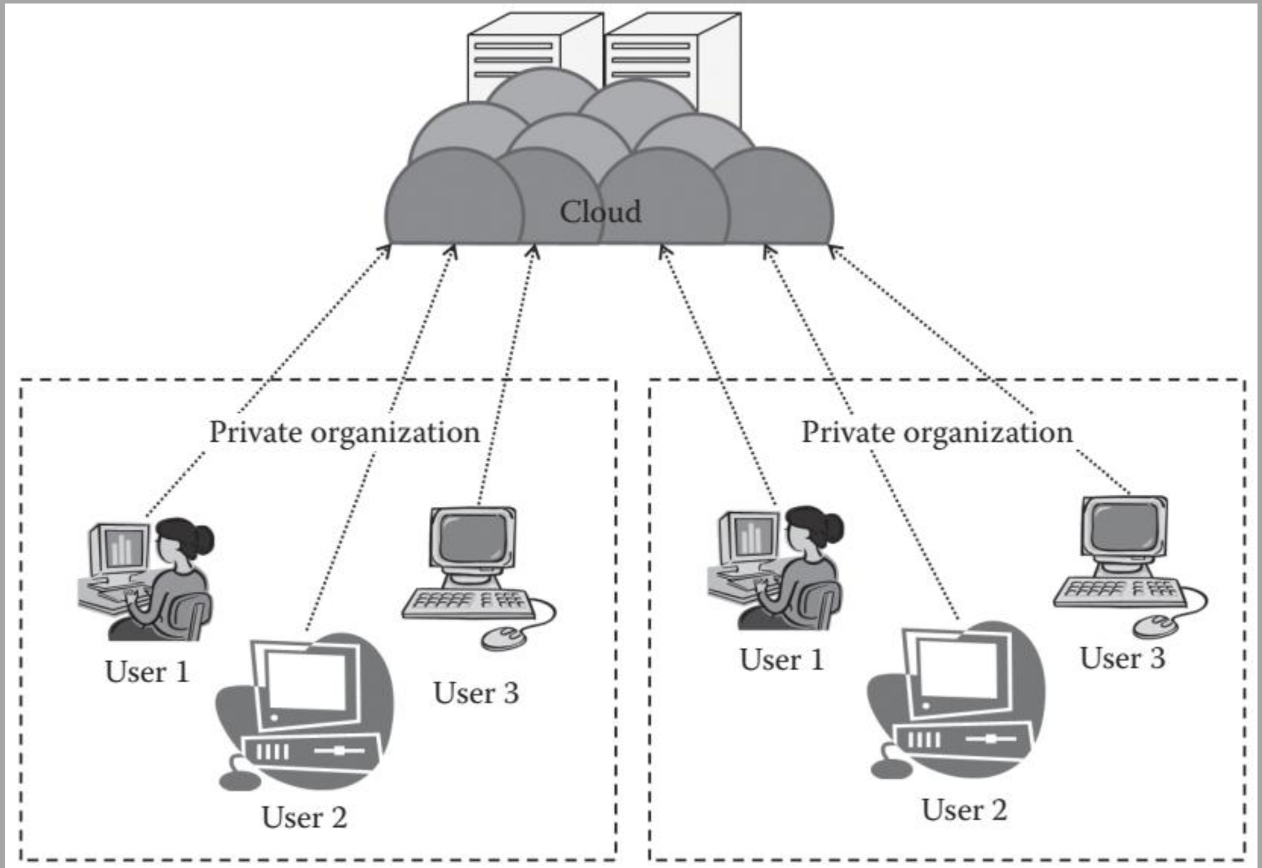
Cloud Deployment Models: Community Cloud

Cloud Deployment Model – Community Cloud



- Cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns
- Managed by the organizations or by a third party

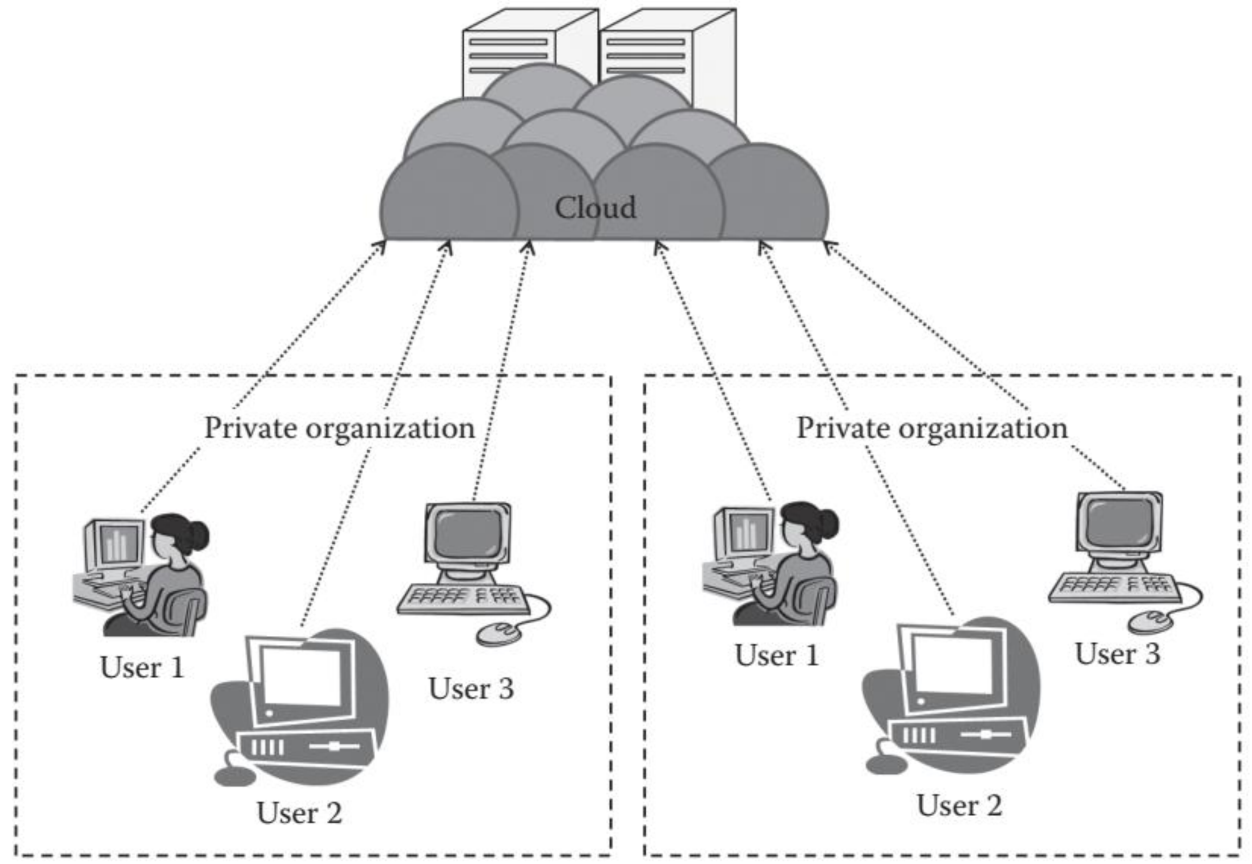
Introduction to Cloud Computing



Cloud Deployment Models: Community Cloud

- The community cloud is the cloud infrastructure that is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations).
- It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.
- It is a further extension of the private cloud. Here, a private cloud is shared between several organizations. Either the organizations or a single organization may collectively maintain the cloud.

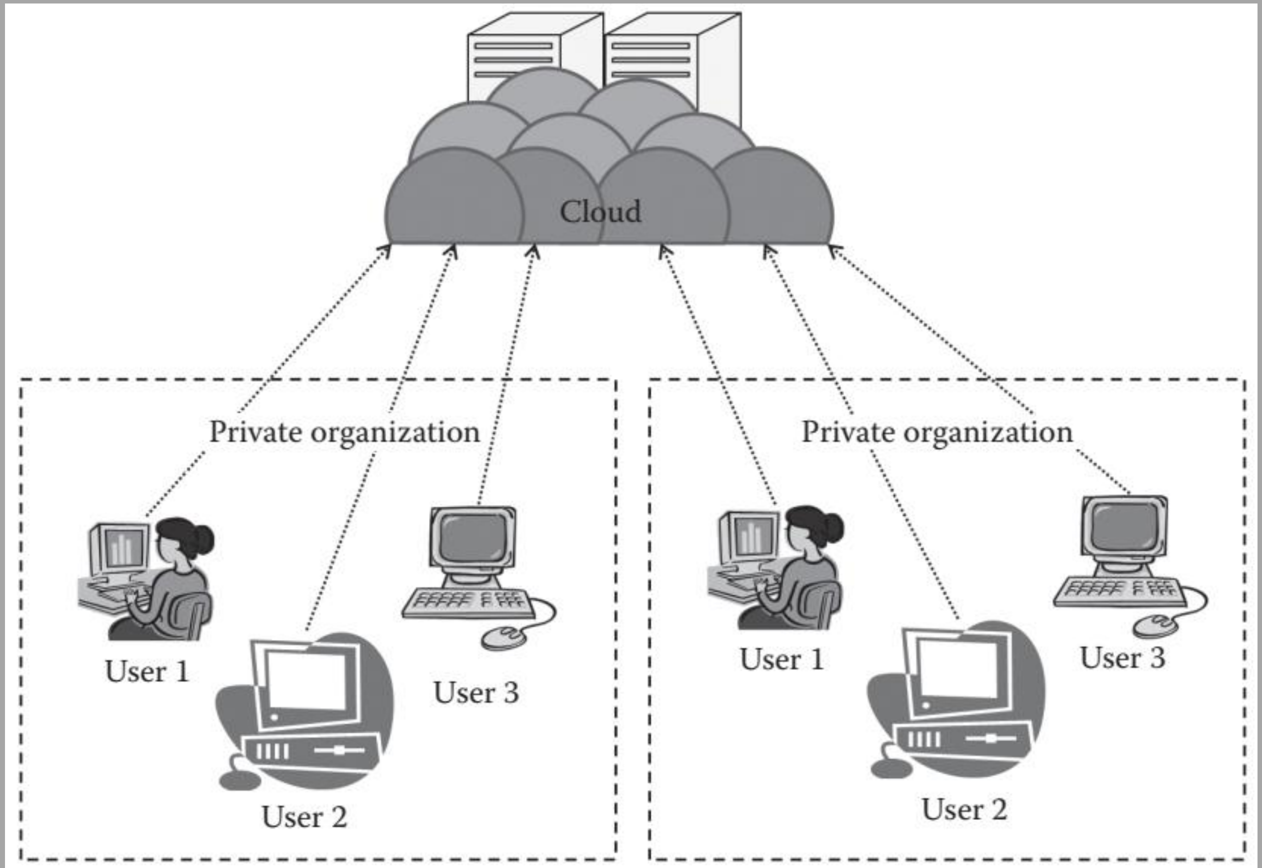
Introduction to Cloud Computing



Cloud Deployment Models: Community Cloud

- The main advantage of the public cloud is that the organizations are able to share the resources among themselves based on specific concerns. Thus, here the organizations are able to extract the power of the cloud, which is much bigger than the private cloud, and at the same time, they are able to use it at a usually less cost.

Introduction to Cloud Computing



Cloud Deployment Models: Community Cloud

Characteristics

- collaborative and distributive control;
- Partially secure/private;
- Cost effective.

Deployments: two types of community cloud:

1) On-premise community cloud;
2) Outsourced community cloud.

Introduction to Cloud Computing

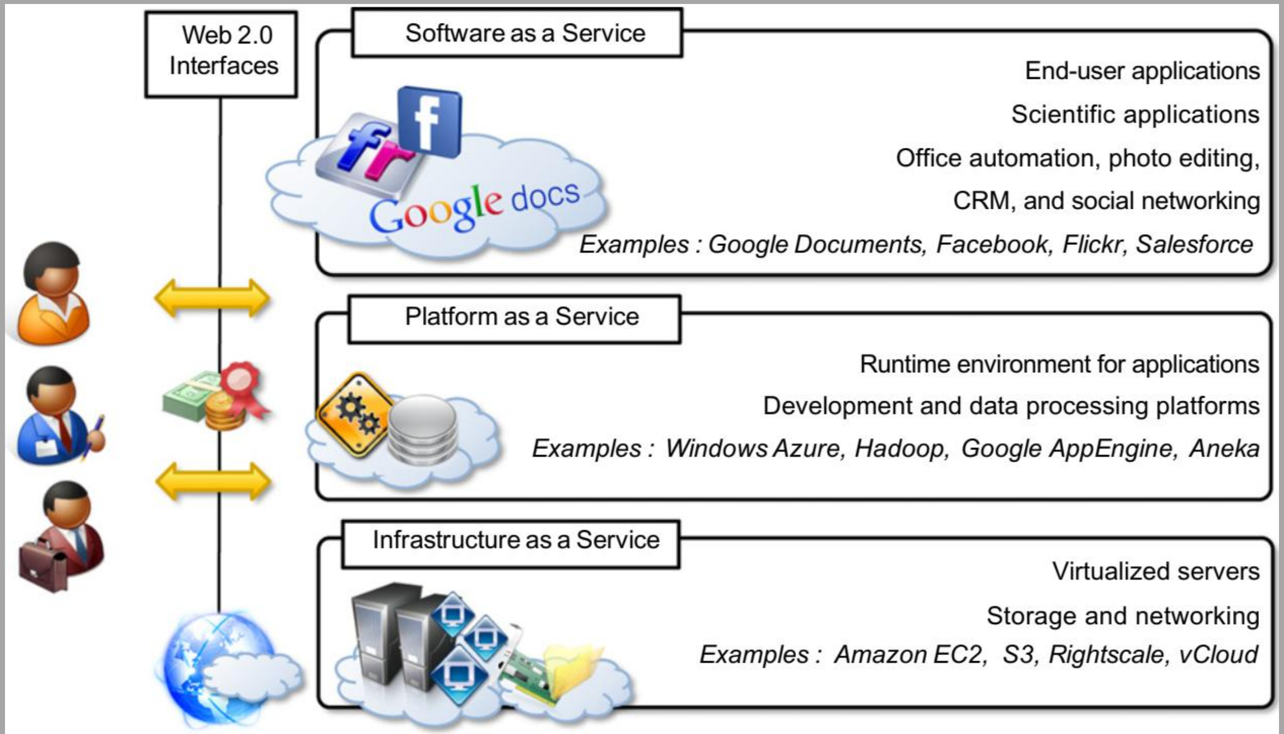
Model	Explanations
IaaS	The customer gets resources such as processing power, storage, network bandwidth, CPU, and power. Once the user gets the infrastructure, he controls the OS, data, applications, services, host-based security, and so on.
PaaS	The customer is provided with the hardware infrastructure, network, and operating system to form a hosting environment. The user can install his applications and activate services from the hosting environment.
SaaS	The customer/user is provided access to an application. He has no control over the hardware, network, security, or OS.

Cloud Computing Service Models

Cloud Computing Reference Model
or
Cloud Computing Service Models

Introduction to Cloud Computing
Cloud Computing Reference Model

- As we have seen in earlier discussions, a fundamental characteristic of cloud computing is the capability to deliver, on demand, a variety of IT services that are quite diverse from each other. This variety in services has created different views of what cloud computing is among users.
- Despite this lack of uniformity, it is possible to classify cloud computing services offerings into three major categories: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS). These categories are related to each other as shown in the figure, which provides an organic view of cloud computing.
- This classification is the **Cloud Computing Reference Model.** It organizes the wide range of cloud computing services into a layered view that shows the computing stack in an ascending order.



Cloud Computing Reference Model

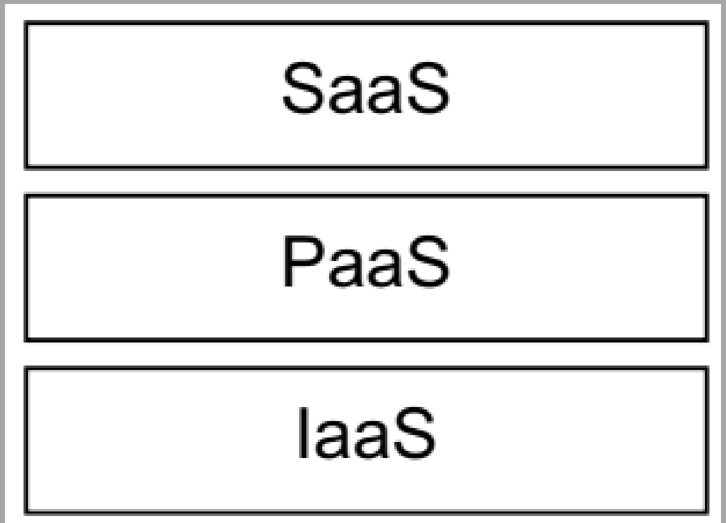
Introduction to Cloud Computing
Cloud Computing Reference Model

*The *-aaS Model*

In the cloud literature, there are three terms that describe the cloud service model:

- Infrastructure as a Service (IaaS),
- Platform as a Service (PaaS), and
- Software as a Service (SaaS).

These three terms form a high level model of how cloud is organized and also represent three different entry points to cloud.



Introduction to Cloud Computing

Cloud Computing Reference Model

*The *-aaS Model:* Infrastructure as Service (IaaS)

- **Infrastructure as Service (IaaS)** is the lowest abstraction layer on top of the actual data center hardware. It exposes resources such as virtual machines, virtual networks, load balancers, and IP addresses.
- **IaaS** brings on-premises concepts to cloud so that you can model your applications in the same way as in an on-premises environment. In other words, if you have an existing on-premises application, you can simply “lift” it from your on-premises data center and “shift” it to cloud—this is where shift-and-lift came from.

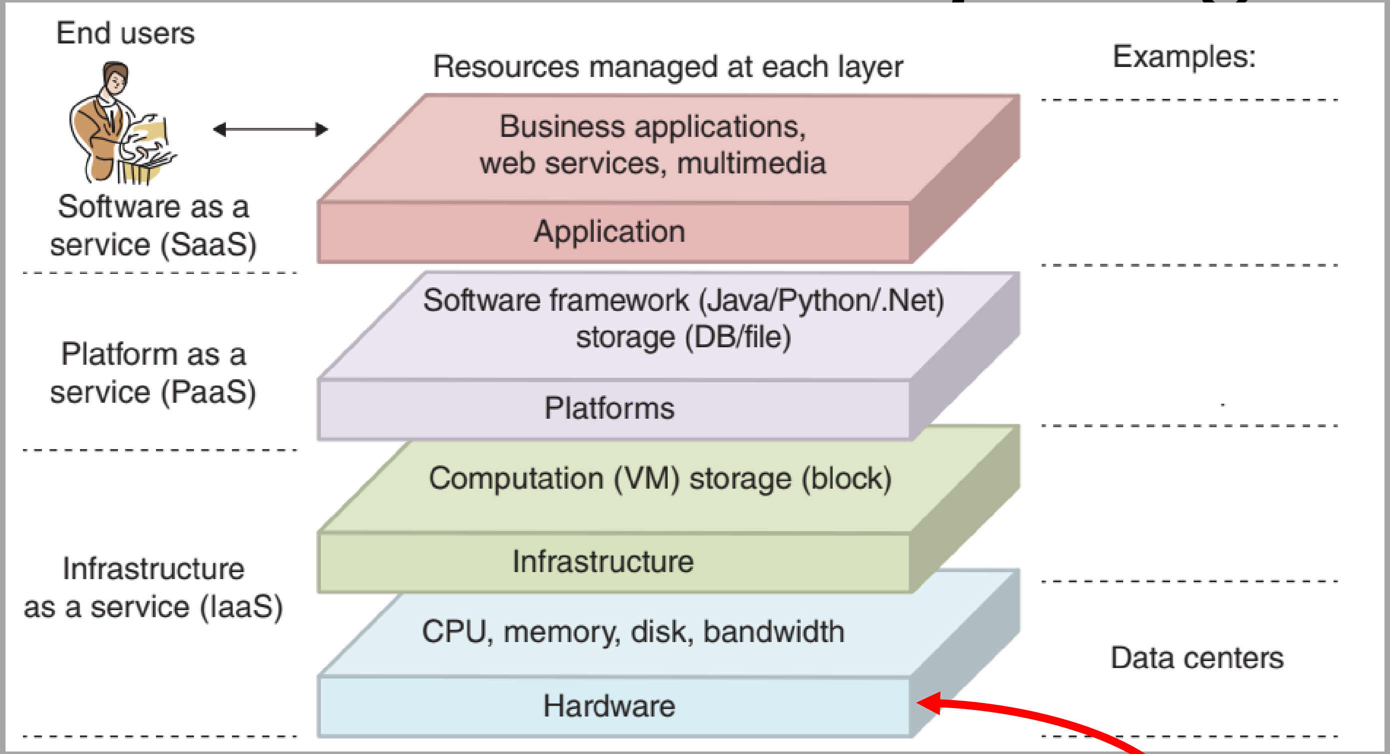
Introduction to Cloud Computing
Cloud Computing Reference Model

*The *-aaS Model:* Infrastructure as Service (IaaS)

- **Platform as a Service (PaaS)** hides infrastructural concepts and provides programming and hosting frameworks on top of which you design and deploy your applications.

Introduction to Cloud Computing

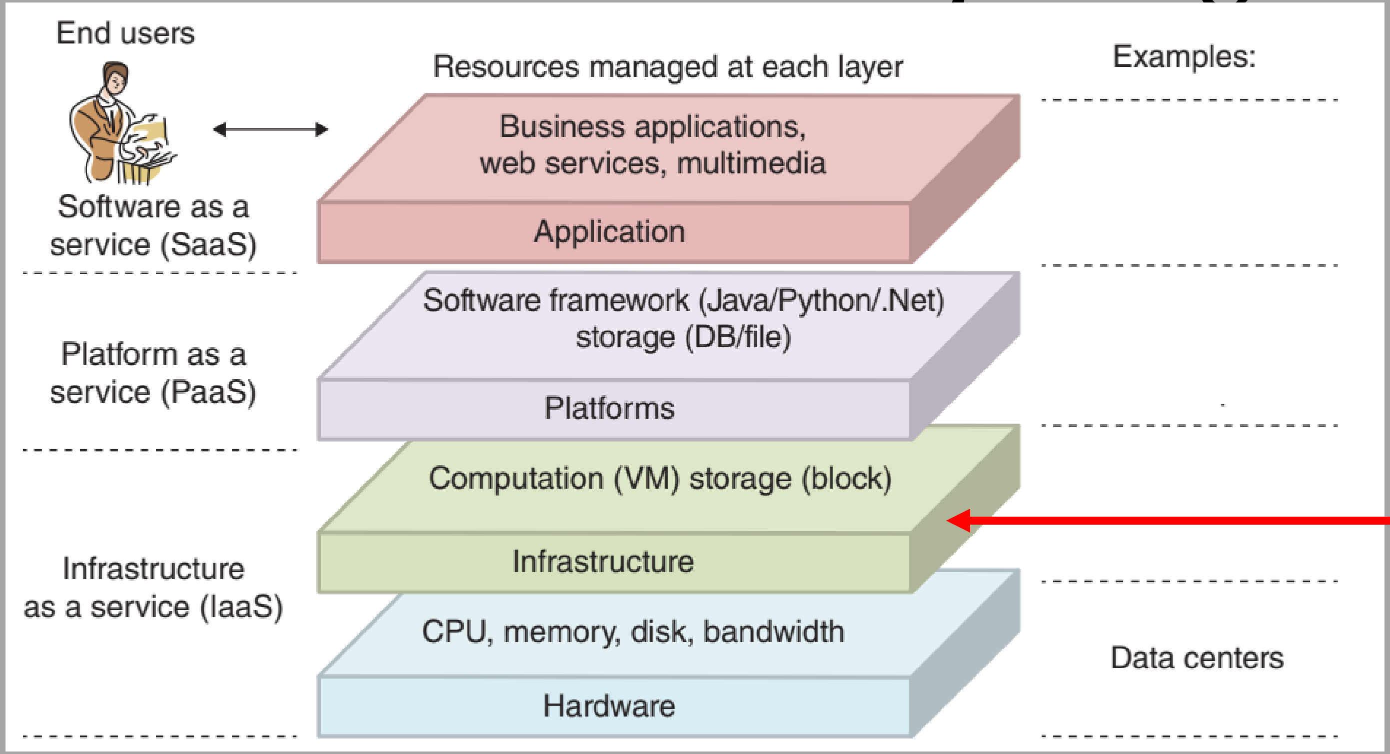
Cloud Computing Reference Model



The hardware layer: This layer is responsible for managing the physical resources of the cloud, including physical servers, routers, and switches, and power, and cooling systems. In practice, the hardware layer is typically implemented in **data centers**. A data center usually contains thousands of servers that are organized in racks and interconnected through switches, routers, or other fabrics.

Introduction to Cloud Computing

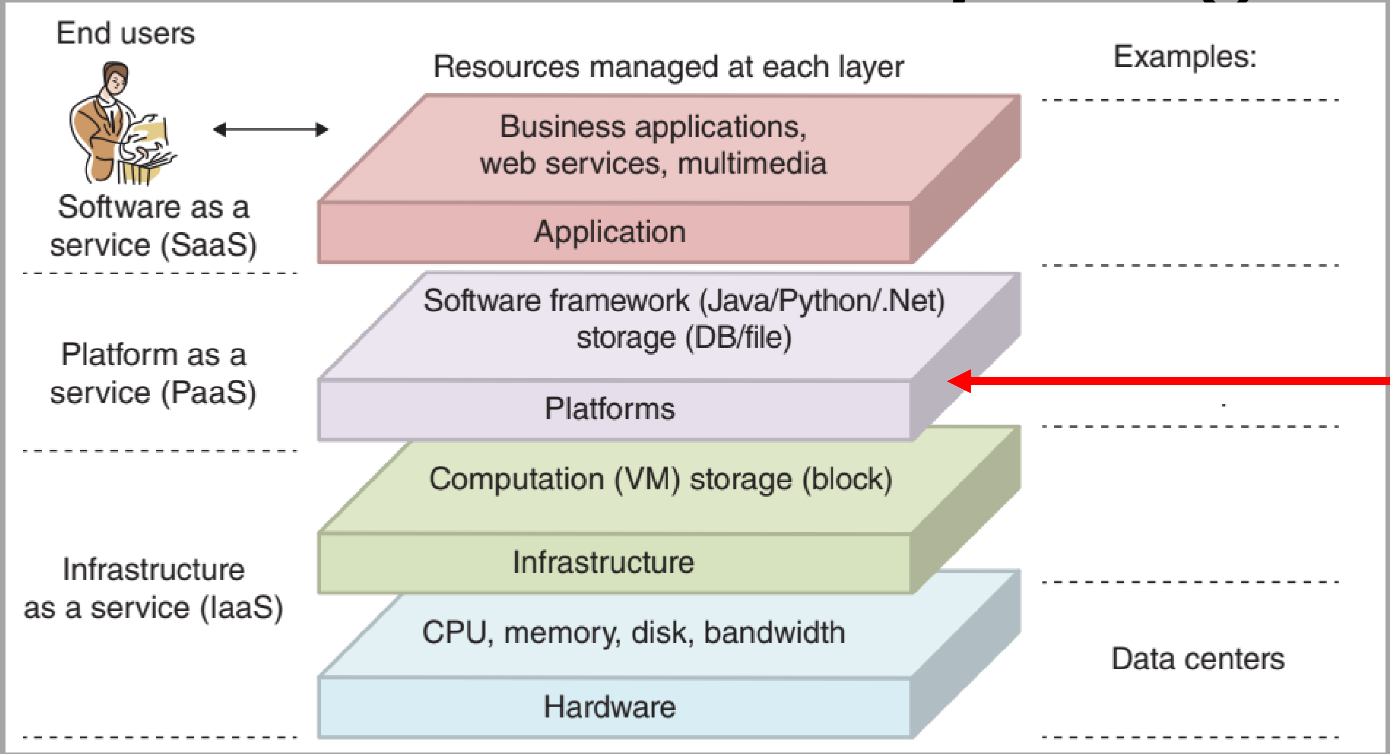
Cloud Computing Reference Model



The infrastructure layer: It is also known as the virtualization layer, the infrastructure layer creates a pool of storage and computing resources by partitioning the physical resources using virtualization technologies. The infrastructure layer is an essential component of cloud computing, since many key features, such as dynamic resource assignment, are only made available through virtualization technologies.

Introduction to Cloud Computing

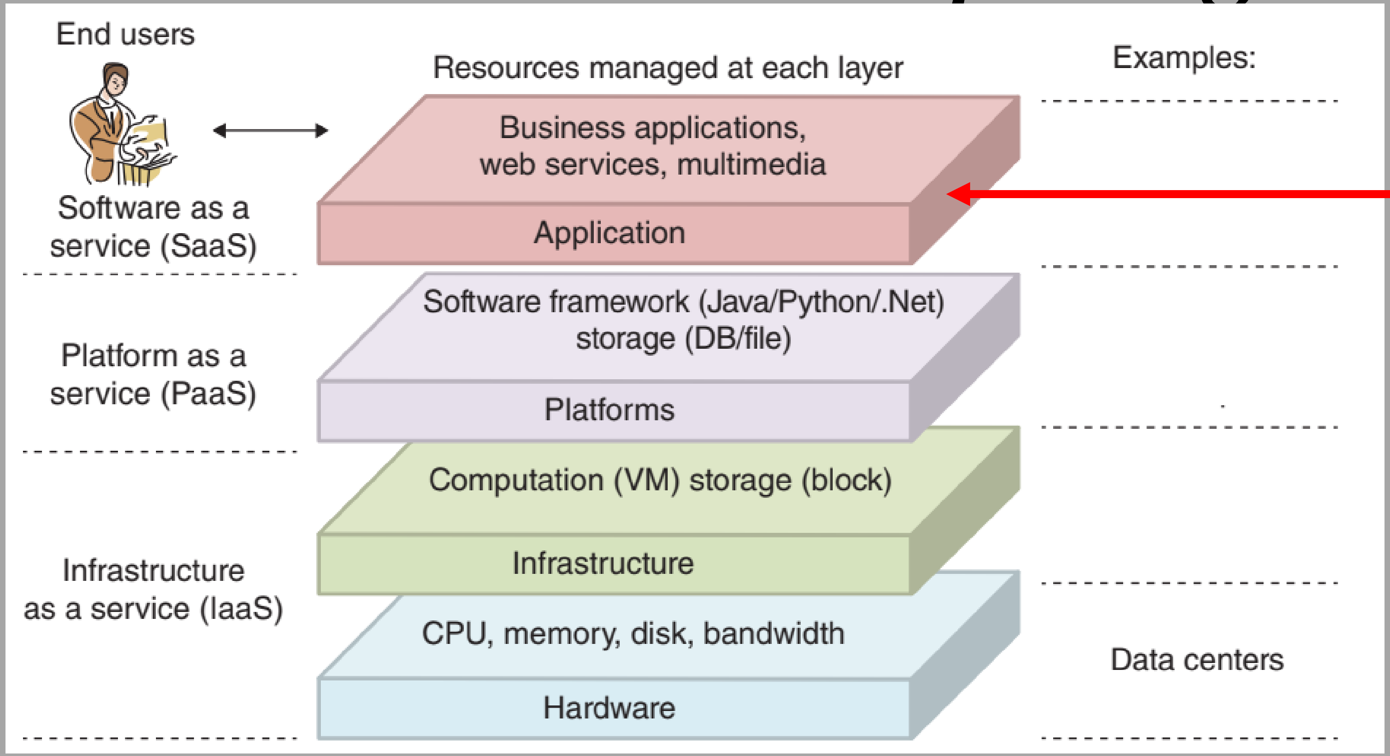
Cloud Computing Reference Model



The platform layer: Built on top of the infrastructure layer, the platform layer consists of operating systems and application frameworks. The purpose of the platform layer is to minimize the burden of deploying applications.

Introduction to Cloud Computing

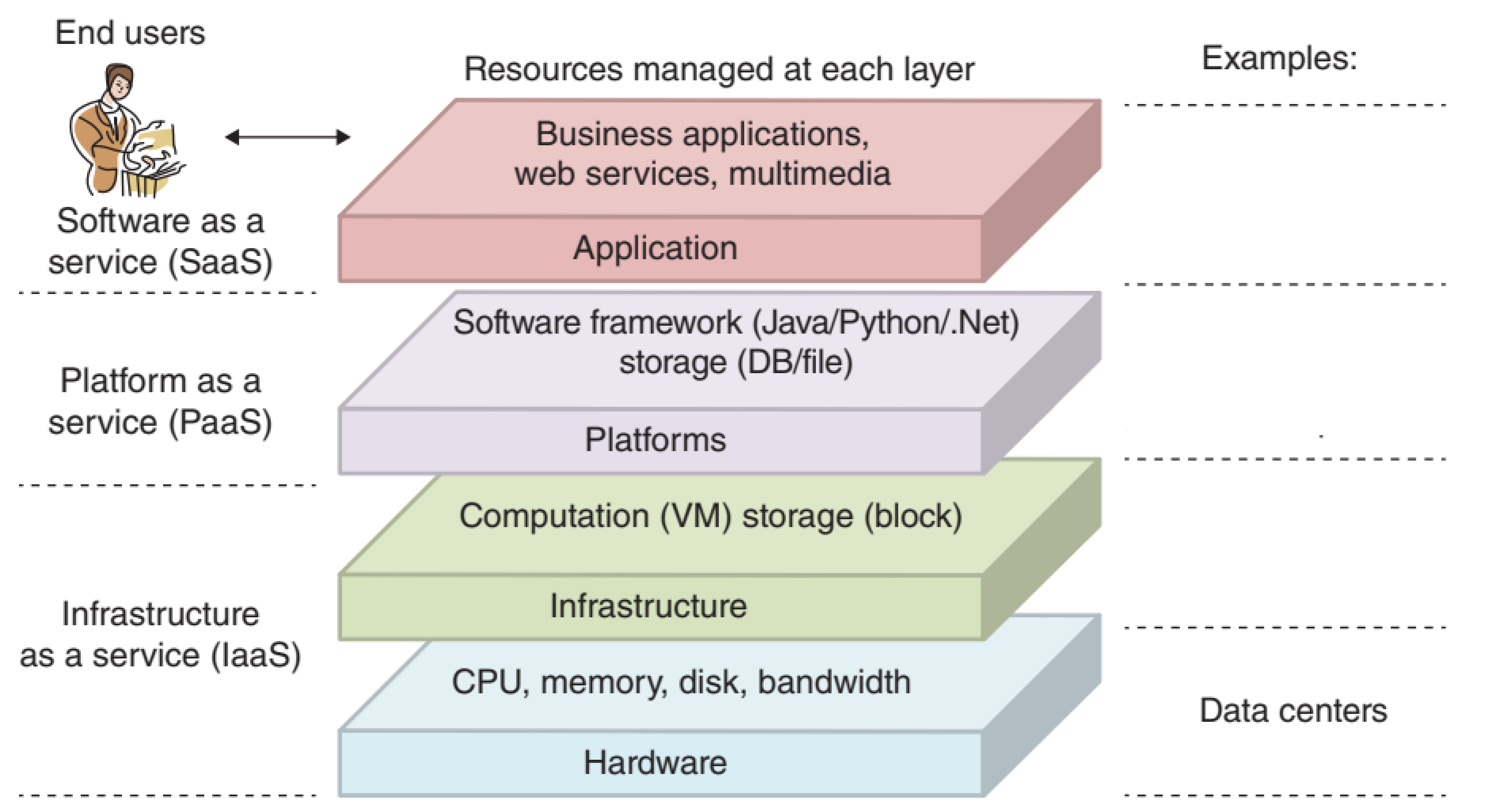
Cloud Computing Reference Model



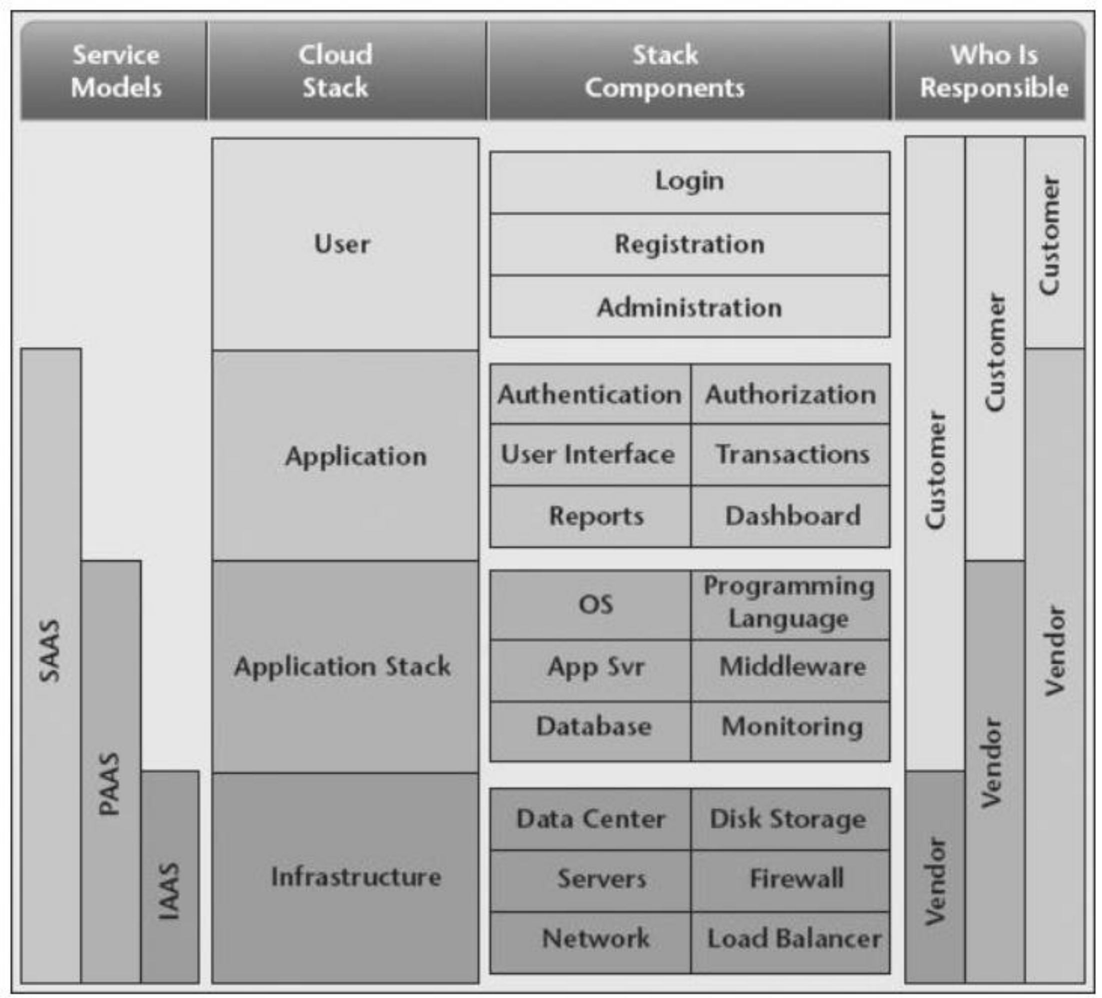
***The application layer*:** At the highest level of the hierarchy, the application layer consists of the actual cloud applications. Different from traditional applications, cloud applications can leverage the automatic-scaling feature to achieve better performance, availability, and lower operating cost.

Introduction to Cloud Computing

Cloud Computing Reference Model



Introduction to Cloud Computing
Cloud Computing Reference Model



Cloud Stack

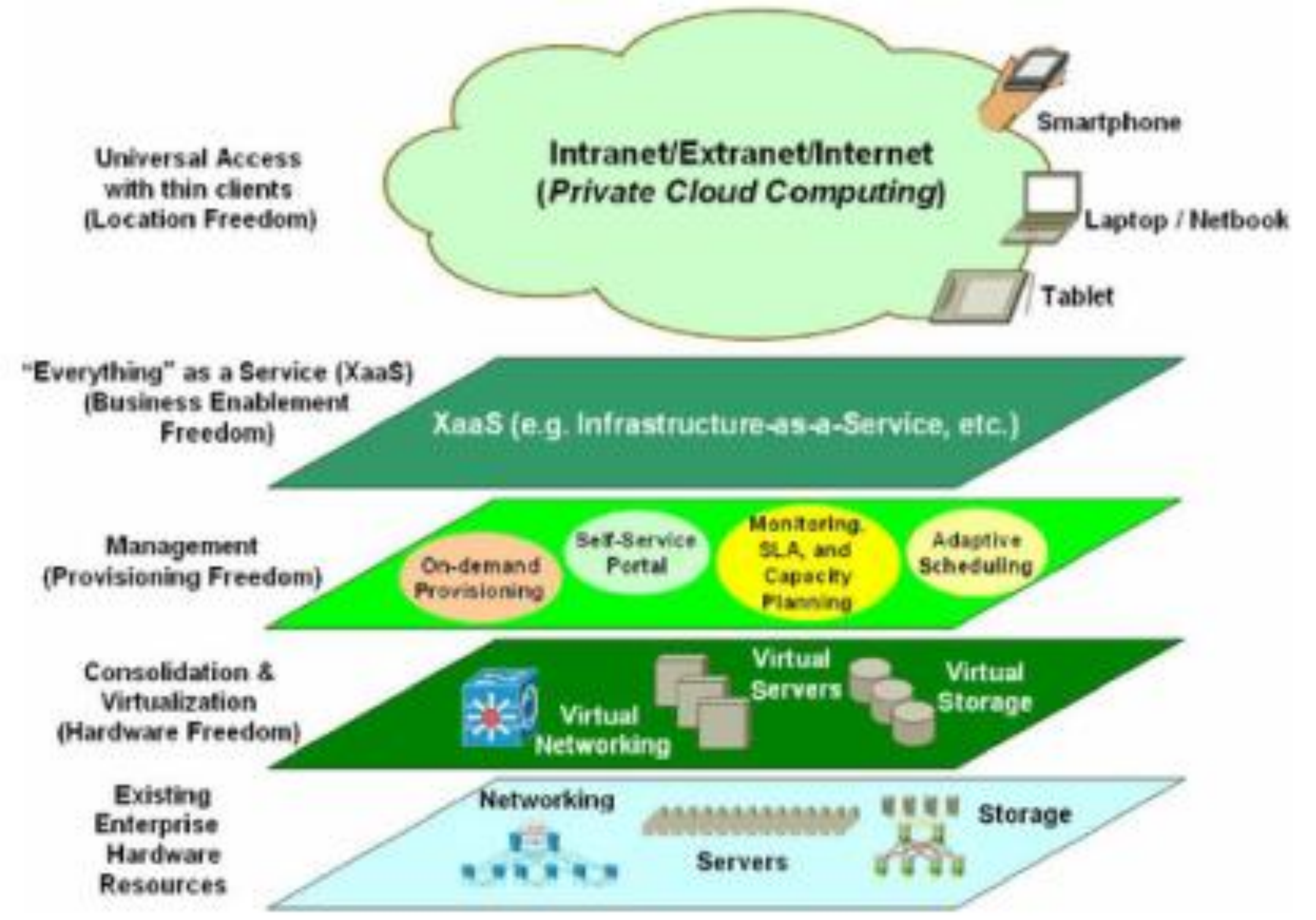
Service-Oriented Infrastructure (SOI)

Introduction to Cloud Computing
Virtualization

Service-Oriented Infrastructure (SOI)

There are so many different problems to be solved in building a next-generation infrastructure that it's useful to organize the approach into layers.

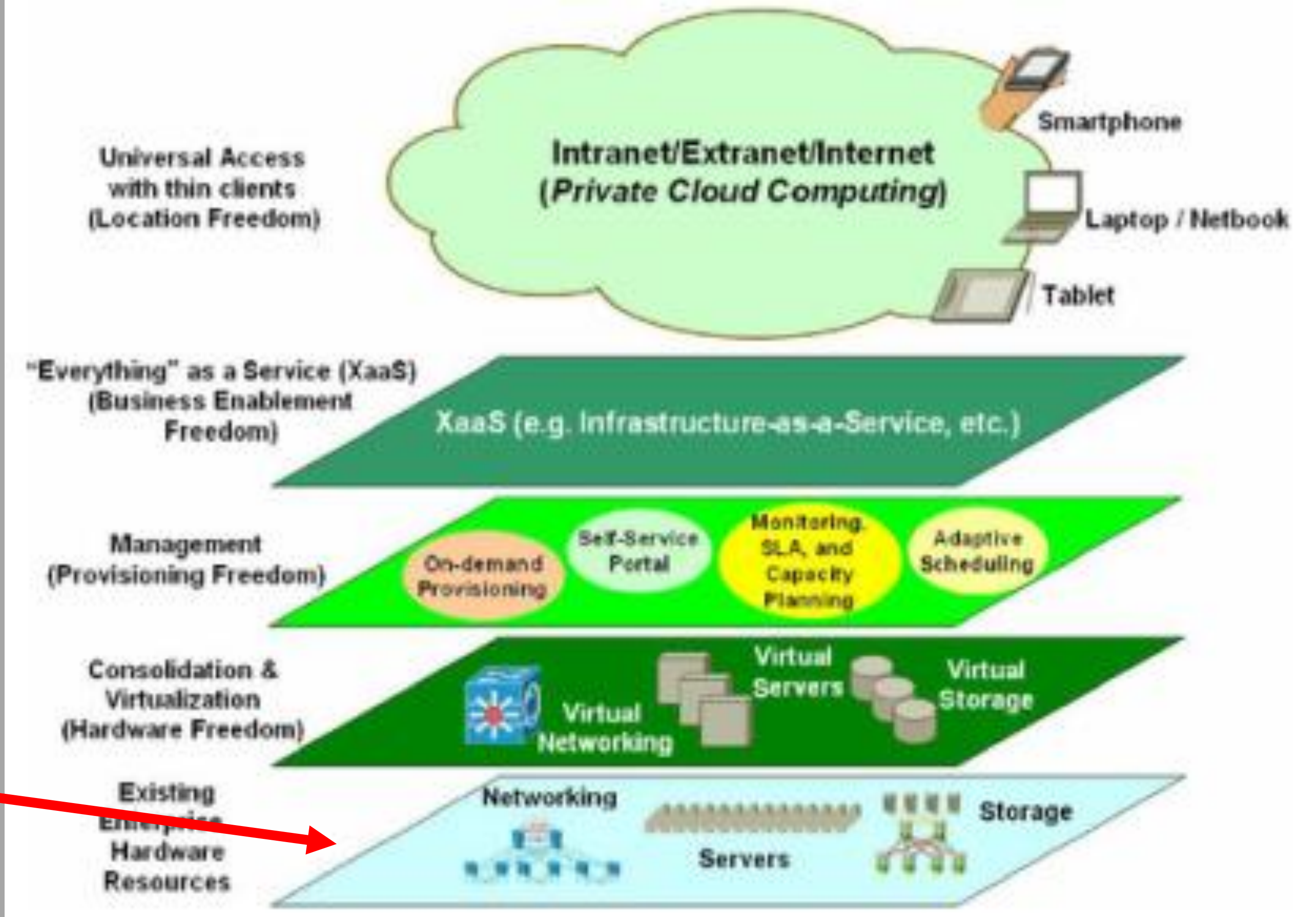
- The top layer supplies various kinds of fantastically powerful, incredibly flexible services to end-users.



Introduction to Cloud Computing Virtualization

Service-Oriented Infrastructure (SOI)

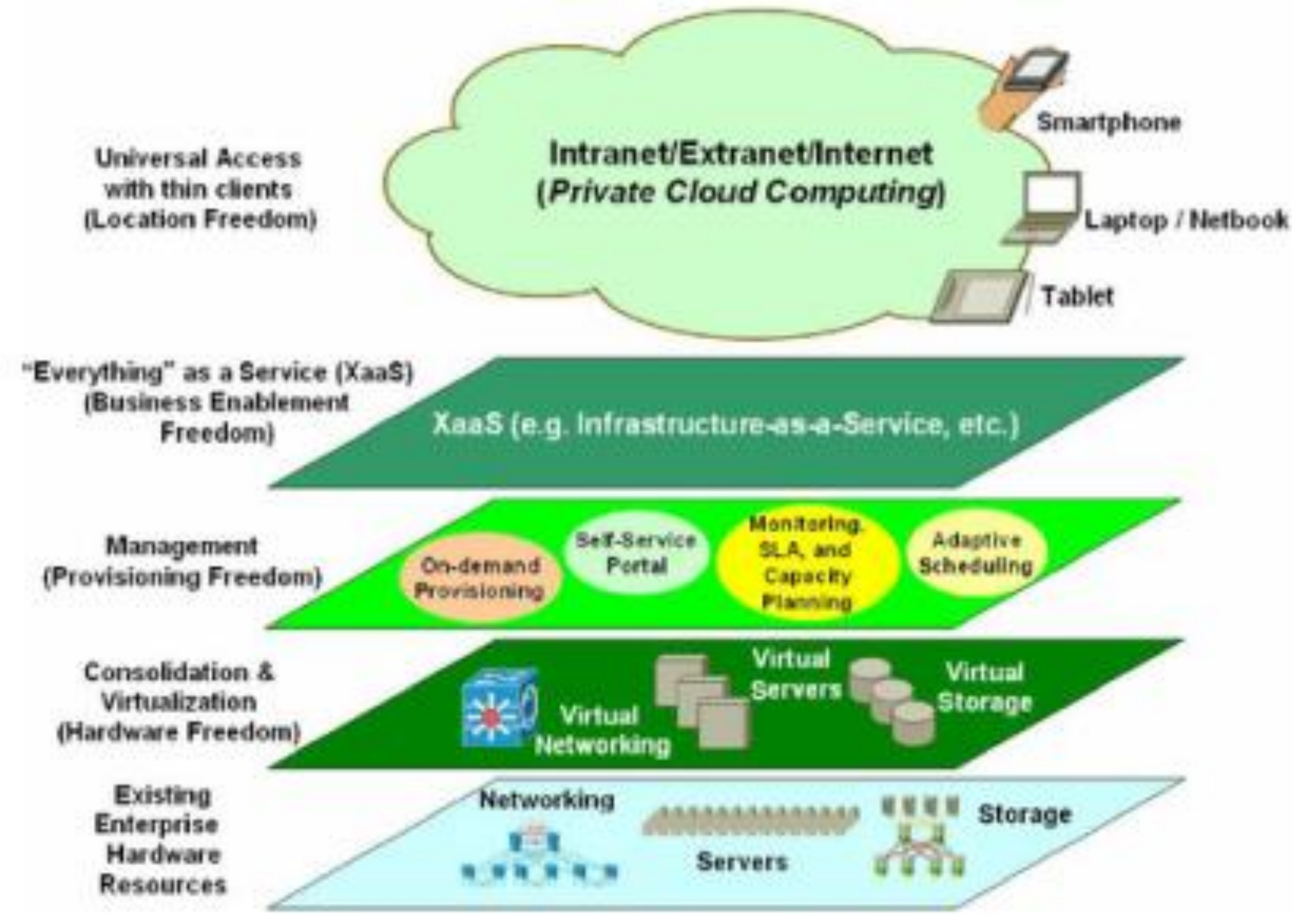
- The bottom layer is a collection of off-the-shelf hardware of various kinds—servers, storage, networking routers and switches, and long-distance telecom services.



Introduction to Cloud Computing
Virtualization

Service-Oriented Infrastructure (SOI)

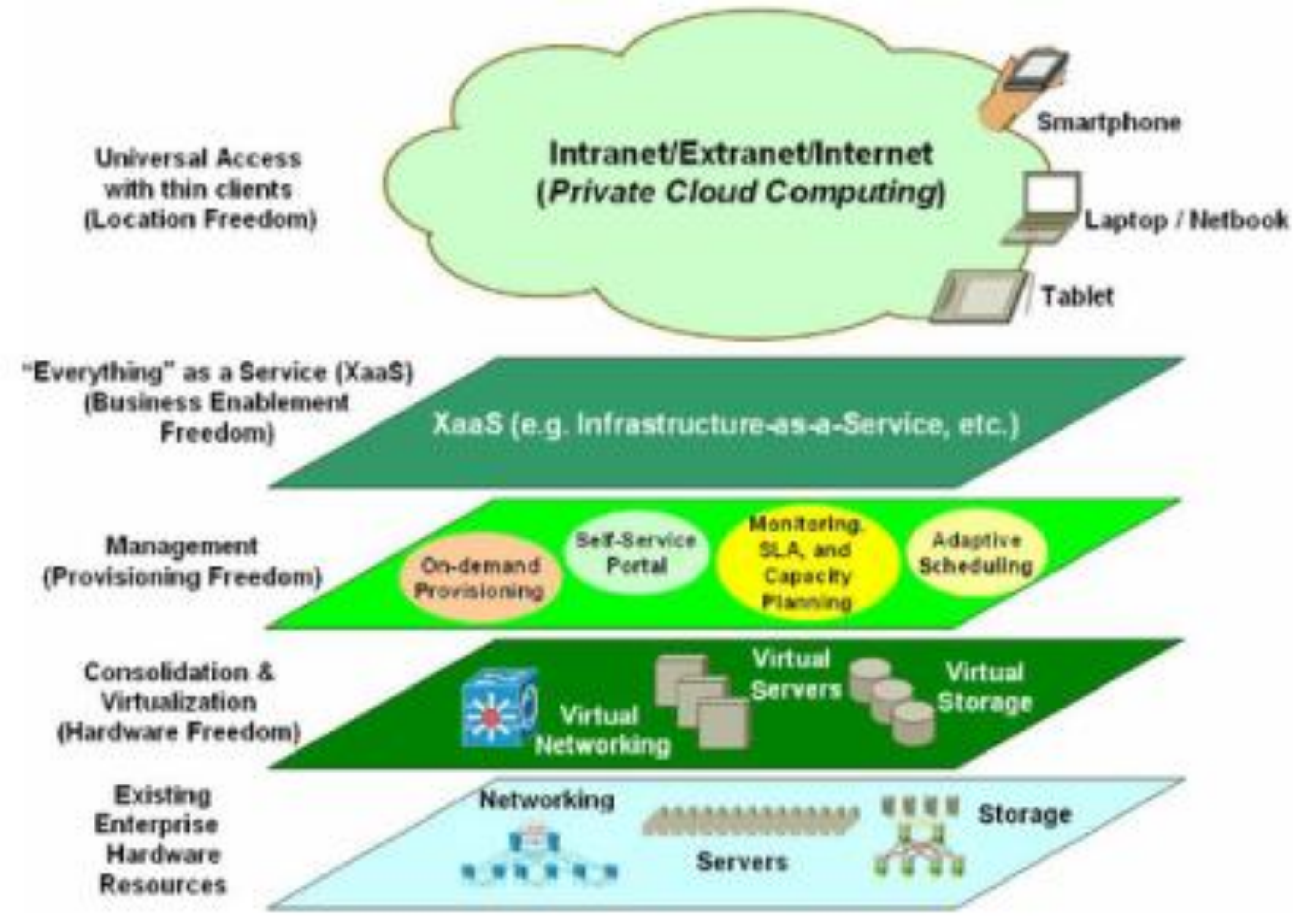
- The intervening layers use the relatively crude facilities of the lower layers to build a new set of more sophisticated facilities.



Introduction to Cloud Computing
Virtualization

Service-Oriented Infrastructure (SOI)

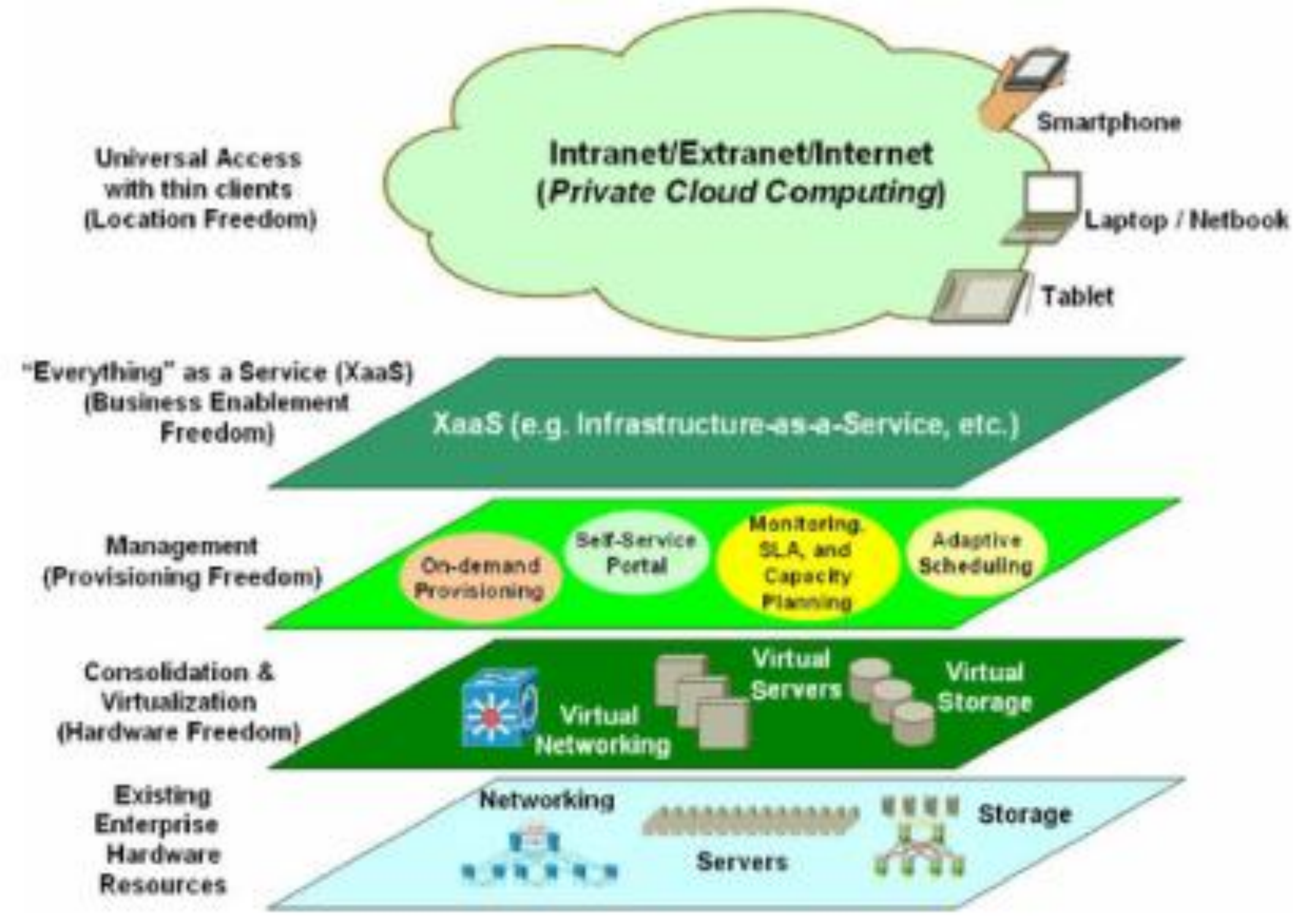
There are so many different problems to be solved in building a next-generation infrastructure that it's useful to organize the approach into layers. The top layer supplies various kinds of fantastically powerful, incredibly flexible services to end-users. The bottom layer is a collection of off-the-shelf hardware of various kinds—servers, storage, networking routers and switches, and long-distance telecom services. The intervening layers use the relatively crude facilities of the lower layers to build a new set of more sophisticated facilities.



Introduction to Cloud Computing
Virtualization

Service-Oriented Infrastructure (SOI)

This particular layered model is called a service-oriented infrastructure (SOI) framework. The layer immediately above the physical hardware is concerned with virtualization—reducing or eliminating the limitations associated with using particular models of computers, particular sizes of disks, and so on. The layer above that is concerned with management and provisioning—associating the idealized resources with the demands that are being placed. A layer above management and provisioning exports these automatic capabilities in useful combinations through a variety of network interfaces, allowing the



Cloud Service Models

Introduction to Cloud Computing
Cloud Computing Reference Model

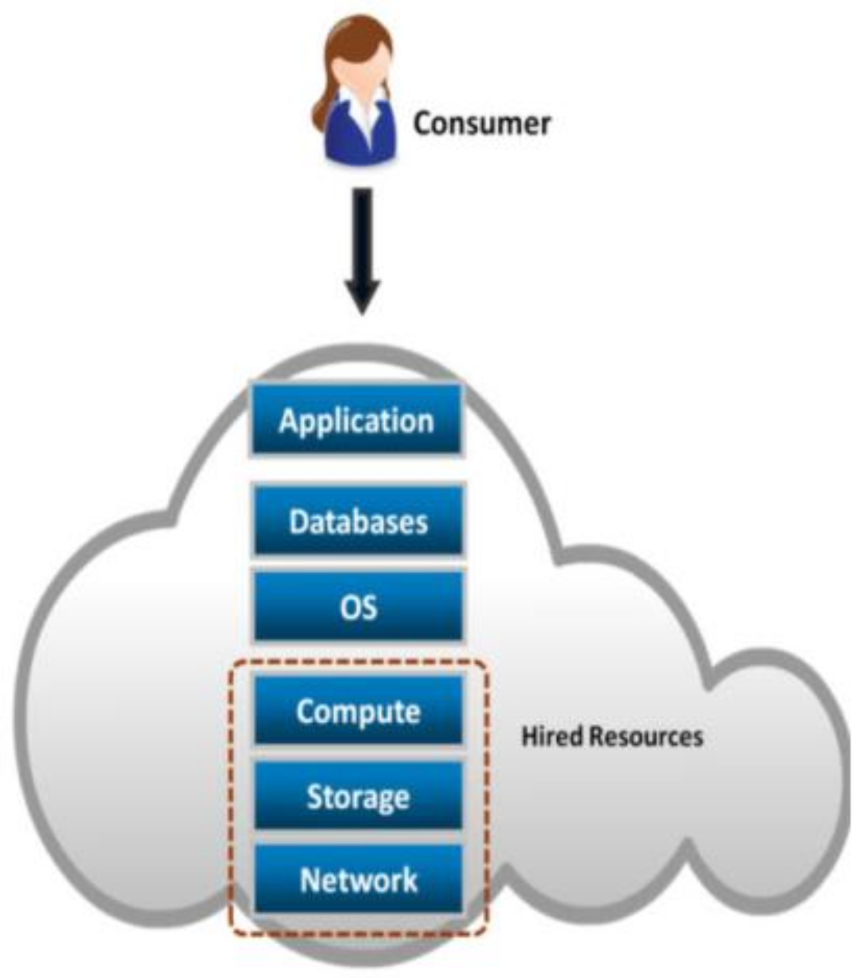
Infrastructure as a Service (IaaS)

Introduction to Cloud Computing
Cloud Computing Reference Model

Cloud computing models:

Infrastructure as a Service (IaaS)

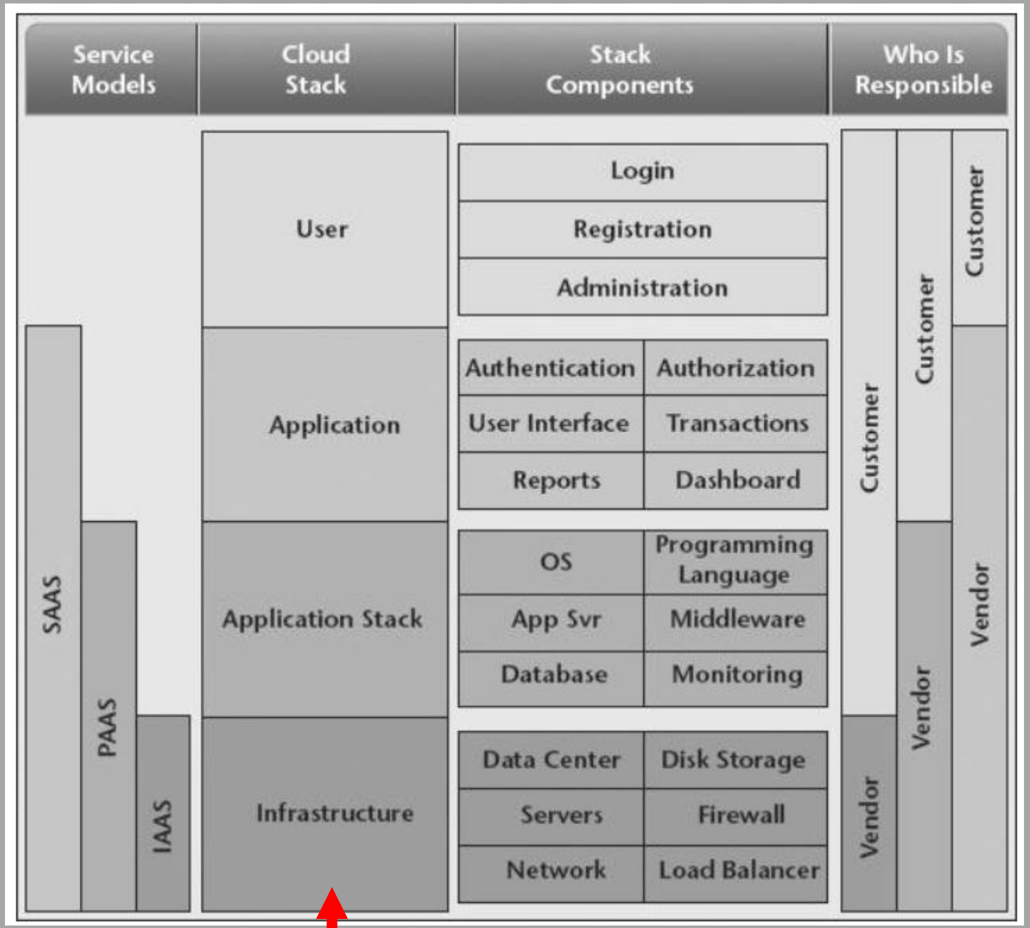
- Provides capability to the consumer to hire infrastructure components such as servers, storage, and network.
- Enables consumers to deploy and run software, including OS and applications.
- Pays for infrastructure components usage, for example, storage capacity, CPU usage, etc.



Introduction to Cloud Computing
Cloud Computing Reference Model

Cloud computing models:
Infrastructure as a Service (IaaS)

- The IaaS model provides the underlying hardware (servers and storage). Clients must install and then manage their own operating system, database software, and support software.



Introduction to Cloud Computing
Cloud Computing Reference Model

Cloud computing models

Infrastructure as a Service (IaaS)

- Infrastructure as a Service (IaaS) contains the basic building blocks for cloud IT and typically provides access to networking features, computers (virtual or on dedicated hardware), and data storage space.
- IaaS provides the highest level of flexibility and management control over IT resources and is most similar to existing IT resources that many IT departments and developers are familiar with today.

Introduction to Cloud Computing
Cloud Computing Reference Model

Cloud computing models

Infrastructure as a Service (IaaS)

In summary, IaaS provides virtual data center capabilities [physical data center and physical infrastructure (servers, disk storage, networking, and so forth)] so service consumers can focus more on building and managing applications and less on managing data centers and infrastructure.

Introduction to Cloud Computing
Cloud Computing Reference Model

Infrastructure as a Service (IaaS): Advantages

Introduction to Cloud Computing
Cloud Computing Reference Model

Cloud computing models

Advantages of Infrastructure as a Service (IaaS)

In the IaaS model, you have maximum control, which means you can achieve the required flexibility as per your application's needs. These are the advantages of using the IaaS model:

1) It offers the most flexibility of all the cloud models.
2) Provisioning of compute, storage, and networking resources can be done quickly.
3) Resources can be used for a few minutes, hours, or days.
4) Complete control of the infrastructure.
5) Highly scalable and fault-tolerant.

Introduction to Cloud Computing
Cloud Computing Reference Model

Advantages of Infrastructure as a Service (IaaS)

The following are the key features and benefits of IaaS:

- **Scalability** Within an IaaS framework, the system can be rapidly provisioned and expanded as needed, either for predictable events or in response to unexpected demand.
- **Cost of ownership of physical hardware** Within IaaS, the customer does not need to procure any hardware either for the initial launch and implementation or for future expansion.

Introduction to Cloud Computing

Cloud Computing Reference Model

Advantages of Infrastructure as a Service (IaaS)

The following are the key features and benefits of IaaS:

- **High availability** The cloud infrastructure, by definition, meets high availability and redundancy requirements, which would result in additional costs for a customer to meet within their own data center.
- **Physical and logical security requirements** Because you're in a cloud environment and don't have your own data centers, the cloud provider assumes the cost and oversight of the physical security of its data centers. Data is also protected by layers of logical network security and user access security.

Introduction to Cloud Computing
Cloud Computing Reference Model

Advantages of Infrastructure as a Service (IaaS)

The following are the key features and benefits of IaaS:

- **Location and access independence** The cloud-based infrastructure has no dependence on the physical location of the customer or users of the system, as well as no dependence on specific network locations or applications or clients to access the system. The only dependency is on the security requirements of the cloud itself and the applications settings used.
- **Metered usage** The customer only pays for the resources they are using and only during the durations of use.

Introduction to Cloud Computing
Cloud Computing Reference Model

Advantages of Infrastructure as a Service (IaaS)

The following are the key features and benefits of IaaS:

- **Potential for “green” data centers** Many customers and companies are interested in having more environmentally friendly data centers that are high efficiency in terms of both power consumption and cooling. Within cloud environments, many providers have implemented “green” data centers that are more cost-effective with the economies of scale that would prohibit many customers from having their own. An additional benefit is that fewer physical machines and lower electricity consumption translate to environmental friendliness – greening friendly.

- **Choice of hardware** Cloud providers offer a variety of processers, each has its own scaling and configuration options.

Introduction to Cloud Computing

Cloud Computing Reference Model

Infrastructure as a Service (IaaS): Disadvantages

Introduction to Cloud Computing

Cloud Computing Reference Model

Cloud computing models : Infrastructure as a Service (IaaS)

Disadvantages of Infrastructure as a Service (IaaS)

Because the IaaS model has more control, it means putting additional effort into maintaining, monitoring, and scaling infrastructure, which leads to the following disadvantages:

1) **Security**: In this case, customers have much more control over the stack, and for this reason, it is highly critical that they have a comprehensive plan in place for security. Since customers manage applications, data, middleware, and the operating system, there are possible security threats.

Introduction to Cloud Computing

Cloud Computing Reference Model

Cloud computing models

Disadvantages of Infrastructure as a Service (IaaS)

2) **Legacy systems**: While customers can migrate legacy applications, the older hardware may need help to provide the needed functionality to secure the legacy applications. Modifications to older applications may be required, potentially creating new security issues unless the application is thoroughly tested for new security vulnerabilities.

3) **Training costs**: As with any new technology, training may be needed for the customer's staff to get familiar with the new infrastructure. The customer is ultimately responsible for securing their data and resources, computer backups, and business continuity.

Introduction to Cloud Computing
Cloud Computing Reference Model

Platform as a Service (PaaS)

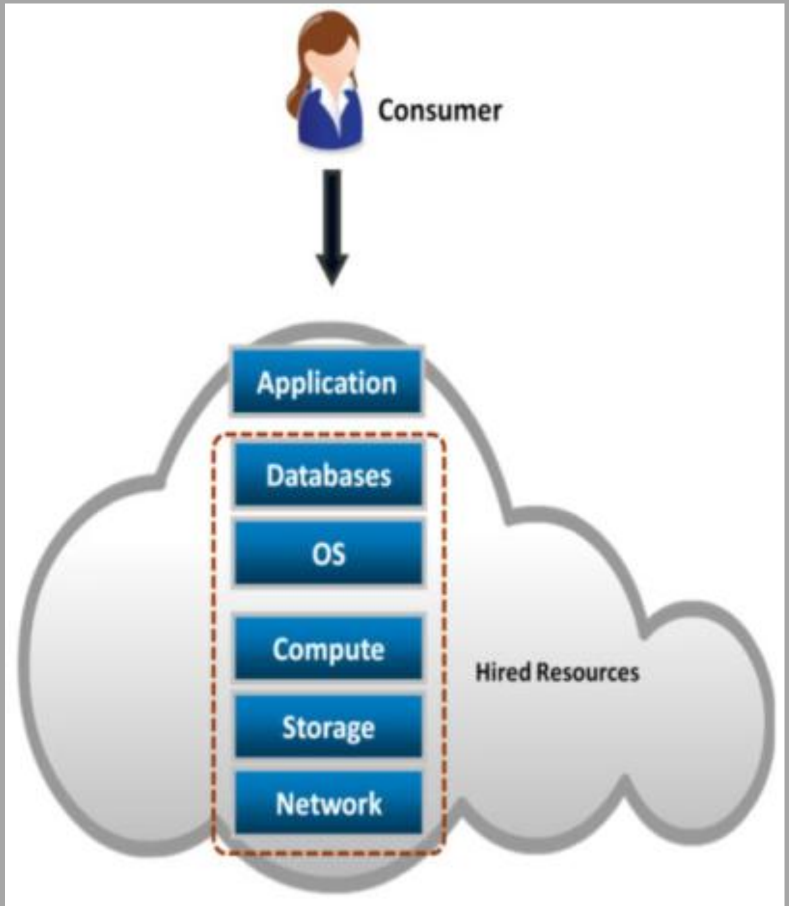
Introduction to Cloud Computing

Cloud Computing Reference Model

Cloud computing models

Platform as a Service (PaaS)

- Capability provided to the consumer to deploy consumer created or acquired applications on the Cloud provider's infrastructure.
- Consumer has control over
  - Deployed applications
  - Possible application hosting environment configurations
- Consumer is billed for platform software components:
  - OS, Database, Middleware



Introduction to Cloud Computing
Cloud Computing Reference Model

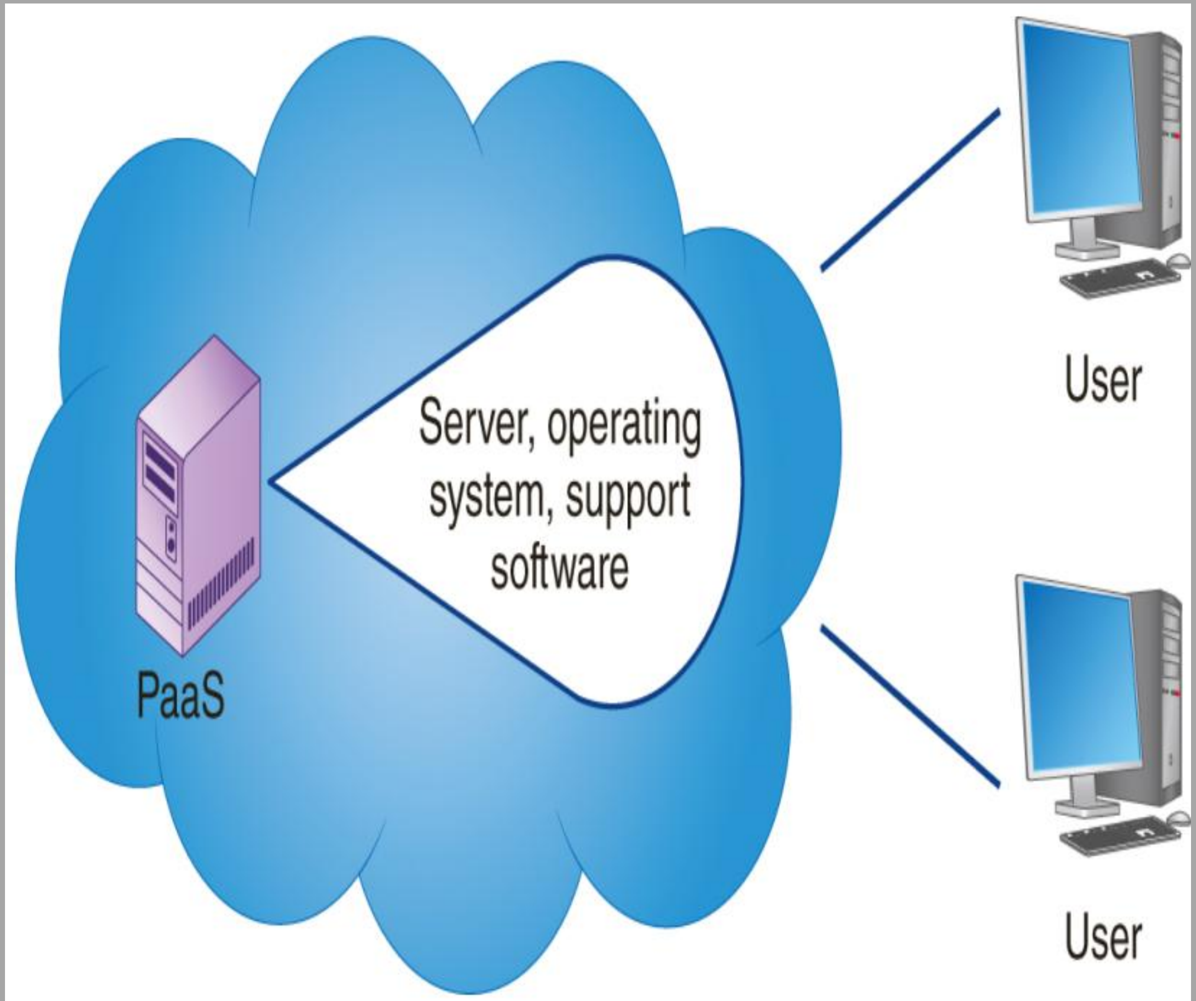
Cloud computing models

Platform as a Service (PaaS)

- Platform as a Service (PaaS) removes the need for the organization to manage the underlying infrastructure (usually hardware and operating systems) and allows it to focus on the deployment and management of applications.
- This results in more efficiency as there is no need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.

Introduction to Cloud Computing

Cloud Computing Reference Model



Cloud computing models

Platform as a Service (PaaS)

- The **platform as a service (PaaS)** model provides the underlying hardware technology, such as one or more servers (physical or virtual servers), operating systems, database solutions, developer tools, and network support, for developers to deploy their own solutions. The hardware and software within a PaaS solution are managed by the platform provider. Developers need not worry about performing hardware or operating system upgrades. Instead, developers can focus on their own applications.

Introduction to Cloud Computing
Cloud Computing Reference Model

Platform as a Service (IaaS): Advantages

Introduction to Cloud Computing
Cloud Computing Reference Model

Advantages of PaaS

PaaS provides many benefits, such as:

1) Cost-effective and continuous development, testing, and deployment of applications, as you don’t need to manage the underlying infrastructure.
2) High availability and scalability.
3) Straightforward customization and configuration of an application.
4) Reduction in development effort and maintenance.
5) Security policy simplification and automation.

Introduction to Cloud Computing
Cloud Computing Reference Model

Platform as a Service (IaaS): Disadvantages

Introduction to Cloud Computing

Cloud Computing Reference Model

Disadvantages of PaaS

PaaS solutions have some limitations as well:

1) **Integrations**: Having multiple parties responsible for the technology stack creates complexity in how integrations must be performed when developing applications.
2) **Data security**: The data will reside in a third-party environment when running applications using a PaaS solution. This poses concerns and risks. There might also be regulatory requirements to be met to store data in a third-party environment. Customers might have policies that limit or prohibit the storage of data *off-site*.

Introduction to Cloud Computing

Cloud Computing Reference Model

Disadvantages of PaaS

3) **Runtime issues**: PaaS solutions may not support the language and framework that your application may require.
4) **Legacy system customization**: Existing legacy applications and services might require more integration work. Instead, complex customization and configuration needs to be done for legacy applications to integrate with the PaaS service properly. The result might yield a non-trivial implementation that may minimize the value provided by your PaaS solution.

Introduction to Cloud Computing

Cloud Computing Reference Model

Disadvantages of PaaS

5) **Operational limitations**: Even though you have control of some of the layers in the PaaS stack, other layers are controlled and maintained by AWS. If the AWS layers need to be customized, you have little or no control over these optimizations.

Introduction to Cloud Computing
Cloud Computing Reference Model

Software as a Service (SaaS)

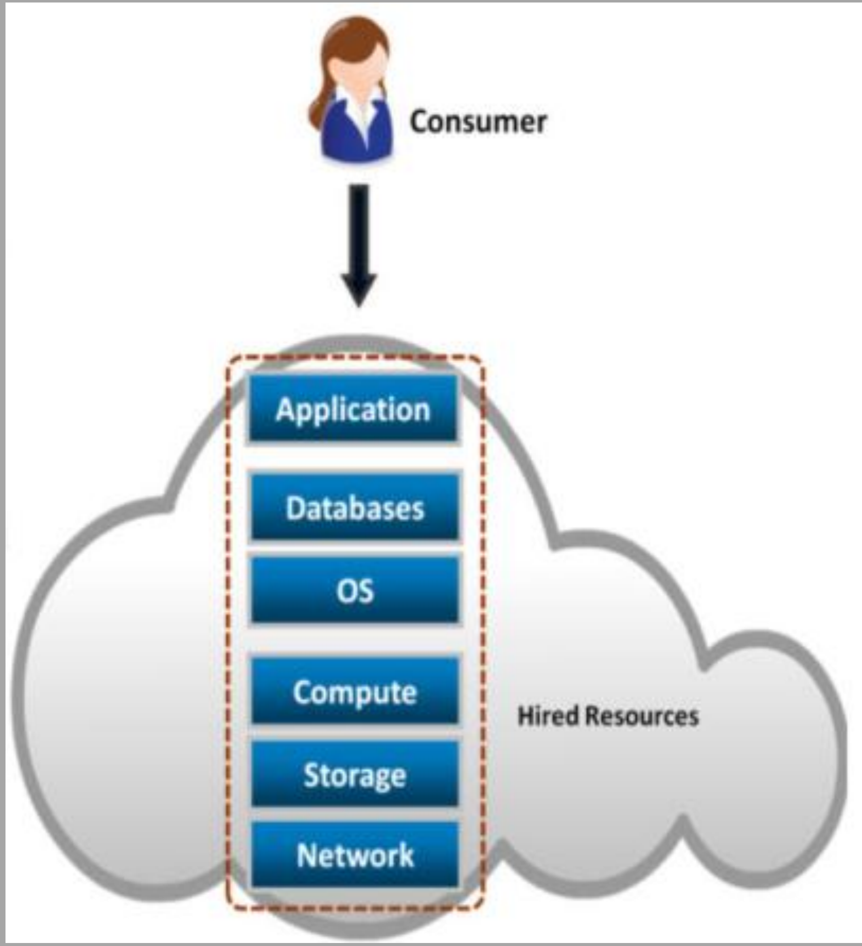
Introduction to Cloud Computing

Cloud Computing Reference Model

Cloud computing models:

Software as a Service (SaaS)

- Capability provided to the consumer to use provider's applications running in a Cloud infrastructure.
- Complete stack including application is provided as a service.
- Application is accessible from various client devices, for example, via a thin client interface such as a Web browser.
- Billing is based on the application usage.



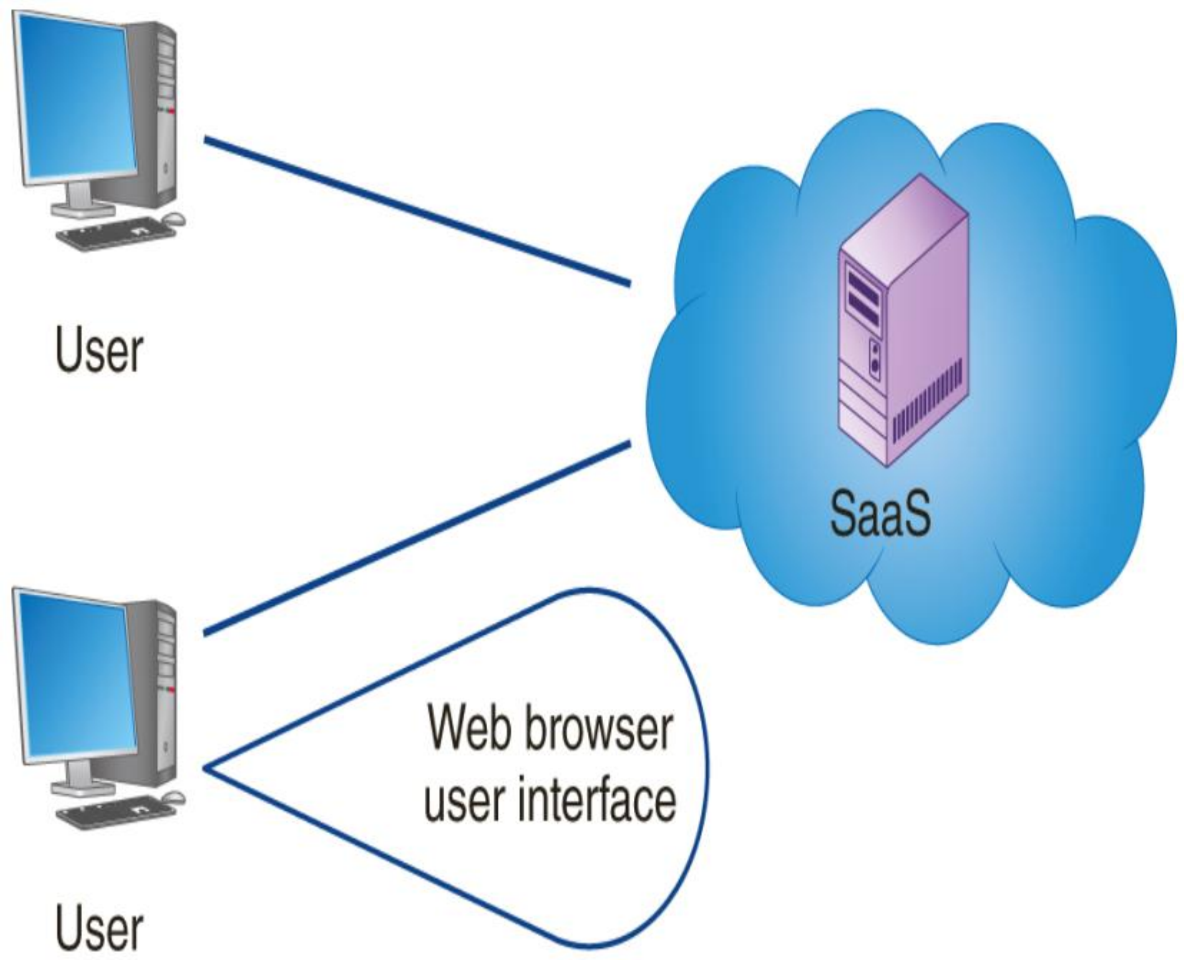
Introduction to Cloud Computing
Cloud Computing Reference Model

Cloud computing models

Software as a Service (SaaS)

- Software as a Service (SaaS) provides a completed product that is run and managed by the service provider. In most cases, Software as a Service refers to end-user applications. With a SaaS offering one does not worry about how the service is maintained or how the underlying infrastructure is managed; one thinks only about how to use that particular piece of software.
- A common example of a SaaS application is web-based email which you can use to send and receive email without having to manage feature additions to the email product or maintain the servers and operating systems that the email program is running on.

Introduction to Cloud Computing
Cloud Computing Reference Model



Cloud computing models:

Software as a Service (SaaS)

- The **software as a service (SaaS)** model provides a cloud-based foundation for software on demand. In general, an SaaS solution is web-delivered content that users access via a web browser. The software can reside within any of the deployment-model clouds.

Introduction to Cloud Computing

Cloud Computing Reference Model

Software as a Service (SaaS): Advantages

Introduction to Cloud Computing
Cloud Computing Reference Model

Advantages of SaaS

SaaS has several advantages:

1) Reducing the time, money, and effort spent on repetitive tasks.
2) Shifting the responsibility for installing, patching, configuring, and upgrading software across the service to a third party.
3) Allowing you to focus on the tasks that require more personalized attention, such as providing customer service to your user base

A SaaS solution allows you to get up and running efficiently. This option, versus the other two solutions, requires the least effort. This option enables companies big and small to launch services quickly and finish a project on time.

Introduction to Cloud Computing

Cloud Computing Reference Model

Software as a Service (SaaS): Disadvantages

Introduction to Cloud Computing

Cloud Computing Reference Model

Disadvantages of SaaS

SaaS solutions have some limitations as well:

1) **Interoperability**: Interoperability with other services may be complex. For example, if you need integration with an on-premises application, it may be more complicated to perform this integration because the on-premises installation uses a different interface, complicating the integration. Further, the on-premises environment is an assortment of technology from different vendors, making it challenging to integrate.

Introduction to Cloud Computing
Cloud Computing Reference Model

Disadvantages of SaaS

2) **Customization**: The convenience of having a vendor such as AWS manage many things for you comes at a price. Opportunities for customization in a SaaS solution will not be as great as with other services that are further down in the stack. For example, an on-premises solution that offers complete control of all levels in the stack will allow full customization.

Introduction to Cloud Computing
Cloud Computing Reference Model

Disadvantages of SaaS

3) **Lack of control**: If your organization requires that you only use a particular approved version of an operating system, this may not be appropriate. For example, there might be a regulatory requirement requiring detailed testing approval of the underlying operating systems, and if the version is changed, a retest and approval are required. In this case, SaaS will most likely not be an acceptable solution.
4) **Limited features**: If the SaaS solution you are using does not offer a feature you require, you might only be able to use that feature if the SaaS vendor provides that feature in the future.

Introduction to Cloud Computing
Cloud Computing Reference Model

Which the suitable cloud service model?